

COMPUTER OPERATIONS & PACKAGES

MODULE

PART 1

DEFINE A COMPUTER

- An electronic machine that works under the control of stored programs to accept, process & output data/information for use by the operator.
- A device that accepts data, processes the data in accordance with a stored program, generates results, and usually consists of input, output, storage, and arithmetic, logic, and control units.
- A functional unit that can perform substantial computation, including numerous arithmetic operations or logic operations, without human intervention during a run.
- It is an electronic device/machine that ***accepts data*** (raw facts & figures) as input and processes (*works on it*) to produce ***information*** (data converted to meaningful form) as output.

The computer is made up of hardware, software and peripheral devices.

Hardware- refers to the tangible, physical and mechanical components of a computer.

Software- refers to the intangible computer components, which are the programs or instructions of the computer. Everything that a computer does, it acts under instructions written out as computer programs. The hardware needs these instructions to function

Computer program:

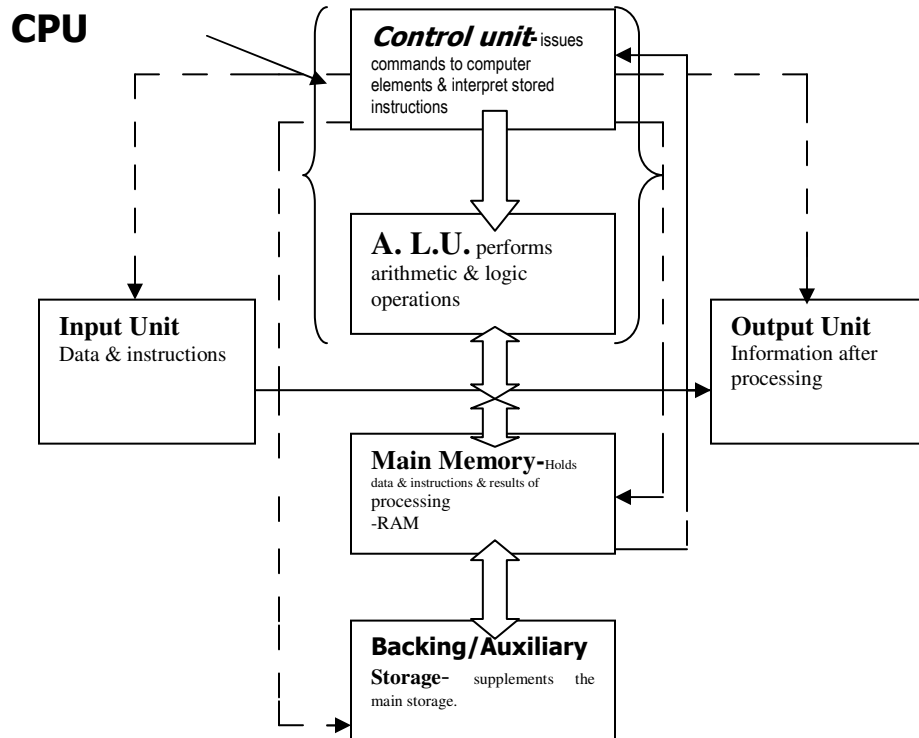
- A set of instructions, written in a specific programming language, which a computer follows in processing data, performing an operation, or solving a logical problem. See also software.
- A computer program is a set of statements or instructions to be used directly or indirectly in a computer in order to bring about a certain result.

NB: Software alone is useless, so is the hardware. The two are inseparable.

Peripheral Devices- These are devices that are external (not part of the computer) but can be connected to a computer. They are not essential for the functioning of the computer but can be connected to the computer for specific reasons. Examples are scanners, modems, printers, speakers, digital cameras, etc.

COMPUTER OPERATIONS & PACKAGES [C.O.P.]

A DIAGRAM SHOWING THE COMPONENTS OF A COMPUTER SYSTEM.



The CPU consists of the ALU & CU

KEY

Data Instructions



--->
Command Flow

DESCRIBE THE FUNCTIONS OF EACH COMPONENT.

Main memory	Auxiliary or backup storage
Stores programs and data while computer is running for current use	Store data or programs for later use
Main memory is fast and limited in capacity	Slow and capacity vary
Cannot retain information when the computer is switched off - Volatile	Non volatile
Memory directly connected to the CPU - not portable	Portable- can be transported from one machine to another.
Primary storage	Secondary storage
PROVIDES FAST ACCESS	ACCESS IS SLOW
Temporary	Permanent
Computer cannot do or work without	Can do or work without

PROCESSING DEVICE

The only processing device on a computer is **The Central Processing unit** known as the **CPU**. It is the main part of the computer as all other devices are built around it. It can be referred to as the heart or brain behind the computer.

Functions of the Central Processing Unit (CPU)

- a) It interprets the instruction in the program according to the type and sequence of processing operations that are to be performed on input data.
- b) Performs the varied arithmetic and logic operations that are necessary to convert input data into output information.
- c) Stores intermediate results of data being processed on its registers.
- d) Directs or controls output to all other devices (input, output, auxiliary storage units), telling each what to do and when to do it.

CPU COMPONENTS

It consists of the control unit (CU) and Arithmetic and Logic Unit (ALU). In most computers the two components - CU and ALU units - are combined in a unit called the microprocessor (CPU).

The Control Unit

The Control Unit manages input/output to and from the main memory and to and from auxiliary storage units and output devices. It interprets instructions in their sequence, It determines what is to be done to follow the instruction and, transmits to the appropriate device directions specifying the work to be done. For example, it advises each input device what data to transfer, when to transfer it, and when to send the data. It arranges for data to move from the main memory to the ALU and spells out the calculating and logic operations that are to be performed, and then arranges to send the processed results to main storage or to an output device for printing or display.

The ALU

The Arithmetic Logical Unit has 2 functions i.e. to perform calculations and logic operations at a fast speed (measured in nanoseconds, i.e. billionths of a second). It consists of registers (special storage locations to hold the data to be processed), and an accumulator to store intermediate results of operations. Logic operations compare values for greater, smaller or equal.

CPU speed

The CPU speed is measured in Megahertz (MHz) being millions of instructions that can be executed per second (MIPS). Each computer has specified, the processor speed.

INPUT DEVICES

These are some devices that feed data into the computer for processing. Some examples include:-

1. KEYBOARD
2. MOUSE
3. TRACKBALL
4. LIGHT PEN
5. TOUCH SCREEN
6. MODEM
7. SCANNER
8. MICROPHONE
9. BAR CODE READER

10. DIGITAL CAMERA
11. VIDEO DIGITISER
12. MAGNETIC INK CHARACTER RECOGNITION (MICR)
13. MAGNETIC STRIPS OR STRIPES READER
14. KEY TO DISK
15. GRAPHICS TABLET
16. OCR
17. OMR
18. VOICE RECOGNITION SYSTEM

Storage devices can be used as input devices

a) **KEYBOARD**

The keyboard is an **input device** with an arrangement of keys, like those on a typewriter, and an additional set of keys that fall into **5 basic categories**:

1. **Function keys**- keys form the first row of keys. They perform special operations in applications e.g. F7 activates Spell Checking Program in word processing and F1 activates the Help facility in any application.
2. **Alphanumeric keys** (like those on typewriter) are arranged in a QWERTY series. First is a row of numbers with some characters like %^&*()@!~ above them, then letters of the alphabet, the spacebar at the bottom, the Shift key on both the left and right side of the Spacebar.
3. **Numeric keypad**- serves 2 functions. At the top of the keypad is the NumLock key, which locks the keypad into Number mode when, pressed. When in Number mode, the keys function as number keys arranged like a standard calculator, otherwise it is in Cursor control mode, which allows you to move the cursor up, down, forward or backward by pressing the appropriate arrow key. When in cursor mode you can also use the home, page up, page down, end, delete and insert keys from Num Lock pad.
4. **Cursor control keys** are the arrow keys as well as Home, End, Ins, Page Up etc. Their major function is to place the cursor in an appropriate position for reading, appending or editing text.
5. **Combination keys** are the **Shift Ctrl** and **Alt** keys. On their own they are ineffective but combined with other keys they work. E.G. The Shift key, can be used to type to type the \$ sign (you would keep Shift pressed as you press the \$ sign key once).

b) **MOUSE**

The mouse is both input and pointing device. It is a hand held device with a rotating ball (**trackball**) underneath and Left & Right buttons, which enable the user to move the cursor (the blinking item on your screen) around the screen and to click on objects. The trackball is made from a ball in a socket, which is used to move the cursor on a screen. The cursor moves in the direction the ball moves as it is rolled.

c) **JOYSTICK**

The joystick is an Input device. A lever input device that can be moved in many directions, e.g. Up, Down, Left or Right. It is normally used with computer games to control the movements of objects on the screen.

d) **LIGHT PEN**

This is an Input device. An electrical device used with a monitor. It has the shape of an ordinary pen. One writes on a monitor with it and it senses the light from the position on the screen and transmits the information to the computer.

e) TOUCH-SENSITIVE SCREEN/ TOUCH PAD

Input device used in public information kiosks. The user simply touches the screen at the point of interest to get what (s)he wants.

f) SCANNER – Input device

It works like a photocopier but copies an image from text to computer mode. There is need for OCR (Optical Character Recognition) software to change the text into digital form, and to be able to modify the scanned image.

g) MICROPHONE

This is Input device. This is used in conjunction with voice recognition programs. It converts words into digital mode for storage. The sound can be played back.

h) DIGITAL CAMERA

This is Input device. It works like a standard camera except that it does not use photographic film. The images are recorded digitally in the camera's memory. Images are transferred onto the computer, and can be printed.

MORE INPUT DEVICES

1. **KEY TO DISK**- An early stand-alone data entry machine that stored data on magnetic disk for computer entry. The key to disk machine was the successor to the key-to-tape machine, which was the first major breakthrough in data entry since the punch card.

2. **BARCODE READER**

Barcodes are different groups of vertical bars that can be read by an optical **scanner**. Barcodes are printed on nearly every product that you can buy. Shops use barcodes because they are cheap to produce and very durable.

3. **CONCEPT KEYBOARD**

This is a flat board that contains a grid of buttons. Each button can be programmed to follow instructions. An overlay sheet is placed on the grid with an explanation for each button. They are used in primary schools with young children. Many modern fast food restaurants have overlays with either a description or picture of the meals that are available to make ordering easier.

4. **DIGITAL CAMERA**

A **digital** camera allows you to take pictures and to store a digital photographic image that can be read by a computer. You can then transfer the images directly from your camera on to your computer.

5. **GRAPHICS TABLET**

This consists of a flat pad (the tablet) on which the user draws with a special pen. As the user draws on the pad the image is created on the screen. Using a graphics tablet a designer can produce very accurate on-screen drawings.

6. **MICR (MAGNETIC INK CHARACTER READER)**

Magnetic ink characters are the strange looking numbers that appear at the bottom of cheques. These characters are used because they are very difficult to forge or damage. Banks use MICR to read the numbers from the bottom of cheques to obtain **data** such as account numbers and bank sort codes.

7. **MAGNETIC STRIP (OR STRIPE) READER**

Magnetic strips are built into many plastic cards such as cheque guarantee cards, cash-point cards and personal identity cards. The magnetic strip on the back of the card can hold the personal details of the card owner and, with the necessary PIN, will allow access to secure information e.g. bank account details. Data stored on the strip is scanned and input into a computer system by a reader.

8. **MIDI (MUSICAL INSTRUMENT DIGITAL INTERFACE)**

A MIDI system allows you to attach a musical instrument to your computer, for example a musical keyboard. You can then store a musical sequence on a computer and then have the computer play it back to you.

9. **OMR (OPTICAL MARK READER)**

An OMR reads marks made by pencil on a printed form into the computer. OMR systems are suited to reading pre-printed forms and check-boxes such as National Lottery number selection sheets and multiple-choice exam papers.

10. **SCANNER**

Scanners are now a cheap and common way of getting images into a computer. They can also be used with **OCR** (Optical Character Recognition) software to scan in text.

11. **TOUCH SCREEN**

A touch screen is a special type of **VDU**, which has a grid of light beams or fine wires criss-crossing the screen. When the screen is touched (usually to choose an on-screen option), the computer senses where you have pressed. Due to the vast amount of information they can store, touch screen operated computers are generally used in the provision of public information and are found in places such as airports.

12. **VIDEO DIGITISER**

A video digitizer takes an image from a video camera or television and converts it so that it can be used by, and stored on, a computer. Video sequences captured using a video digitizer is often used in multimedia presentations.

13. **VOICE RECOGNITION SYSTEM**

A microphone is used with a voice recognition system. This can be used with a word processing program to enter text. It can also be used as part of security systems - only certain digitally stored voices have access.

TEXT INPUT DEVICES	• Chorded keyboard ,GKOS keyboard ,Keyer ,Telegraph key (~20 WPM Morse code) • Vibroplex (30–80 WPM Morse) ,Keyboard, Typing ,Computer keyboard • Handwriting recognition ,Optical character recognition ,Speech recognition
POINTING DEVICES	• Light pen,Light gun,Cyberglove ,Touch screen ,Head pointer ,Eye gaze/eye tracking • Computer mouse ,Trackball ,Touchpad ,Pointing stick ,Graphics tablet (or digitizing tablet) ,Stylus
GAMING DEVICES	• paddle, Power Pad, Joystick ,Gamepad (or joypad)
IMAGE, VIDEO INPUT DEVICES	• Digital camcorder ,Webcam, Digital video recorder,Image scanner • 3D scanner ,Digital camera
AUDIO INPUT DEVICES	• Microphone (see also speech recognition) ,Digital audio recorder ,Digital dictaphone

OUTPUT DEVICES

1. LOUD SPEAKER

Allows a user to hear sounds and music from the computer provided the system has installed a sound card.

2. MODEM

An Input/ Output electronic device with circuits which changes data to be transmitted from its machine form (digital form) into a form suitable for transmission over the telephone line (analogue form). At the receiving end it converts data vice-versa. It is only necessary when one needs to communicate information over a long distance and when connected to the internet.

3. HEADPHONES

Headphones give sound output from the computer. They are similar to speakers, except they are worn on the ears so only one person can hear the output at a time

4. PLOTTERS

A plotter can be used to produce high quality, accurate, A3 size or bigger drawings. They are usually used for Computer Aided Design (**CAD**) and Computer Aided Manufacture (**CAM**) applications such as printing out plans for houses or car parts.

5. SPEAKER

A speaker gives you sound output from your computer. Some speakers are built into the computer and some are separate

6. VDU (Visual Display Unit)

The VDU is output device. It displays images and text in colour or black and white as it. It displays results of processing. It is also called a monitor.

7. PRINTERS

It produces output on paper i.e. hard copy or on other print-key media. The 2 types of printers are **impact** and **non-impact**.

Impact printers

The printing device being a series of pins have a direct impact on the paper. The pins form characters on the paper by pressing a printing element and an inked ribbon against paper. An example of such a printer is the Dot Matrix. The printers are reasonably fast, inexpensive and can produce both text and graphics. They also accommodate various paper sizes but are however noisy.

Non-impact printers

These are quieter printers and examples include LaserJet and inkjet printers. The **Laser printers** are a more popular choice for business use. They use a printing technique similar to that used in a photocopying machine. A laser beam is used to form an image onto the paper using toner. The print quality is high. Lasers are fast, and quiet. They print graphics but are more expensive.

The ink-jet printers are mainly colour printers which mix ink depending on the image colour and spray it onto the paper. The toner has the 4 basic colours. They are slow, expensive and have a high print quality.

What is the difference between impact and non-impact printers?

Impact	Non impact
Makes noise when printing	Quiet printers
Print head makes contact with the paper	No contact <i>creates an image without striking a ribbon against the paper. Creates images on a surface without contacting it.</i>
Strike ribbon against the paper	Use a laser beam which form an image onto paper using toner
Slow printer	Fast printers
Use ribbons or ink	Use toner
Has too many moving parts hence unreliable	Has fewer moving parts hence they are reliable.
Impact printers are relatively inexpensive, and the cost of printing is also low because ink ribbons are inexpensive.	Very expensive computers

Give **two** advantages of non-impact printers over impact printers.

- Have high print quality
- They are fast and do not make noise
- They are reliable because they have less moving parts
- See above [non impact section]

Give **three** examples of impact printers & **three** examples of non-impact printers.

Impact printers	Non impact printers
Dot matrix	Laser
Daisy wheel printers	Inkjet
Line printers	LED
Chain printers	Thermal
	Magnetic
	Electrostatic
	Xerographic

NON-IMPACT PRINTERS

1. Laser printer - A type of printer that utilizes a laser beam to produce an image on a drum. The light of the laser alters the electrical charge on the drum wherever it hits. The drum is then rolled through a reservoir of toner, which is picked up by the charged portions of the drum. Finally, the toner is transferred to the paper through a combination of heat and pressure.

Advantages

- Monochrome or four color
- High-quality print
- Capable of printing an almost unlimited variety of fonts
- Quiet operation

Disadvantages

- Color laser printers tend to be about five to ten times as expensive as monochrome
- Cannot print on multiple-copy paper

2. Ink-jet printer - A type of printer that works by spraying ionized ink at a sheet of paper. Magnetized plates in the ink's path direct the ink onto the paper in the desired shapes. Ink-jet printers are capable of producing high quality print approaching that produced by laser printers

Advantages

- Inexpensive way to print full-color document
- Easily portable due to smaller mechanical parts than laser printers
- Quiet operation

Disadvantages

- Slow output
- Require a special type of ink that is apt to smudge on inexpensive copier paper
- Cannot print on multiple-copy paper

3. Thermal transfer printers - A type of printer that applies heat to a ribbon and the image or barcode is then transferred to labels or tags. This technology produces a higher quality, longer lasting image on the tag or label.

Advantages

- 1 to 2 years media shelf life
- Media is not heat sensitive
- Medium to high image quality
- Wide range of available types
- Recommended for industrial use

Disadvantages

- Special ribbons required
- Average operational cost
- Cannot print on multiple-copy paper
- Single-pass ribbons only

4. Direct thermal printers - A type of printer that applies heat directly to specially treated paper, labels or tags to form the image or barcode.

Advantages

- No ribbon required
- Low operational cost

Disadvantages

- Short media shelf life
- Very heat sensitive media
- Limited image quality
- Limited number of available types
- Not recommended for use in industrial, outdoor, or harsh environments
- Cannot print on multiple-copy paper

IMPACT PRINTERS

1. Dot-matrix printer - A type of printer that produces characters and illustrations by striking pins against an ink ribbon to print closely spaced dots in the appropriate shape.

Advantages

- Can print to multi-page forms-
- Moderately priced
- Multi-pass ribbons (saves money)

Disadvantages

- Noisy operation
- Low-quality output
- Slow print speed
- Multi-pass ribbons (decrease in quality with each pass)

2. Daisy wheel Printers

- Round disk found on older computer printers that extends a portion of the wheel making contact with ink ribbon that makes contact with paper creating the character. This procedure is repeated for each key pressed.
- **Daisy Wheel printer** - A printer that uses a wheel with all the characters on it to produce output. The wheel spins to the desired character and makes an imprint, then spins to go to the next character. It sounds like a little machine gun.
- These printers generally aren't used much anymore due to the decreased cost of other printers that can produce graphics as well as text. The one good thing about a daisy-wheel printer is that the text is generally very crisp, but even that advantage is nullified now with laser printers that can produce very crisp characters as well.

Advantages

-Best print quality of impact printers

Disadvantages

-Very slow

-Only one font can be used at a time because you must change out the Daisy Wheel to change fonts.

-Cannot print graphics

3. Chain and band printers - Uses characters on a band or chain that is moved into place before striking the characters onto the paper.

Advantages

Very fast

Disadvantages

-Very loud

-Very expensive

You have been delegated the task of selecting a new printer from a range of different options currently on the market. The machine has to serve a small network of microcomputers in your office. Software used includes a word processing package, database, spreadsheet, and presentation packages. Assuming that cost is not a major constraint, list 10 features of the printers, which you would investigate in your evaluation.

1. **Print quality**, for example the print from dot matrix printers varies with the number of pins and even the best are inferior to laser printers.
2. **Character formation**, whether made from dots as in matrix printer, or full characters as with daisywheel or laser printer
3. **Multi-part**, i.e. whether several copies can be printed at once, as with an impact printer.
4. **Fonts** or type style; early computer users were satisfied with any quality whatsoever, but the advances into areas such as word processing and desk top publishing have stimulated a demand for varying type faces or fonts within the same document
5. **Letter quality**, whether the quality is equivalent or superior to that of a good typewriter and so acceptable for the word processing of mail. Some printers have a choice of two print modes, draft quality and letter quality.
6. **Speed**, in terms of characters per second.
7. **Graphics** capabilities.
8. **Flexibility**, for example is it suitable for one task only, such as good letter quality but not graphics, as with daisywheel printers, or good graphics but slow speed and indifferent letter quality, as with some dot matrix printers.
9. **Sheet feeding**, in terms of single sheets, friction feed or sprocket feed.
10. **Capital cost** or what it costs to buy.
11. **Running cost** or what it costs in ribbons, print toner, special stationery & maintenance.

12. **Compatibility** with standard packages; many programs support a range of printers by generating the correct control characters for fonts, features like italics and underline, and for graphics.
13. **Robustness**; some very cheap printers are ideal at home but would not last very long in an office.
14. **Reputation of the manufacturer** as a supplier of printers.

STORAGE DEVICES

These are devices that store information permanently. They are also known as Auxiliary storage devices or Secondary storage devices. They include:-

a) THE HARD DISK

A fixed device for storing information within the computer system unit. The disk is not moveable and is also referred to as the C:\ drive. The amount of disk space a computer has determines the power of a computer. It provides fast access known as Random access and has high-speed storage.

b) FLOPPY DISKETTE

An external form of storage used for back up. Information can be deleted and rewritten and re-saved. The storage device is common for microcomputers. The disks are either High density or double density disks because data is stored more densely/ compactly than on regular density disks. Common size of disk is 3¹/₂ inches.

ADVANTAGES OF FLOPPY DISKETTES

1. They are cheaper than Hard disks and CD-ROMS
2. They are portable hence used to transfer data or information from one computer to another
3. They offer direct access and they are faster than magnetic tapes

DISADVANTAGES OF FLOPPY DISKETTES

1. They are prone to viruses
2. They are difficult to handle
3. They can easily be corrupted
4. They have a relatively short life span
5. They hold relatively small amounts of data
6. They are not always 100% reliable

CARE AND HANDLING OF A FLOPPY DISKETTE

1. Don't put the diskette near any magnetic field.
2. Don't put the diskette near any electronic motor.
3. Don't expose the diskette to direct sunshine or heat.
4. Store the diskette in a cool dry dust-free environment.
5. Don't spill liquid onto the diskette.
6. Don't use any diskette which has had liquids spilled on it.
7. Don't bend or fold a diskette.
8. Don't staple labels onto the diskette.
9. Use a soft pencil or felt-tip pen when writing on diskette labels, as the pressure of a ballpoint may leave indentations on the magnetic surface.
10. Don't touch the magnetic surface (the black plastic like plate) of the disk, as your fingerprints may hinder the drive in reading from and writing to the disk .

c) CATRIDGE TAPE

Data is recorded in the form of magnetized spots on an iron oxide coating of a plastic tape. Gaps are used to separate individual data records. The tape is similar to audiotapes. Access of information from the tape is in sequential form and thus slow.

d) CD-ROM (Compact Disk Read Only Memory)

Beams of laser light are used to write on optical disks by magnetic means. After writing, the disk can only be read from and not written to. Access to information is direct and very fast.

e) WORM (Write Once, Read Many)

WORM Disks allow microcomputers with the proper drive known as a CD Rewriters (or a burner) to record own data on a blank CD. Once the recording is done, the disk can only be read from and cannot be rewritten.

Erasable optical disks are also available and are known as EPROM. The major advantage of optical disks is their storage capacity. The information on one optical requires multiple floppies.

f) DVD (Digital Versatile Disks)

A form of optical disks similar to CD ROMS but has much more storage capacities of up to 3.9GB. They support audio and video as well as data on the same disk.

g) ZIP DISKS

These are high capacity removable diskettes. They have gained favor for back-up purposes. Some zip disks use magnetic technology and others use laser. They have a capacity of about 100MB.

Storage devices differ in terms of capacity, cost and access methods. For any device to be used there is a need for the device drive on the computer e.g. A floppy disk drive.

DISK CARE

- 1) Do not expose them to sun or a hot environment.
- 2) Keep them away from moisture or water.
- 3) Keep them away from magnetic fields i.e. Phones, tops of some monitors, magnetized items etc..
- 4) Do not touch the actual tape.
- 5) Keep them away from dust.
- 6) Do not place heavy objects on them.
- 7) Write on a floppy disk label only with a felt-tip pen, or write before sticking the label
- 8) Avoid using rubber bands, paper clips and erasers on floppy disks

HARDWARE CARE

- 1) Make sure there is enough space for the machines to breathe. (Fresh air is necessary to make the fan keep the electronics cool).
- 2) Do not block air vents with books.
- 3) Keep the computer dry.
- 4) House the machine in a dust free environment e.g. 1ST floor carpeted-room (Carpets suck dust) and, use a vacuum cleaner to clean the carpet. (Dust is the biggest killer of hard drives).
- 5) Do not expose the machine to extreme temperatures.
- 6) Properly shut down the machine before turn-off.
- 7) Pull the plug out of socket when there is thunder and lightning.
- 8) Do not move the system when the computer is on. The hard drive can be damaged.
- 9) Use of UPS- Uninterruptible Power Supply in case of power failures,
- 10) Use of Surge Protectors against too high voltages.

DEFINE HARDWARE & GIVE EXAMPLES.

- Hardware comprises all of the physical parts of a computer, as distinguished from the data it contains or operates on, and the software that provides instructions for the hardware to accomplish tasks.

DEFINE SOFTWARE & GIVE EXAMPLES.

- Computer software (or simply software) refers to one or more computer programs and data held in the storage of a computer for some purpose.
- Program software performs the function of the program it implements, either by directly providing instructions to the computer hardware or by serving as input to another piece of software.

TYPE OF COMPUTERS/COMPUTER CATEGORIES/MAJOR CLASSES

- Supercomputers
- Mainframes
- Minicomputers
- Microcomputers

Supercomputers

- ☑ Describes a category of extremely powerful computers specifically designed for high-speed numeric computation.
- ☑ The computers can process hundreds of millions of instructions per second.
- ☑ High capacity computers capable of executing instructions faster than main frame & are extremely expensive.
- ☑ The speed of executing these instructions generates extreme heat, and therefore the computers require special cooling system to operate.
- ☑ Uses include weather forecasting, scientific and mathematical research, and design of high-speed aircraft and space exploration.

Mainframes

- ☑ Are large, powerful computers that are physically larger than micros and minis and usually have one or more central processors with faster instruction processing speeds?
- ☑ They typically process hundreds of millions of instructions per second.
- ☑ Mainframes have large primary storage capacities.
- ☑ Many mainframe models have the ability to service hundreds of users at once.
- ☑ Some commercial organizations require large amounts of data to be processed in the list possible time.
- ☑ Mainframes allow one to perform such functions
- ☑ Uses of mainframes include: data warehousing, commercial airline ticketing & reservations, government record keeping & financial servicing
- ☑ Application categories: Host computers, Database servers, and Transaction processors.

Minicomputers

- ☑ Are larger and more powerful than most microcomputers but are smaller and less powerful than most mainframe computer systems.
- ☑ Serve in industrial process-control manufacturing plant computers and play a major role in CAM.
- ☑ Also take the form of powerful technical workstations for CAD applications.
- ☑ Often used as *front-end* processors/computers to help mainframe computers control data communications networks with large numbers of data entry terminals.
- ☑ Also used as powerful *Network servers* to help manage large interconnected LANs that tie together many workstations
- ☑ Downsizing is a term that was commonly used in the early nineties when smaller computers, capable of much of the functions of mainframes, began capturing the computer market.
- ☑ Minicomputers are suitable for mid-sized companies that require the computing power of mainframes to be efficient and at the same time be cost effective.
- ☑ Users of mini-computers would be smaller banks & financial institutions, supermarket chains and departmental stores.
- ☑ Application categories: Network servers, and Application systems

Microcomputers

- ☑ We refer to a microcomputer as a personal computer or PC.
- ☑ Microcomputers categorized by size include: *Desktops*, *Laptops* for example.

COMPUTER OPERATIONS & PACKAGES [C.O.P.]

- ☑ Most microcomputers are single user computers.
- ☑ The late nineties have seen a huge influx of microcomputers whose sole aim was to provide affordable computing power to the general public.
- ☑ They are now used as Network servers. The demand for microcomputers and the changes in microchip technology have produced cheaper computers that are affordable to students and the general public.
- ☑ Can support CAD.
- ☑ Users of microcomputers range from students, who use them for word processing, to a salesperson. Who depend on the microcomputer for information?
- ☑ Application categories: PCs, Network servers.
- ☑ **Network servers** are powerful microcomputers that controls & coordinates communication & resource sharing in LANs of interconnected PCs & other devices.

The main differences among the above computer types are in:

- Processing speed,
- Memory capacity
- Number and capabilities of peripheral devices that can be attached to the computer
- Usage.

COMPUTER APPLICATIONS

- a) **Business-** for administration, payroll and accounting.
- b) **Government-** keeping database records.
- c) **Education-** Record keeping, accounting and training in the classroom.
- d) **Health-** Record keeping, life-support systems.
- e) **Sport-** fitness monitoring.
- f) **Art and Design-** desktop Publishing, technical graphics, Computer Aided Design.
- g) **Homes-** PCs for home accounts, games, e-mail, online shopping, home education etc.
- h) **Banks-** Record keeping and updating, on-line Automated Teller Machines (ATMs).
- i) **Supermarkets-** Electronic point Of Sale Machines (EPOS) – tills.

FACTORS TO CONSIDER WHEN YOU WANT TO PURCHASE OR BUY A COMPUTER

1. **COST; how much can I afford to pay for a computer?** Prices of personal computers range from a few hundred to thousands of dollars. Faster and more feature-rich PCs are usually more expensive.
2. **AREA OF USE; where will my new PC be used?** If you will be using it only in your home or office, a desktop **computer** will be suitable. However, if you will need to take it with you, you should consider purchasing a laptop (notebook) **computer**.
3. **APPLICATIONS PACKAGES; Which application packages will I run on my computer?** Make a list of applications for which you plan to use your PC. For example, will you use your PC to prepare letters and reports? Analyze numeric and financial data? Prepare visual presentations? Access the Internet? Listen to music? Create and work with graphics?
4. **DURATION; How long will I keep this computer?** Try to estimate the length of time (years) you will use your **computer** before BUYING the next one. If you expect use your PC for several years or if you think you may later want to add new applications, CONSIDER one that is expandable, so you can add new components, such as a modem, printer, or add-on boards. Make sure the PC has expansion slots so you can connect these devices to your PC.

5. **MANUFACTURER AND SELLER REPUTATION. Check out the manufacturers and seller's reputations.** Talk with friends, co-workers, classmates, teachers, and others about their PCs. Ask about the performance of their PCs and get recommendations from persons you trust. Eventually you may need to have your PC serviced. Ideally, the vendor has a service department that can repair your PC. If not, you may need to locate a third-party to provide this service. Reputable computer manufactures include:
1. Hewlett-Packard 2. Apple 3. Sun Microsystems 4. Dell 5. Panasonic 6. Sony 7. Toshiba 8. Shuttle 9. IBM 10. Samsung Electronics 11. Hitachi 12. Fujitsu Siemens
6. **WARRANTIES & GUARANTEES**
7. **PC ARCHITECTURE.** PC architecture refers to the design and construction of the PC and its system unit, and not all architectures are the same. For example, the architecture of an Apple Macintosh differs from that of an IBM or compatible PC. Therefore, software written for an Apple Macintosh PC may not run on an IBM or IBM-compatible PC. Although some users prefer a Macintosh PC, more software is available for IBM and IBM-compatible PCs.
8. **MICROPROCESSOR SPEED.** Selecting the right microprocessor is extremely important. Processing speed, measured in megahertz (MHz) or gigahertz (GHz), is probably the first consideration. The higher the number of MHz or GHz, the faster the processor will manipulate data. If speed is important, consider choosing a microprocessor with a speed of 2 GHz or more. PCs containing microprocessors with speeds up to 10 GHz and higher are available.
9. **MAIN MEMORY (RAM).** Main memory (RAM) is needed for the temporary storage of programs and data while the data is being processed. Some application software requires a considerable amount of RAM to function properly, and newer software versions usually require more RAM than older versions. Typical PCs now come with 2 gigabytes (GB), 5GB, or more of RAM. Make certain the PC has sufficient RAM to run the software you will be using.
10. **SECONDARY STORAGE (ROM).** What type(s) and amounts of secondary storage are you likely to need? Typical computers come with a floppy disk drive, cd drive, USB port/ drive and a hard disk drive already installed. A hard disk drive contains one or more rigid storage platters and provides for the permanent storage of considerably more data. However, the disk itself cannot be removed from the drive.

The STORAGE CAPACITY of a hard disk is an important consideration because it is used to store all system and application software. Typical hard disk capacities are 100, 200, 500GB or 2TB and more. Be certain the PC you are considering has sufficient secondary capacity for your needs
CD-ROM or DVD drives provide high storage capacities. Most software publishers store their programs on CD-ROM or DVDs because of the high capacity and because DVD/CD-ROMs operate faster than floppy disks.
If you will use your PC to play movies, your purchase should include a DVD (digital video disk) drive. If you will work with large files, consider purchasing a computer that includes a DVD/CD-RW drive. A DVD/CD-RW disk is a reusable high-capacity disk that allows you to store huge amounts of data and to erase data no longer needed.
11. **PORTS.** The number of ports (slots) available inside the system unit determines the number of add-on boards that can be installed inside the system unit. For example, you cannot connect a printer to your computer unless your computer is equipped

with an appropriate port (slot). Be sure the PC contains the necessary slots for the add-on boards and peripheral devices you will be using with your computer.

12. **INPUT DEVICES.** Typical input devices are a keyboard and a mouse, although other kinds of input devices are available. Most keyboards and mice operate similarly. However, there are slight differences in how each "feels" to the user. Before buying a PC, you may want to test the keyboard and mouse for comfort and ease of use. Some sellers will allow you to exchange the keyboard or mouse that comes with the computer for a different one.
13. **OUTPUT DEVICES.** Output devices produce output in either *soft copy* or *hard copy* form. Most PCs come with a monitor (for soft copy output), but you may have to purchase a hard copy device, such as a printer, separately.
14. **MONITORS.** There are wide differences among PC monitors, with resolution being perhaps the most important variable.

Resolution refers to the clarity of the text and images being displayed. Before making a purchase, carefully evaluate the monitor's resolution. Many vendors allow you to choose from monitors with varying resolutions. A resolution of 1,024 by 1,024 is considered high-resolution, in which text and images display exceptionally clearly. High-resolution monitors are typically more expensive.

Monitor size is another important consideration. Viewing areas range from 15 diagonal inches to 21 inches and higher. Larger monitors are usually more expensive, but may be worth the extra cost. For example, an individual with weak vision may prefer a larger monitor. If your desktop space is limited, consider purchasing a flat-panel monitor, which occupies less space. Flat-panels are more expensive than standard monitors, however.

NECESSARY CONDITIONS FOR AN IDEAL COMPUTER ROOM ENVIRONMENT.

1. Cleanliness [books, manuals not encouraged] because they bring dust & caught fire easily.
2. Air conditioners, which control the room and dehumidates it.
3. It should be dust free; dust is the biggest "killer" of hard drives.
4. No carpets, they caught fire more easily.
5. You are not allowed to eat, smoke or drink in the computer room.
6. No magnetic material should be in the computer room.
7. The computer room should be located in such a way that computers are not exposed to direct sunlight or hot environments, use curtains if otherwise.
8. The computer room should be fitted with burglar bars, alarm systems, or guarded 24hours to ensure security.
9. The computer room should have surge protectors or suppressors to guard against electrical faults and high voltages.
10. The computer room should have UPS [uninterrupted power supply] in case of power failures, which may be fatal.
11. Computer room should be well ventilated, to allow for air circulation,
12. Foreign media, such as floppy diskettes, flash disks should not be allowed in order to minimize virus spread.
13. Always switch off machines if not in use.
14. Avoid overloading circuits.
15. All cabling and wiring should not be exposed; it should be covered and should not criss cross the room.

16. Leave enough space between computers and the walls.
17. Do not use chalk the computer room , dry wipe markers are encouraged.

DUTIES & RESPONSIBILITIES OF COMPUTER PERSONNEL

ROLES OF A Database Administrator (DBA)

- a) The design of the database
- b) After the initial design, the DBA must monitor the performance of the database, and if problems surface (such as a particular report taking an unacceptably long time to produce), appropriate changes must be made to the database structure.
- c) Keeping users informed of changes in the database structure that will affect them; for example, if the size or format of a particular field is altered or additional fields added
- d) Maintenance of the data dictionary for the database, and responsibility for establishing conventions for naming tables, columns, indexes 7 so on.
- e) Implementing access privileges for all users of the database; that is, specifying which items can be accessed and / or changed by each user.
- f) Allocating passwords to each user.
- g) Providing training to users in how to access and use the database.
- h) Manage the organization of
 - i. -data resources
 - ii. -database plans
 - iii. -design
 - iv. -operations
 - v. -training
 - vi. -user support
 - vii. -security & Maintenance
- i) Maintain data consistency and security
- j) Approve access to data stored
- k) Approve access procedures
- l) ABILITY to delete, add, modify –existing data must be tightly controlled.

1. **COMPUTER SUPPORT SPECIALISTS** provide technical assistance, support, and advice to customers and other users. This occupational group includes *technical support specialists* and *help-desk technicians*. These troubleshooters interpret problems and provide technical support for hardware, software, and SYSTEMS. They answer telephone calls, analyze problems by using automated diagnostic programs, and resolve recurring difficulties. Support specialists may work either within a company that uses computer SYSTEMS or directly for a computer hardware or software vendor. Increasingly, these specialists work for help-desk or support services firms, for which they provide computer support to clients on a contract basis.
2. **TECHNICAL SUPPORT SPECIALISTS** answer telephone calls from their organizations' computer users and may run automatic diagnostics programs to resolve problems. Working on monitors, keyboards, printers, and mice, they install, modify, clean, and repair computer hardware and software. They also may write training manuals and train computer users in how to use new computer hardware and software. In addition, technical support specialists oversee the daily performance of their company's computer SYSTEMS and evaluate software programs with regard to their usefulness.
3. **HELP-DESK TECHNICIANS** assist computer users with the inevitable hardware and software questions that are not addressed in a product's instruction manual. Help-desk technicians field telephone calls and e-mail messages from customers who are seeking guidance on technical problems. In responding to these requests for guidance, help-desk technicians must listen carefully to the customer, ask questions to diagnose the nature of the

problem, and then patiently walk the customer through the problem-solving steps. Help-desk technicians deal directly with customer issues, and companies value them as a source of feedback on their products. These technicians are consulted for information about what gives customers the most trouble, as well as other customer concerns. Most computer support specialists start out at the help desk.

4. **NETWORK ADMINISTRATORS AND COMPUTER SYSTEMS ADMINISTRATORS** design, install, and support an organization's local-area network (LAN), wide-area network (WAN), network segment, Internet, or intranet system. They provide day-to-day onsite administrative support for software users in a variety of work environments, including professional offices, small businesses, government, and large corporations. They maintain network hardware and software, analyze problems, and monitor the network to ensure its availability to system users. These workers gather data to identify customer needs and then use the information to identify, interpret, and evaluate system and network requirements. Administrators also may plan, coordinate, and implement network security measures.
5. **SYSTEMS ADMINISTRATORS** are the information technology employees responsible for the efficient use of networks by organizations. They ensure that the design of an organization's computer site allows all of the components, including computers, the network, and software, to fit together and work properly. Furthermore, they monitor and adjust the performance of existing networks and continually survey the current computer site to determine future network needs. Administrators also troubleshoot problems reported by users and by automated network monitoring SYSTEMS and make recommendations for enhancements in the implementation of future servers and networks.
6. In some organizations, **COMPUTER SECURITY SPECIALISTS** may plan, coordinate, and implement the organization's information security. These workers may be called upon to educate users about computer security, install security software, monitor the network for security breaches, respond to cyber attacks, and, in some cases, gather data and evidence to be used in prosecuting cyber crime. The responsibilities of computer security specialists has increased in recent years as there has been a large increase in the number of cyber attacks on data and networks. This and other growing specialty occupations reflect an increasing emphasis on client-server applications, the expansion of Internet and intranet applications, and the demand for more end-user support.
7. **COMPUTER PROGRAMMERS** write, test, and maintain the detailed instructions, called programs that computers must follow to perform their functions. Programmers also conceive, design, and test logical structures for solving problems by COMPUTER. Programmers write programs according to the specifications determined primarily by COMPUTER software engineers and systems analysts. After the design process is complete, it is the job of the Programmer to convert that design into a logical series of instructions that the COMPUTER can follow. The Programmer codes these instructions in a conventional programming language such as COBOL; an artificial intelligence language such as Prolog; or one of the most advanced object-oriented languages, such as Java, C++, or ACTOR. Different programming languages are used depending on the purpose of the program.
8. **DATA PREPARATION STAFF**
Are those people responsible for the translation of data into machine readable form.
9. **COMPUTER SCIENTISTS** work as theorists, researchers, or inventors. Their jobs are distinguished by the higher level of theoretical expertise and innovation they apply to complex problems and the creation or application of new technology. Those employed by academic institutions work in areas ranging from complexity theory to hardware to programming-language design. Some work on multidisciplinary projects, such as developing and advancing uses of virtual reality, extending human-computer interaction, or designing robots. Their counterparts in private industry work in areas such as applying theory; developing specialized languages or information technologies; or designing programming tools, knowledge-based systems, or even computer games.

10. **DATA ENTRY AND INFORMATION PROCESSING WORKERS** help ensure the smooth and efficient handling of information. By keying in text, entering data into a computer, operating a variety of office machines, and performing other clerical duties, these workers help organizations keep up with the rapid changes that are characteristic of today's "Information Age." In addition to the job titles discussed below—such as word processors, typists, and data entry keyers—data entry and information processing workers are known by various other titles, including electronic data processors, keypunch technicians, and transcribers.
11. **DATA ENTRY KEYERS** usually input lists of items, numbers, or other data into computers or complete forms that appear on a computer screen. They also may manipulate existing data, edit current information, or proofread new entries into a database for accuracy. Some examples of data sources include customers' personal information, medical records, and membership lists. Usually, this information is used internally by a company and may be reformatted before other departments or customers utilize it.
12. **DATA CONTROL** is our customer services area for all agencies that utilize Information Technology's services for reports of any kind. This section receives and distributes all incoming data for appropriate processing; they set up batch reporting for all agencies and submit schedules for processing to Computer Operations. After processing has occurred, they check and distribute reports to all customers. This area interacts with agency customers and other Information Technology areas to resolve processing problems. This area maintains all tape media including updating the tape inventory system and it also has responsibility for off site tape storage. Updating the Information Technology web page using HTML and other software tools is done by Data control. Data control is also responsible for all data preparation prior to centralized processing. Assisting the paper peripheral area and reception area are also functions of this section.
13. **COMPUTER OPERATORS** are responsible for the operation of all enterprise, Unix, NT, and Windows 2000 servers, Unix and personal computers that are utilized to service the County's customer base. This area receives all hardware service calls, assists in terminal and communication equipment installation and software/hardware trouble diagnosis. Processing of all batch production work as well as all paper peripheral processing are also functions of Computer Operations.
14. **NETWORK MANAGER**, managing a network infrastructure and providing technical support and advice to the client organisations. This type of post would usually require the postholder to be educated to degree level or equivalent standard and to have considerable experience of network management and protocols including TCP/IP. Additional requirements might include experience in the management of industry standard network operating systems such as Novell NetWare, UNIX and MS Windows NT & 2000 server, together with skills in the use of database applications, such as Oracle, email server applications such as MS Exchange 5.5 or 2000 and PC applications such as MS Office & XP.

Research – important area

1. FIND OUT THE DUTIES AND RESPONSIBILITIES OF THE FOLLOWING

- | | |
|----------------------------------|-----------------------------------|
| a) Computer programmer | g) Data control staff |
| b) System analyst | h) IT manager |
| c) System administrator | i) Data processing manager |
| d) Computer librarian | j) Computer operators |
| e) Data preparation staff | k) Data capturing clerk |
| f) Network administrator | |

2. Write notes on

DATA PROCESSING METHODS

- | | |
|-------------------------------|----------------------------------|
| a) Manual data processing | f) Distributed data processing |
| b) Mechanical data processing | g) Centralized data processing |
| c) Electronic data processing | h) Transaction processing |
| d) Automatic data processing | i) Decentralized data processing |
| e) Online data processing | j) Time sharing data processing |

- k) Batch or offline data processing

3. Advantages and disadvantages of computers

Ask if you are not sure

DATA SECURITY /PROTECTION

❖ Data Security/Protection

- Refers to keeping data safe from various hazards or dangers like natural hazards, deliberate corruption or destruction of software & hardware by malicious or terrorist acts, illegal access to data by hackers [people who break into the system] & accidental destruction of data by hardware or software failure [operator error].

MEASURES TO ENSURE DATA SECURITY

1. **Data Encryption/Decryption-** Data is coded before transmission over a WAN & decrypted only if you have the key & code to decrypt the data on the receiving end. The process of transforming a message in ordinary language i.e. **plain text** to produce what is called **cipher text** which is then send along a communication line/link. The receiving computer uses another transformation to decode the message.
 - a. Definitions of **decryption** on the Web:
 - The process of converting encrypted content back into its original form, often the process of converting cipher text to plaintext. Decryption is the opposite of encryption.
 - Definitions of **encryption** on the Web:
 - Any procedure used in cryptography to convert plaintext into cipher text in order to prevent anyone except the intended recipient from reading that data. There are many types of data encryption, and they are the basis of network security. Common types include Data Encryption Standard and public-key encryption.
 - Definitions of **cipher text** on the Web:
 - Data that has been transformed by encryption so that its semantic information content (ie, its meaning) is no longer intelligible or directly available.

2. Firewall- a firewall is a software program designed to prevent unauthorized access to a PC or network through a connection to the Internet. It works by monitoring all data sent to and from the PC and comparing the data with a set of user-defined security criteria. Any data that does not meet that criteria is blocked. Firewalls also process encrypted data. They verify the validity of the user. User would require access to the firewall before they can transmit data. Latest firewalls also have the ability to detect virus software in packets of data that is sent through the network. Firewalls disallow data transmission if it detects strains of virus on the data being transmitted.

3. **Username & Passwords-** Passwords restrict only authorized personnel/users to have access to the data or computer rooms [cards]. Passwords bring in an aspect of accountability, if a file is deleted, the person with the password will be reported as the culprit through logging. Passwords can be forgotten. Widely used by companies to protect their equipment & data
4. **Authorized entry to computer installations-** Most installations have card readers, voice recognition systems or keypads on the doors that prohibit entry to unauthorized personnel.
5. Backing-up files on external disks periodically.
6. Keeping backup copies of files in a different location and making about 3 generations of backup.
7. Saving work frequently
8. Avoiding viruses.
9. Restricting access to the computer room. Access could be only through codes, voice recognition etc.
10. Installing fire alarms.
11. Lining computer rooms with fire resistant material.
12. Placing the computer room in upper floors of a building to avoid burglaries.

13. Having a security guard 24 hours a day.

Summary - Keeping data secure

Measures that can be taken to keep data safe include:

- Making regular back-ups of files. (**Back up** copies should be stored safely in fireproof safes or in another building.)
- Protecting yourself against viruses by running anti-**virus software**.
- Using a system of passwords so that access to data is restricted.
- Safe storage of important files stored on removable disks - eg locked away in a fireproof and waterproof safe.
- Allowing only authorized staff into certain computer areas, eg by controlling entry to these areas by means of ID cards or magnetic swipe cards.
- Always logging off or turning terminals off and if possible locking them.
- Avoiding accidentally deletion of files by write-protecting disks.
- Using data **encryption** techniques to code data so that it makes no apparent sense

❑ **BACK UP**

- Copying or saving data to a different location. One can restore the backup copy if data is lost or damaged.
- To create a copy of a disk's contents on another location for safe keeping. Since hard drives are not infallible, it is recommended that you backup its contents regularly.

DISASTER PLANNING

- Many companies have comprehensive emergency plans so that even after suffering a severe:
 - ☑ **Bomb**
 - ☑ **Fire damage**
 - ☑ **Natural disaster**, the company can be up and running within a day or two.
- A disaster recovery service/programme provides for example
 - ☑ **Office space**
 - ☑ **Computer facilities**
 - ☑ **Phones**
 - ☑ **Desks**, at an emergency **BACKUP** site.
- A completely up to date **copy of the company Database** may be permanently maintained at this site with all transactions being transmitted to this **remote site** updating the database, copying etc.
- The data should be **backed up/stored** to one or more media to ensure recovery in case of disaster.
- The back up procedures and media should be tested periodically to assess their effectiveness.

Definitions of **disaster plan** on the Web:

- The documented policies and procedures intended to either prevent damage, minimize damage, or recover from damage to record materials.

Definitions of **disaster recovery plan** on the Web:

- The document that defines the resources, actions, tasks and data required to manage the business recovery process in the event of a business interruption. The plan is designed to assist in restoring the business process within the stated disaster recovery goals.
- Part of an overall contingency plan. The plan for a process whereby an enterprise would restore any loss of data in the event of fire, vandalism, natural disaster, or system failure.

MAJOR THREATS TO INFORMATION SYSTEMS/COMPUTERS

Key threats to data security

- Data can be lost or damaged during a system **crash** - especially one affecting the hard disk.
- Data can become **corrupt** as a result of faulty disks or **disk** drives, or power failures.
- **Data** can be lost by accidentally deleting or overwriting files.
- Data can be lost or become corrupted by computer **viruses**.

COMPUTER OPERATIONS & PACKAGES [C.O.P.]

- Data can be **hacked** into by unauthorized users and deleted or altered.
- Data can be destroyed by terrorist activities, war, bombs and fire.
- Data can be deleted or altered by unpleasant employees wishing to make money or seek revenge on their employers.

Explain 5 major threats to information systems; for each threat describe its impact and preventative measure you would adopt to prevent it. [15].

THREAT	IMPACT	CONTROL
Hardware failure	• You are grounded • You cannot access the system • Data is incorrectly/ incompletely processed	• Have spares in the warehouse. • Do regular servicing • Have a disaster plan • Run hardware diagnostics frequently
Electrical faults	• Loss of data • Disk crushes • Damage to hardware	• Have standby generators • Install UPS [uninterrupted power supply] • Install Surge suppressors/ protectors
Software failures	• Operations are grounded	• Buy software from reputable vendors. • Do regular back ups • Software diagnostics
Natural disasters/ physical threats	• Absolute destruction	• Relocate backups to other areas free from disasters
Viruses	• Deletion & corruption of files • Reformatting of documents • System may fail to work	• Install a reliable anti-virus software • Never download unknown e-mail attachments • Scan unknown diskettes
Hackers, Computer crime	• Security is breached • Security is bypassed • Destruction of files • Data is stolen	• Change passwords regularly • Employ a security guard • Install a firewall • Prevent unauthorized access to computer facilities.
War and Terrorist activity		
Human error		

Definitions of **computer abuse** on the Web:

- The willful or negligent unauthorized activity that affects the availability, confidentiality, or integrity of computer resources. Computer abuse includes fraud, embezzlement, theft, malicious damage, unauthorized use, denial of service, and misappropriation

Definitions of **hacking** on the Web:

- Unauthorized use, or attempts to circumvent or bypass the security mechanisms of an information system or network. **Hacking** means illegally accessing other people's computer systems for destroying, disrupting or carrying out illegal activities on the network or computer systems.

DATA INTEGRITY/RELIABILITY

- Refers to the correctness AND The accuracy of data after being transmitted or processed
- Data in the computer system may become incorrect, corrupted or of poor quality in many different ways & at any stage during data processing.

❖ **Consequences of system failure:**

- Loss of business due downtime
- Delays
- Air traffic control system could well have catastrophic results

COMPUTER CRIME/FRAUD

❖ **Hacking i.e.**

Hackers are users of computer systems that gain unauthorized access into a computer system.

Cracking of ineffective security systems so as to gain unauthorized access to commercially sensitive or confidential personal files

❖ **Computer Fraud/Crime**

- Criminal actions accomplished through the use of computer systems, especially with the intent to defraud, destroy, damage, or make unauthorized use of computer resources.
- **E.g.** improper transfer of funds from one account to another

TYPES OF COMPUTER CRIME/FRAUD

- a) **Intellectual Crime** – cracking into a computer system for the purpose of transferring or obtaining funds is a typical e.g.
- b) **Destruction of property** – destroying a computer property
- c) **Theft** – Software piracy which is unauthorized copying of software. Hardware & software being taken away without the knowledge/consent of its owners.
- d) Fraud achieved by the manipulation of computer records.
- e) * Spamming wherever outlawed completely or where regulations controlling it are violated.
- f) * Deliberate circumvention of computer security systems.
- g) * Unauthorized access to or modification of programs (see software cracking and hacking) & data.
- h) * Intellectual property theft, including software piracy.
- i) * Industrial espionage by means of access to or theft of computer materials.
- j) * Identity theft where this is accomplished by use of fraudulent computer transactions.
- k) * Writing or spreading computer viruses or worms.
- l) * Salami slicing is the practice of stealing money repeatedly in extremely small quantities.
- m) * **DNS** Denial-of-service attack, where company websites are flooded with service requests and their website is overloaded and either slowed or crashes completely.
- n) * Making and digitally distributing child pornography

MEASURES TO COUNTER OR COMBAT COMPUTER CRIME [COMPUTER SECURITY]

- Careful vetting of employees
- Separation of duties
- Use of passwords
- Security manager software-to monitor attempts to access the system whether successful or not.
- Educating staff
- Prevention of unauthorized access to the computer operations room
- Data Encryption/Decryption

COMPUTER VIRUS

What is it?

- An executable computer program written intentionally to alter the way a computer operates without permission, to do harm to the computer.
- **Program viruses** e.g. cascade, pacman 1508, override.
- **Boot viruses** e.g. pasta, pentagon, Michelangelo
- **Internet Viruses** e.g. Trojan horse, Stealth

DEFINITION2.

A virus is a computer program designed to enter your computer and tamper with your files without your knowledge. Once the program containing the virus is open, the activated virus can not only infect other programs and documents on your computer, it can duplicate and transmit itself to other computers that are connected to yours, just like a physical virus can move from one human host to another.

Viruses began in the late 1980s as personal computers and electronic bulletin boards became more common. Back then, operating systems, word processing programs and other programs were stored on floppy disks. Hidden Viruses were programmed onto these disks; as the disks were transferred from person to person, the virus spread.

WHO CREATES VIRUSES?

Where do Viruses come from? Every virus is created by an author with a different motive—but all virus builders feel their actions are justified. For some, a killer virus is the ultimate technical challenge, like climbing a mountain. For others, creating Viruses is a form of self-expression. Some disgruntled employees, consumers or citizens turn to virus building as revenge for perceived injustices. And though it's a frightening thought, some Viruses are built and aimed by legitimate (but disreputable) businesses to weaken competitors. Other virus authors want to make their mark in Internet history; these writers get a thrill out of seeing their virus cause damage enough to attract news headlines both online and on the evening news.

What Viruses Do? /Signs & symptoms of an infected computer.

- Reduced memory or disk space
- Files are overwritten or damaged
- Hard drive may be erased
- Data is modified /corrupted.
- Change files & date stamp
- Drive lights blink without reason
- Longer times are experienced when loading programs
- Slower system operation.

How computers are infected with Viruses?

- Use of infected floppy diskettes on uninfected computers. Floppy disks or other media that users can exchange.
- Downloading unknown files, when your computer is connected to the Internet.
- Opening an E-mail attachment. **Email attachments.**
- Accepting unknown program installations from the internet
- Use of network files/direct connection/through networks.
- Software piracy [buying software from unauthorized agents]. **Pirated software.**
- Booting a PC from an infected medium.
- Executing an infected program.
- Opening an infected file.
- Shareware.
- Viruses may be sent by opening email attachments,

- clicking on spam,
- visiting corrupt websites and links online,
- Opening spreadsheets or even the original method—infected disks.
- But the Internet is now the superhighway for virus transmission.

CHARACTERISTICS and ATTRIBUTES OF COMPUTER VIRUSES

The demonstrated characteristics of computer viruses include several remarkable items, including size, versatility, propagation, effectiveness, functionality, and persistence.

- 1) **Size.** The sizes of the program code required for computer viruses has been demonstrated to be surprisingly small. This has facilitated the ability of these programs to attach themselves to other applications and escape notice for long periods of time.
- 2) **Versatility.** computer viruses have appeared with the ability to generically attack a wide variety of applications. Many do not even require information about the program they are infecting.
- 3) **Propagation.** Once a computer virus has infected a program, while this program is running, the virus is able to spread to other programs and files accessible to the computer system. The ability to propagate is essential to a virus program.
- 4) **Effectiveness.** Many of the computer viruses that have received widespread publicity have had far-reaching and catastrophic effects on their victims. These have included total loss of data, programs, and even the operating systems.
- 5) **Functionality.** A wide variety of functions has been demonstrated in virus programs. Some virus programs merely spread themselves to applications without otherwise attacking data files, program functions, or operating systems activities. Other virus programs are programmed to damage or delete files and systems. The effectiveness of these programs is enhanced through the use of several phases of operation, in which the virus propagates through a system or lies dormant until triggered by a specified event. This allows the virus program increased time to spread before the victim system's user becomes aware of its presence.
- 6) **Persistence.** Even after the virus program has been detected, recovery of data, programs, and even system operation has been difficult and time consuming. In many cases, especially in networked operations, eradication of viruses has been complicated by the ability of the virus program to repeatedly spread and reoccur through the networked system from a single infected copy.

Attributes of computer viruses

- Auto replicating [self replicating].
- Attaches itself to a program or file
- It infects as it travels
- Reproduces itself
- Distribute itself
- Copies itself
- Duplicate copies of itself
- It spreads
- It is software
- It can destroy
- It hides

DESTRUCTIVE EFFECTS OF COMPUTER VIRUSES

- Cause damage to data
- Extract data from machines for spying or theft
- Slow machine performance
- Corrupt data

- Damage software, hardware and files
- Damage the computer by damaging programs
- Reformatting of the hard disk
- Deletion of files

ANTIVIRUS TOOLS/PACKAGES/SOFTWARE

- 1) Dr Solomon's Antiviral Package
- 2) Norton Antivirus Package
- 3) AVG Antivirus
- 4) Microsoft Antiviral Package
- 5) McAfee VirusScan
- 6) F-PROT **anti-virus package**
- 7) Protector Plus 2000 for
Windows ver 7.2.H03
- 8) **Anti Trojan Elite** ver 3.7.5
- 9) Panda **Antivirus** platinum 7
ver 7.06
- 10) AntiSpy ver 4.4.2
- 11) avast! 4 Professional Edition
ver 4.6.652
- 12) **DrWeb® Anti-Virus,**
Home Edition ver 4.30a
- 13) **Dr.Web Anti-Virus for**
Workstations ver 4.30.a
- 14) File Sharing Sentinel ver 1.0.1
- 15) Real Password Protector 2005
ver 2.0
- 16) Code(Red) Hunter v1.5 ver v1.5
- 17) **Digital Patrol** ver 5.00.12
- 18) BitDefender
- 19) Kaspersky
- 20) F-Secure Anti-Virus
- 21) PC-cillin
- 22) ESET Nod32
- 23) CA Antivirus
- 24) Norman Virus Control
- 25) AntiVirusKit
- 26) AVAST!
- 27) Panda Titanium
- 28) Windows OneCare
- 29) ViRobot Expert
- 30) PCTools AntiVirus
- 31) WinAntiVirus
- 32) CyberScrub AntiVirus
- 33) The Shield AntiVirus

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SCANNER – examine every file on a specific disk drive, looking/searching for known virus “signatures”. A “signature” is a string of software code that identifies a virus program. Every virus has a unique signature

MEASURES/FACILITIES PROVIDED BY AN ANTIVIRUS PACKAGE.

1. **Scanning** – searching for viruses
2. **Cleaning** of the system (removal of viruses)
3. **Rearrangement** of corrupted data.
4. **Repairing** of corrupted data files.
5. **Quarantine** – separation of infected files from uninfected files.
6. **Prevention** includes creating user awareness, implementing hygiene rules, using disk authorization software, or providing isolated 'quarantine' PCs.
7. **Detection** involves the use of anti-virus software to detect, report and (sometimes) disinfect viruses..
8. **Containment** involves identifying and isolating the infected items.
9. **Recovery** involves disinfecting or removing infected items, and recovering or replacing corrupted data.

PRECAUTIONS/SAFETY GUIDES AGAINST VIRUSES

- Install a reliable antiviral package
- Avoid Software piracy thus buy software from authorized dealers.
- Never download unknown files from the internet
- Boot the system from virus free disks
- Using **write protect** tabs on floppy disks
- Scan every floppy diskette for viruses before use. Unknown floppy diskettes should not be used

COMPUTER LANGUAGES

(a) Machine Language

- A computer language made up of binary digits (0s and 1s)
- The most fundamental way to program a computer, using instructions made up entirely of strings of 0s and 1s.
- The language is far from English and any mathematical notation.
- Machine language is not portable and is machine dependent meaning it cannot be moved from one machine to the other.
- Errors are very difficult to remove and programs are difficult to write using machine language.

(b) Low Level Language

- Low level means the language can better communicate with the system.
- LLL are closer to machine code
- They use mnemonics i.e. ADD to add SUB to subtract etc
- Easier for a programmer to write a program in LLL than machine language.
- Programs have to be translated from source code into machine language

(c) High Level Language

- A programming language that utilizes macroinstructions and statements that closely resembles human language or mathematical notation to describe the problem to be solved or the procedure to be used.
- Programs are much easier to write / read because they are next English
- [English like statements].

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- A programming language where each instruction corresponds to several machine code instructions. A compiler or interpreter must translate instructions in this language before they can be processed.

MACHINE LANGUAGE

ADVANTAGES	DISADVANTAGES
• A very powerful language • No translation is needed	• Much harder to code for programmers because it's a series of 1s & 0s. • Not portable • Machine Dependent • Errors are difficult to remove

LOW LEVEL LANGUAGE

ADVANTAGES	DISADVANTAGES
• Easier to use than machine language	• Machine dependent • Harder to code than HLL • Not portable

HIGH LEVEL LANGUAGE

ADVANTAGES	DISADVANTAGES
• It is a portable language • Programs are much easier to read/write coz it's close to English language. • Programs are problem oriented. • Errors are removed before translation to Machine code [during compiling].	• Slower in execution • Programs tend to be wordy hence they take long to compile. • Requires more memory.

❖ **LANGUAGE TRANSLATOR PROGRAMS**

- A **Translator** is a program that translates a source program/code into an equivalent object code [machine code] before the program is executed by the computer.
 - **Compiler**
 - Translates a complete program written in HLL into machine code before it is executed.
 - Produces a translated document.
 - E.g. COBOL compiler.
 - **Assembler**
 - Translates a source program written in assembly language into machine code.
 - **Interpreter**
 - A computer program that translates one statement at a time [source code] and generates the machine code for that statement only & executes it there & there before moving to the next line/statement.
 - Interpreter translates line per line and does not produce a translated document.
 - E.g. Pascal Interpreter, BASIC Interpreter.

Compare & contrast Interpreter & Compiler

- 1) Compiler- fast execution of program.
- 2) Interpreter- Slowed down execution of program.
- 3) Compiler- compiler time overhead.
- 4) Interpreter- No compiler time overhead
- 5) Compiler- Stand alone executables

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- 6) Interpreter- Dependency on interpreter
- 7) Compiler- Compiler is complex [time, space]
- 8) Interpreter- Relative simplicity of interpreter

List the advantages & disadvantages of interpreters & compilers.

1. A compiled program will always run faster than an interpreted one
2. Object code generated by compilers may be saved on disk & run as many times as required without being compiled.
3. A compiler gives more helpful error messages.
4. Compiler gives the line number where the error has occurred.
5. An interpreter will find the error at the first instance whereas the compiler will list all errors after compiling.
6. A compiler will check the syntax of the entire program whereas the interpreter only checks the syntax of statements that are executed during that particular run of a program.
7. Interpreters are very convenient for program development, since making modifications does not mean the whole program has to be reloaded & recompiled which takes considerable time.
8. Interpreter is cheaper than compiler & is simpler to use
9. Many interpreters will allow a program to run up to a point where an error occurs, let the programmer fix the error & then continue to run the program from that point.
10. With an interpreter debugging is easier since the code is executed line by line.
11. Line by line interpretation can be slow
12. Interpreter can test your code as you write it
13. With an Interpreter there is no need to create object code & link programs.

GENERIC SOFTWARE AND THE ORGANISATION OF DATA

Define Software

Are basically programs, which are in a computer system?

Computer programs and procedures concerned with the operation of a computer system.

Definitions of **software** on the Web:

- Written coded commands that tell a computer what tasks to perform. For example, Word, PhotoShop, Picture Easy, and Photo Deluxe are software programs.
- Coded instructions (programs) that make a computer do useful work.

TYPES OF SOFTWARE

SYSTEM SOFTWARE

- Programs that control and support operations of a **computer** system.
- System software includes a variety of programs such Operating Systems, DBMS, Communication control programs. Service and utility programs, and programming language translators.
- **Definitions of system software**
A term for a complicated set of programs that act together to allow a computer, and other programs, to function. Different from an page layout application that the user may have, system software is what allows the page layout application to interact with the monitor display, the RAM, the hard drive and the rest of the computer.

OPERATING SYSTEM

- The main control of a computer system.
- It is a system of programs that controls & coordinates the execution of computer programs and may provide scheduling, debugging, input/output control, system accounting, compilation, storage assignment, data management and related services.

Definitions of **operating system** on the Web:

- The software that the rest of the software depends on to make the computer functional. On most PCs this is Windows or the Macintosh OS. Unix and Linux are other operating

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systems often found in scientific and technical environments.

- A computer program that controls the components of a computer system and facilitates the operation of applications. Windows Me, Windows XP, Linux, and MacOS are common operating systems.

APPLICATION SOFTWARE

- Programs that specify the information processing activities required for the completion of specific tasks of computer users. E.g. Electronic *Spreadsheet* and *word processing* programs or inventory or payroll programs.
 - Software that is designed and written for a specific personal, organizational, or processing task, such as graphics software.
 - These programs accomplish the specialized tasks of the user, while operating system software allows the computer to work. A computer-aided dispatch system is application software, as is each word processing program.

FREEWARE

- Public domain software that is freely available in magazines and on the internet at no charge to users though manuals may be offered at a cost. Freeware may or may not have copyrights and may or may not have distribution restrictions.

SHAREWARE

- Refers to software that is widely distributed in much the same way as freeware **BUT** is always copyrighted and includes a software license that requires its users to pay for the privilege of using the software beyond a certain trial period. If you decide to use it, you are expected to pay for the license. There is usually a disabling system in the program after a certain period of time i.e. a month.

OPERATING SYSTEM

- Are programs that create an environment, which enables us to use computers?
- Is an integrated system of program that manages the operations of the CPU, controls the input/output and storage resources and activities of the computer system, and provides various support services as the computer executes the application programs?
- Is a set of software routines that communicate between application programs and the hardware?
 - All communications go through the operating system.
 - E.g. when a user issues a command to print, the O/S receives the command & processes it according to priorities with the use of interrupts.
 - Once the processor is free to execute the command, the O/S sends the message to the processor & receives the output, which is then directed to the printer.

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Types /EXAMPLES of operating systems

There are several types of operating system, defined according to whether they can simultaneously manage information measuring 16 bits, 32 bits, 64 bits or more.

OPERATING SYSTEM	PROGRAMMING	SINGLE USER	MULTI USER	SINGLE TASK	MULTI TASK
1. MS DOS	16 bits	X		X	
2. Windows3.1	16/32 bits	X			not pre-emptive
3. Windows95/98/Me	32 bits	X			cooperative
4. WindowsNT/2000	32 bits		X		pre-emptive
5. WindowsXP	32/64 bits		X		pre-emptive
6. Unix / Linux	32/64 bits		X		pre-emptive
7. MAC/OS X	32 bits		X		pre-emptive
8. VMS	32 bits		X		pre-emptive
9. Windows Vista					
10. Windows 7					

BOOTING (also known as "booting up") is a bootstrapping process that starts operating systems when the user turns on a computer system. A **boot sequence** is the initial set of operations that the computer performs when power is switched on. The **bootloader** typically loads the main operating system for the computer.

Rebooting

Hard reboot

- A *hard reboot* (also known as a *cold reboot*, *cold boot* or *cold start*) is when power to a computer is cycled (turned off and then on) or a special reset signal to the processor is triggered. This restarts the computer without first performing any shut-down procedure. (With many operating systems, especially those using disk caches, after a hard reboot the filesystem may be in an "unclean" state, and an automatic scan of on-disk filesystem structures will be done before normal operation can begin.) It may be caused by power failure, be done by accident, or be done deliberately as a last resort to forcibly retrieve the system from instances such as a critical error or virus-inflicted DoS attack. It can also be used by intruders to access cryptographic keys from RAM, in which case it is called a cold boot attack.

Soft reboot

- A *soft reboot* (also known as a *warm reboot*) is restarting a computer under software control, without removing power or (directly) triggering a reset line. It usually, though not always, refers to an orderly shutdown and restarting of the machine.
- The Control-Alt-Delete key combination on the original IBM PC was designed to allow a soft reboot for a quicker and more convenient (and, some argue^[who?], less stressful on system components) restart than powering the computer completely down then back up.
- This kind of reboot will not usually reset the hard disks, so that they have time to update their write cache to permanent storage. Hard disks will also keep their configuration (like C/H/S adjustments, HPA, DCO, internal passwords...) over these reboots.
- The Linux kernel has optional support for the kexec system call, which transfers execution to a new kernel and skips hardware or firmware reboot. The entire process is done independent of the system firmware. Note that the kernel being executed does not have to be a Linux kernel.

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- Sometimes, the computer does not respond and in that case it can be restarted by pressing **Ctrl+Alt+Del** keys. The process of starting the computer using these keys is called **warm booting** or **soft booting**.

Random reboot

- *Random reboot* is a non-technical term referring to an unintended (and often undesired) reboot for which the cause is not immediately evident to the user. Such reboots may occur due to a multitude of software and hardware problems, such as triple faults.
- As Windows XP/Vista has an option to skip its Blue Screen of Death (Blue Screens of Death in Windows XP/Vista offer no option of pressing any key and seeing if the computer continues functioning) and immediately restarts the computer in the event of a fatal error, users can be mistaken in thinking a Windows XP/Vista computer suffers from random rebooting.
- Also, If a Wii game has a disc failure or the Wii game disk is dirty, it will reboot to the Wrist Strap screen.

Errors

- In Windows, when an error occurs in the boot process, a Blue Screen of Death or a Black Screen of Death may occur. On UNIX and Unix-like operating systems, such as Linux, a fatal error in the boot process may cause a kernel panic.

FACTORS ONE SHOULD CONSIDER WHEN

- **Making comparisons of the microcomputer operating systems available in the organization**
- **Making microcomputer operating system acquisition decision.**
- ❖ **Is the OS a single or multitasking operating system?** Many microcomputer OS are single tasking, meaning they allow only one person to run one program at a time. Multitasking and multi-user OS permit more than one program to run and more than one person to use the system at one time. Multitasking OS may allow a user to receive a fax message at the same time one is searching a database.
- ❖ **Is it a task switching operating system?** A task switching OS lets one load more than one program at a time and allows switching between the programs. The task that one is working with is the foreground task and the suspended tasks are background tasks.
- ❖ **What utilities are available with the OS?** Some of the most commonly used utilities on microcomputer OS are delete, copy contents of one diskette to another, format a diskette, delete files stored in a diskette, and compress data.
- ❖ **Is the OS hardware dependent?** An OS may be limited to running on specific hardware or may have versions that allow it to run on different types of computer systems.
- ❖ **What user interface is offered by the operating system?** Graphical User Interface and Command Driven User Interface are examples of user interfaces. *See Interface section.*

Types of operating systems

- Multiprocessing
- Multitasking
- Multi-user
- Multi programming
- Single user

Modes of operation/ Operating system techniques

Computers can operate in many different ways requiring different and complex **operating systems**.

Real time processing

- ❖ When the computer has to react within a guaranteed time to an input, a real time operating system is used. For example, the engine management system within a car uses a real time operating system in order to react to all the things going on within the engine. A real time operating system does not necessarily have to be fast. It simply has to be quick enough to

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respond to inputs in a predictable way. Embedded computers often contain an RTOS as many are used to control something.

- ❖ Computers operating in real time are often dedicated to the **control of systems** such as industrial processes, planes and space flights. Another example of real time processing is computer games. For example, when an action is selected in a game, the data is fed back to the computer and the systems updated.

Multi-programming

- ❖ Multi-programming is a method of operating such that several programs appear to be running at once.
- ❖ The jobs are usually given **priorities**. The **operating system** switches jobs in and out of processor time according to priority. For example, while one job is being allocated printer time, another will be being processed in memory. The **processor** is so fast that it seems that many jobs are being processed at the same time.

Multi-tasking

- ❖ This isn't just about running more than one application at the same time. Multi-tasking allows multiple tasks to run **concurrently**, taking turns using the resources of the computer.
- ❖ This can mean running a couple of applications, sending a document to the printer and downloading a web page.
- ❖ However, operating systems are not truly multi-tasking - in order to do this, you would need parallel processors, which are now appearing on personal computers.

Multiprocessing

- ❖ An operating system technique which can manage multiple instruction processors, assigning each to operate upon a different instruction stream (program or job) concurrently.
- ❖ **Multiprocessing** is the use of two or more central processing units (CPUs) within a single computer system. The term also refers to the ability of a system to support more than one processor and/or the ability to allocate tasks between them.^[1] There are many variations on this basic theme, and the definition of multiprocessing can vary with context, mostly as a function of how CPUs are defined (multiple cores on one die, multiple chips in one package, multiple packages in one system unit, etc.).
- ❖ *Multiprocessing* sometimes refers to the execution of multiple concurrent software processes in a system as opposed to a single process at any one instant. However, the terms multitasking or multiprogramming are more appropriate to describe this concept, which is implemented mostly in software, whereas multiprocessing is more appropriate to describe the use of multiple hardware CPUs. A system can be both multiprocessing and multiprogramming, only one of the two, or neither of the two.

Multi-access or multi-user

- ❖ A multi-access (or multi-user) system is one where a number of users are able to use the same system together.
- ❖ Modern personal computers can allow multi user access i.e. more than one person can make use of the computer resources at the same time via a LAN.
- ❖ The central processor deals with users in turn; clearly the more users, the slower the **response time**. Generally, however, the processor is so fast that the response time at the most is a fraction of a second and the user feels they are being dealt with immediately.
 - **Multi-user** is a term that defines an operating system or application software that allows concurrent access by multiple users of a computer. Time-sharing systems are multi-user systems. Most batch processing systems for mainframe computers may also be considered "multi-user", to avoid leaving the CPU idle while it waits for I/O operations to complete. However, the term "multitasking" is more common in this context.
 - An example is a UNIX server where multiple remote users have access (such as via Secure Shell) to the UNIX shell prompt at the same time. Another example uses multiple X Window sessions spread across multiple terminals powered by a single machine - this is an example of the use of thin client.
 - Management systems are implicitly designed to be used by multiple users, typically one system administrator or more and an end-user community.

Single-user

- Is most commonly used when talking about an operating system being usable only by one person at a time, or in reference to a single-user software license agreement. Multi-user operating systems such as UNIX sometimes have a single user process available for emergency maintenance.

Batch processing

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- ❖ A batch processing system is where programs or **data** are collected together in a batch and processed in one go. Typically the processing of payrolls, electricity bills, invoices and daily transactions are dealt with this way.
- ❖ This method of operation lends itself to jobs with similar inputs, processing and outputs where no human intervention is needed. Jobs are stored in a queue until the computer is ready to deal with them. Often batch processed jobs are done overnight.

ROLES (functions) OF THE OPERATING SYSTEM

The operating system has various roles:

1. **Management of the processor:** the operating system is responsible for managing allocation of the processor between the different programmes using a **scheduling algorithm**. The type of scheduler is totally dependent on the operating system, according to the desired objective.
2. **Management of the random access memory:** the operating system is responsible for managing the memory space allocated to each application and, where relevant, to each user. If there is insufficient physical memory, the operating system can create a memory zone on the hard drive, known as "**virtual memory**". The virtual memory lets you run applications requiring more memory than there is available RAM on the system. However, this memory is a great deal slower.
3. **Management of input/output:** the operating system allows unification and control of access of programmes to material resources via drivers (also known as peripheral administrators or input/output administrators).
4. **Management of execution of applications:** the operating system is responsible for smooth execution of applications by allocating the resources required for them to operate. This means an application that is not responding correctly can be "killed".
5. **Management of authorisations:** the operating system is responsible for security relating to execution of programmes by guaranteeing that the resources are used only by programmes and users with the relevant authorisations.
6. **File management:** the operating system manages reading and writing in the file system and the user and application file access authorisations.
7. **Information management:** the operating system provides a certain number of indicators that can be used to diagnose the correct operation of the machine.

Additional Functions of the operating system:

- Controls and coordinates the computer system
- Scheduling and loading of jobs to provide continuous processing. So that no parts of the computer system are idle for any significant period of time.
- It is a platform for application programs
- Provides user interface
- Acts as an interface between hardware and application programs
- Provides error correctional routines
- Control over selection and operation of input-output devices & file handling.
- Records details of processing for control analysis
- File management, control the creation, deletion, saving and access of files
- Memory management, allocates memory to various jobs or tasks
- Multiprogramming, multiprocessing, multitasking, timesharing, housekeeping & handling errors.
- Resource management manages processor time, memory space, disk space and peripheral devices.
- Reports errors when they occur.
- *Task management.* Manage the accomplishment of the computing tasks of end users. It may involve multitasking capability where several computing tasks can occur at the same time.
- Minimizes the amount of human intervention required during processing.

COMPUTER INTERFACE

- Refers to the type of interaction between two elements.

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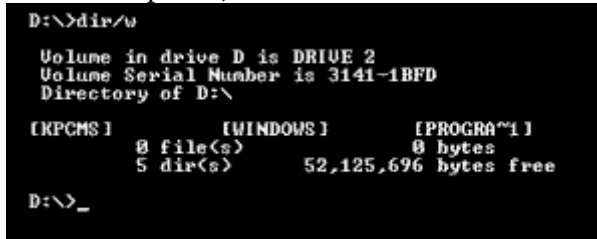
- Is part of the operating system that allows you to communicate with it or the computer so that you can load programs, access files, and accomplish other tasks?
- When a person interacts with a computer he uses a User Interface, which can be a screen, keyboard, mouse, scanners or other input devices.

Definitions of **interface** on the Web:

- The interaction between the computer and the user or the control of the flow of data between a computer and its peripherals.
- ❑ Basically the interface can be
 - ☑ Command driven- include operating systems and database languages that provide users with flashes of text and in which users would generally type in their command to perform an action.
 - ☑ Menu driven
 - ☑ GUI- point and click
 - ☑ WIMP interface

1. Command line interfaces (MS-DOS BASED OPERATING SYSTEM)

A command line **interface** allows the user to interact with the computer by **typing in commands**. The computer displays a prompt, the user keys in the command and presses enter or return. A **COMMAND-LINE INTERFACE (CLI)** is a mechanism for interacting with a computer operating system or software by typing commands to perform specific tasks. This text-only interface contrasts with the use of a mouse pointer with a **graphical user interface (GUI)** to click on options, or menus on a text user interface (TUI) to select options.



```
D:\>dir/w

Volume in drive D is DRIVE 2
Volume Serial Number is 3141-1BFD
Directory of D:\

[KPCMS]      [WINDOWS]      [PROGRAM~1]
      0 file(s)                0 bytes
      5 dir(s)              52,125,696 bytes free

D:\>_
```

Features of a command line interface include:

In the early days of personal computers, all PCs used command-line interfaces.

- Commands - usually abbreviated - must be typed correctly and in the right order or the command will not work.
- Experienced users who know the commands can work very quickly without having to find their way around menus.
- An advantage of command driven programs is that they do not need the memory and processing power of the latest computer and will often run on lower spec machines.
- Command driven programs do not need to run in Windows.
- A command line interface can run many programs, for example a batch file could launch half a dozen programs to do its task.
- An inexperienced user can sometimes find a command driven **program** difficult to use because of the number of commands that have to be learnt.

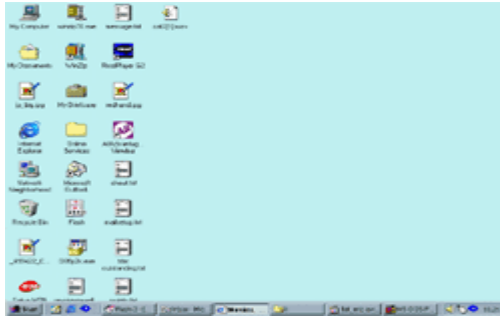
An example of a common command-driven interface is MS-DOS, the original operating system for all Microsoft-compatible PCs. The MS-DOS command to *display all files on drive a* would be: **dir c:\.**

2. Graphical user interfaces (WINDOWS BASED OPERATING SYSTEM)

Graphical user interface is sometimes shortened to **GUI**. The user chooses an option usually by pointing a mouse at an icon representing that option. A **GRAPHICAL USER INTERFACE (GUI)** is a type of user interface item that allows people to interact with programs in more ways than typing such as computers; hand-held devices such as MP3 Players, Portable Media Players or Gaming devices; household appliances and office equipment with images rather than text

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commands. A *GUI* offers graphical icons, and visual indicators, as opposed to text-based interfaces, typed command labels or text navigation to fully represent the information and actions available to a user.



Features of GUIs include:

- They are much easier to use for beginners.
- They enable you to easily exchange information between software using cut and paste or 'drag and drop'.
- They use a lot of memory and processing power. It can be slower to use than a CLI if you are an expert user.
- They can be irritating to experienced users when simple tasks require a number of operations.

When discussing user interfaces, it is important to note that Windows XP, OS/X and Linux are all **graphical user interfaces**.

A **good user interface** should:

- be attractive and pleasing to the eye
- allow the user to try out different options easily
- be easy to use
- use suitable colours for key areas
- use words that are easy to understand aimed at the type of user
- have online help

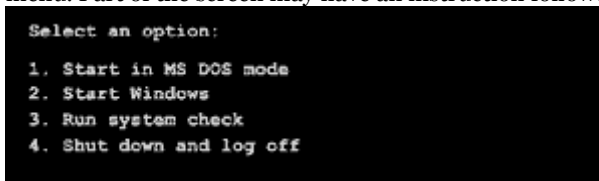
It also should consider the **needs** of the users. For example, young children are likely to prefer pictures to words and people with disabilities may benefit from particular input or output devices

3. WIMP interface

- Windows, Icons, Menus and Pointer - a method by which a user can interact with a computer without having to learn lots of special words - it provides a much more intuitive 'view' of the computer and its facilities than non-WIMP operating systems provide
- "Windows, /images/, Menus and Pointer", or "Windows, /images/, Mouse and Pull-down menus" This is a graphical interface which makes access to computers much more straightforward and intuitive than the main alternative, the command-line interface. The WIMP interface has been used on Apple Macintosh computers since they were invented, and are coming into use on PCs through Microsoft Windows and on Unix and other workstations through the X Window system.

4. Menu driven interfaces

The user is offered a **simple menu** from which to choose an option. One menu often leads to a further menu. Part of the screen may have an instruction followed by a numbered list of options to choose from:



A **full screen menu** takes up most of the screen.

A **menu bar** is the set of options at the top of the screen. When an option is chosen a **pull down menu** may be offered.

Features of menu driven interfaces include:

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- They are easy to use as the user does not have to remember sets of commands.
- They are user friendly - you can often guess your way around the options.
- They can be irritating if there are too many levels of menus to move around - with a command line interface you can go to the option required immediately.

Dialog box

In graphical user interfaces, a **dialog** (or **dialogue**) **box** is a special window, used in user interfaces to display information to the user, or to get a response if needed. They are so-called because they form a dialog between the computer and the user—either informing the user of something, or requesting input from the user, or both. It provides controls that allow you to specify how to carry out an action.

Scrollbar

A **scrollbar** (Also known as a handle in the very first GUIs) is a graphical object in a GUI with which continuous text, pictures or anything else can be scrolled including time in video applications, i.e., viewed even if it does not fit into the space in a computer display, window, or viewport.

Taskbar

In computing, **taskbar** is a desktop display bar used to launch and monitor applications. Microsoft incorporated a taskbar in Windows 95 and it has been a defining aspect of Microsoft Windows's graphical user interface ever since. Other desktop environments, such as KDE and GNOME, also include a taskbar. Other operating systems use terms such as *Panel* or *Dock*.

Toolbar

In a graphical user interface on a computer monitor a **toolbar** is a GUI widget on which onscreen buttons, icons, menus or other input or output elements are placed.

WYSIWYG

WYSIWYG, is an acronym for *What You See Is What You Get*. The term is used in computing to describe a system in which content displayed during editing appears very similar to the final output,^[2] which might be a printed document, web page, slide presentation or even the lighting for a theatrical event.

The phrase was originally a catch phrase popularized by Flip Wilson's drag persona "*Geraldine*" (from Rowan & Martin's Laugh-In in the late 60s and then on *The Flip Wilson Show* until 1974), who would often say "What you see is what you get" to excuse her quirky behavior.

Meaning

WYSIWYG implies a user interface that allows the user to view something very similar to the end result while the document is being created. In general WYSIWYG implies the ability to directly manipulate the layout of a document without having to type or remember names of layout commands. The actual meaning depends on the user's perspective, e.g.

- In Presentation programs, Compound documents and web pages, WYSIWYG means the display precisely represents the appearance of the page displayed to the end-user, but does not necessarily reflect how the page will be printed unless the printer is specifically matched to the editing program, as it was with the Xerox Star and early versions of the Apple Macintosh.
- In Word Processing and Desktop Publishing applications, WYSIWYG means the display simulates the appearance and precisely represents the effect of fonts and line breaks on the final pagination using a specific printer configuration, so that a citation on page 1 of a 500-page document can accurately refer to a reference three hundred pages later.
- WYSIWYG also describes ways to manipulate 3D models in Stereochemistry, Computer-aided design, 3D computer graphics and is the brand name of Cast Software's lighting design tool used in the theatre industry for pre-visualisation of shows.

Menu bar

A **menu bar** is a region where computer menus are housed. Its purpose is to house window- or application-specific menus which provide access to such functions as opening files, interacting

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with an application, or help. Menu bars are typically present in graphical user interfaces with windows.

File Edit View History Bookmarks Tools Help

Menu bar from Mozilla Firefox.

SYSTEM SOFTWARE- consists of programs that manage and support a computer system and its information processing activities. These programs serve as a vital software interface between computer system hardware and the application programs of end users.

System software is computer software designed to operate the computer hardware and to provide a platform for running application software.

The most basic types of system software are:

- The computer BIOS and device firmware, which provide basic functionality to operate and control the hardware connected to or built into the computer.
- The operating system (prominent examples being Microsoft Windows, Mac OS X and Linux), which allows the parts of a computer to work together by performing tasks like transferring data between memory and disks or rendering output onto a display device. It also provides a platform to run high-level system software and application software.
- Utility software, which helps to analyze, configure, optimize and maintain the computer.

In some publications, the term *system software* is also used to designate software development tools (like a compiler, linker or debugger).

Computer purchasers seldom buy a computer primarily because of its system software. Rather, system software serves as a useful (even necessary) level of infrastructure code, generally built-in or pre-installed.

In contrast to system software, software that allows users to do things like create text documents, play games, listen to music, or surf the web is called application software.

Types of system software programs

System software helps use the operating system and computer system. It includes diagnostic tools, compilers, servers, windowing systems, utilities, language translator, data communication programs, database systems and more. The purpose of system software is to insulate the applications programmer as much as possible from the complexity and specific details of the particular computer being used, especially memory and other hardware features, and such accessory devices as communications, printers, readers, displays, keyboards, etc.

Specific kinds of system software include:

- Loaders
- Linkers
- Utility software
- Desktop environment / Graphical user interface
- Shells
- BIOS
- Hypervisors
- Boot loaders
- Database Management Systems(SQL, NoSQL)

If system software is stored on non-volatile memory such as integrated circuits, it is usually termed firmware.

3 functional categories of system software

🚩 **System management programs**

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- ✓ Programs that manage the hardware, software, and data resources of the computer system during its execution of the various information processing jobs of users.
- ✓ The most important system management programs are operating systems and operating environments, followed by telecommunications monitors & Database Management Systems.

🚧 System support programs

- ✓ Programs that support the operations and management of a computer system by providing a variety of support services.
- ✓ Major support programs are system utilities, performance monitors, & security monitors.

🚧 System development programs

- ✓ Programs that help users develop information system programs and procedures and prepare user programs for computer processing.
- ✓ Major development programs are language translators, programming tools and CASE [computer-aided software engineering] packages.

APPLICATION PACKAGES/SOFTWARE

- Are programs that are or were developed to solve specific user problems?
- Word processors and spreadsheets fall into this category.
- Software application packages can be custom built or they can be bought off the shelf.

#	APPLICATION SOFTWARE		EXAMPLES
1	Word processing software	A collection of software that permits the user to create, edit and print documents. Most word processing packages come with spell-checkers, grammar checkers and thesaurus programs for creating drawings and table of contents.	• Ms word, Word perfect, Word star, • Ms works word processor ,WordPro
2	Spreadsheet software	Allows managers to prepare budgets, tax analysis, sales and profit projections. Managers can design templates, which contain formulas used to create columns or row averages. Spreadsheets allow managers to simulate various solutions. A user may enter a number of different values and see the effect on the results such as profit margins	• Ms Excel, Lotus 1-2-3
3	Database management software	Allows user to prepare reports based on data found in different records. DBMS is an important managerial decision support tool for managers at all levels of the organization.	• Ms Access, DbaseIV, Oracle
4	Payroll		• Pay plus, Pay well
5	Accounting		• Pastel, Tetra 2000
6	Presentation graphics software	Lets users or managers prepare slides containing charts, text and images. Presentation graphics software usually provides libraries of clip art images that can be cut and pasted into a slide to make the slide more attractive and informative	• Ms PowerPoint, Lotus freelance graphs
7	Desktop Publishing		• Ms publisher, Aldus
8	Multimedia		• Internet explorer Netscape
9	Communication		• Ms Outlook, Ms Exchange
10	Design		• AutoCAD, CAM

State five advantages of using a word processor (e.g. MS Word) rather than a manual typewriter in an office [10].

- Ability to correct mistakes
- Ability to view documents on screen before printing
- Ability to incorporate diagrams
- Ability to move parts of text to other sections of the document.
- Ability to save or store documents for future reference or use.
- Ability to alter or change document layout.
- Ability to print a lot of copies [no retyping]
- Ability to insert tables
- Ability to format document [font, paragraphs, bullets etc] Spelling and grammar checker
- Import files
- Mail merge
- WYSIWYG capability

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- Creation of templates
- Automatic creation of index and table of contents
 - ther - Good reasoning [2 marks each for 5 & 1 mark each for more than 5]

a) Explain what is meant by the term 'word processing package'. [3 marks]

- ❖ Is a program or set of programs used to enter, edit, format, store and print documents. A document may be anything from a single memo to a complete book.
- ❖ An application package that involves the use of computers to manipulate text data in order to produce office communications in the form of documents.
- ❖ Any of many popular application programs designed for composing, revising, printing, and filing written documents.
- ❖ **Word Processing is** the efficient and effective production of written communications at the lowest possible cost through the combined use of systems management procedures, automated technology, and accomplished personnel. The equipment used in word processing applications includes but is not limited to the following: dictation and transcription equipment, automatic repetitive typewriters, visual display text editing typewriters, keyboard terminals, etc.

b) Give any 2 examples of a word processing package'. [2marks]

- ❖ MS WORD
- ❖ WORD STAR
- ❖ WORD Perfect
- ❖ MS WORKS WORD

c) Briefly describe any 5 important features of a word processor. [5 marks]

- 1) Spelling and grammar checker
- 2) Import files
- 3) Mail merge
- 4) WYSIWYG capability
- 5) Creation of templates
- 6) Automatic creation of index and table of contents

d) List any 10 facilities that spreadsheets offer. [10 marks]

1. Format cells, rows and columns, specifying for example, the alignment of text, number of decimal points, height and width of a cell.
2. Copy cell contents to other locations
3. Determine the effect of several different hypothetical changes of data
4. Insert, move or delete rows and columns
5. Use functions such as SUM, AVERAGE, MAX, MIN in formulae
6. Create a simple database and sort or query the data to produce a report
7. Write macros to automate common procedures
8. Create templates i.e. spreadsheets with formats and formulae already entered, into which new figures may be inserted.
9. Create multi-dimensional spreadsheets using several sheets, and copy data from one sheet to another
10. Create many different types of charts and graphs

Shortcomings of spreadsheets

While spreadsheets are a great step forward in quantitative modeling, they have deficiencies. At the level of overall user benefits, spreadsheets have four main shortcomings.

- Spreadsheets have significant reliability problems. Research studies estimate that roughly 94% of spreadsheets deployed in the field contain errors, and 5.2% of cells in unaudited spreadsheets contain errors.
- The practical expressiveness of spreadsheets is limited. Several factors contribute to this limitation. Implementing a complex model requires implementing detailed layouts, cell-at-a-time. Authors have difficulty remembering the meanings of hundreds or thousands of cell addresses that appear in formulas.

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- Collaboration in authoring spreadsheet formulas is difficult because such collaboration must occur at the level of cells and cell addresses. By comparison, programming languages aggregate cells with similar meaning into indexed variables with names that indicate meaning. Although some spreadsheets have good collaboration features, authoring at the level of cells and cell formulas remains a significant obstacle to collaboration in authoring spreadsheet models. On the other hand, many people collaborate on entering numerical data and many people can use the same spreadsheet.
- Productivity of spreadsheet modelers is reduced by the cell-level focus of spreadsheets. Even conceptually simple changes in spreadsheets (such as changing starting or ending time or time grain, adding new members or a level of hierarchy to a dimension, or changing one conceptual formula that is represented as hundreds of cell formulas) often require large numbers of manual cell-level operations (such as inserting or deleting cells/rows/columns, editing and copying formulas, re-laying out worksheets). Each of these manual corrections increases the risk of introducing further mistakes.

These four deficiencies in high-level benefits have deeper causes that, ironically, flow directly from the signature strength of spreadsheets (that they capture the structure of models in terms of WYSIWYG sheet layout for authors and report users).

- Spreadsheets capture model logic in terms of sheet layout, especially contiguous layout of cells in a table. Spreadsheets have weak or nonexistent methods to capture higher level structures such as named variables, segmentation dimensions, and time series.
- Formulas are subordinated to the cell layout. This forces the sheet layout to carry the structure of the model, not variables and formulas that relate variables. This also causes a large proliferation of cells, formulas and cell-level tasks even when only a few basic concepts are involved in a model. This forces authors to think and work at the level of cells instead of at the level of the natural concepts and structures of the model.
- Formulas expressed in terms of cell addresses are hard to keep straight and hard to audit. Research shows that spreadsheet auditors who check numerical results and cell formulas find no more errors than auditors who only check numerical results.
- Proliferation of error-prone manual cell-level operations contributes to all four of the high-level problems listed above.

List 5 factors to consider when buying application software. [10].

1. **Cost.** This includes the original cost of the package, technical support, and upgrades.
2. **Portability,** can it be installed on one type of hardware or a variety
3. **Relevance of the software to the task at hand.**
4. **Compatibility with existing hardware & software.** Will the package run on existing hardware? Can files be easily transferred from existing systems without re-keying? Can files created in the package be exported to other systems in use in the company?
5. **It should be on a media compatible with your computer** i.e. if you purchase software on CD media your computer should have compact disk drive.
6. **Sophistication,** simple programs are usually easy to learn, cost less and are appropriate for those who use the software infrequently.
7. **Should include enough documentation** such as installation instructions, system requirements e.g. Pentium 4, reference manual, registration information, supplier details and user manual.
8. **Standard software,** which is well supported locally i.e. used by most people in that area
9. **Is the application upgradeable and** does it support future upgrades.
10. **Easiness of installation**
11. **Technical support.** Is support available? Is it very costly? Often, technical support contract can add 50% or more to the price of a package, but without it no support at all will be given by the manufacturer.

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12. **Easy of learning.** Are tutorials supplied? Are books on the software available in bookshops? Are training courses available?
13. **Easy of use and user- friendliness.** Is it easy to use, for example using pull down menus, icons, helpful error messages when you do something wrong?
14. **Is the application from a reputable supplier** e.g. Microsoft.
15. **Do you have enough resources on your PC to install it** e.g. it may require 256MB RAM, 80GB HDD. Memory requirements; software packages vary in the amount of memory they require. Will the system require expenditure on more powerful hardware?
16. *Does the version of the product that I am considering run on my personal computer system? Is it compatible with my platform (PC, Macintosh, Sun, and so on)? with the processor used in my computer (486, Pentium, Motorola 6040, PowerPC, and so on)? and with the operating system that I am running (DOS, Windows, NeXT Step, UNIX)?*
17. *Is the product compatible with other programs, such as extensions and device drivers, that I have on my personal computer system?*
18. *Does the product do all that I want it to do? Is it simple enough to use immediately but powerful enough to provide the capabilities that I will want as my familiarity with the program increases?*
19. *Is the product an up-to-date, current version or release?*
20. *Does the product have a clear, step-by-step tutorial? Is the product easy to learn?*
21. *Is the product accompanied by clear, easy-to-follow instructions, or documentation?*
22. *Does the manufacturer provide technical assistance? Is a toll-free technical assistance number included in the software documentation? Does the manufacturer have a bulletin board or Internet address for technical assistance questions? Does the manufacturer charge for technical assistance? for product revisions or upgrades?*
23. *Can I find instruction, or training, in the use of the product?*
24. *Is the product from a respected, well known manufacturer?*
25. *Does the program have any known problems, or **bugs**, and will these affect my use of the program?*
26. *Does the product come with a warranty, and what are the conditions of that warranty?*
27. *Does the product cost more or less than comparable products on the market?*

Benefits of application software to the user.

- Installation should be easy, quick and should be provided by the vendor/supplier.
- Good documentation is usually provided with the application for reference purposes.
- Onsite assistance from the supplier or vendor
- Provision of cheaper future upgrades upon acquisition of an application provided you have completed the registration information.
- Value for money

Applications software comes in different 'flavors' - or types:

- **Utility programs** - these can be part of an **operating system**, but are also available for separate purchase. Examples include virus checkers, disk defragmenters, back up utilities and disk checker software.
- **Generic** - general purpose software that is not written for any particular type of business. Examples of this include word processors and **spreadsheets**.
- **Integrated** - a collection of software that has a common set of commands/icons. Usually they include word processors, spreadsheets and graphics software, but they can contain **databases** as well. They tend to be cheaper than purchasing each application separately.
- **Specific** - software written for a defined purpose. Accountancy software is a good example of this, and can be bought by anyone.
- **Bespoke** - also known as 'tailor made'. A company may need a particular piece of **software** that it cannot purchase because it does not exist yet - or it does not like what is available. The company can commission the software to be specially written for them. This tends to be expensive.

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CLASSIFICATION OF SOFTWARE APPLICATIONS

1. Commercial / generic Packages/ Off the shelf software

- Commercial software is computer software sold for profit; such software represented, until recently, the vast majority of all software used.
- These are bought from a vendor.
- Also known as off the shelf packages e.g. MS Office, Pastel.
- Also known as work enhancement applications
- **Generic or Off the shelf** software has the advantage of cost effectiveness due to larger market & the ability to incorporate available technology in the shortest possible time.

These increase the productivity & efficiency in the work place i.e.

- Word Processing packages
- Spreadsheet packages
- Database /Information retrieval Packages
- Graphics packages
- Data communications Software

Advantages of an off the shelf package

1. It is cheaper than custom-written software. The development costs of the package may be millions of pounds, but the customer may be able to buy it for a few hundred pounds, since sales are made to thousands of other customers.
2. It is immediately available and already thoroughly tested so is unlikely to have major bugs in it
3. Documentation is usually available in the form of reference manuals, user guides and tutorials
4. Training courses may be available from third party trainers.
5. Technical support is usually available from the manufacturers via a Web site or telephone line [at a price].
6. Other users of the package can be consulted as to its suitability before purchase.
7. Upgrades are usually available every year or two

Disadvantages

1. The package may not do exactly what you want it to do;
2. It may not run on the firm's existing hardware
3. It may not interface with other software already in use in the organization.

2. Homemade / Customized packages/ Tailor-made Software

- Customized software is software designed around a company or other organization's processes to make those processes more efficient.
- It is distinct from market software [**commercial or generic**], which approaches problems in a general way so that the product can be sold to more than one customer.
- Has been written by the vendor specifically for a user's needs. Also known as in-house packages, tailor-made packages.
- E.g. most payroll packages, billing software, product design applications
- Is that application that has been written for a specific user's needs by a software vendor?
- It is non-standard software.

Disadvantages

1. Being customized, the non-standard software may not be used by any other organization.
2. It has huge support costs after implementation.
3. Dependency syndrome with the supplier. There is heavy reliance on the developers of the software for its continued existence & maintenance
4. Future support may be difficult if the supplying company goes under, shuts down, or liquidate or if its development staff leaves.

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5. May not be upgradeable and may not run if moved to a new hardware platform.
6. You are tied to a supplier of which you may not be willing to have a maintenance contract with.
7. **Written or Custom-built** software performs a defined function and is expensive to build.
8. Custom built is expensive to maintain
9. Custom built software has reduced lifespan due to high speed of technological advancement in the field of computers.

3. Managerial applications

Help to make management decisions

- Decision support systems
- Database Management Systems Report Writing Tools

MAKE NOTES ON:

- ❖ **How application packages solve problems?**
- ❖ **Typical problems which can be solved by application packages**
- ❖ **Customization of packages using macros**

Utility software

Utility software (also known as *service program*, *service routine*, *tool*, or *utility routine*) is computer software designed to help manage and tune the computer hardware, operating system or application software by performing a single task or a small range of tasks. Some utility software has been integrated into most major operating systems.

Examples

- **Disk storage utilities**
 - - **Disk defragmenters** can detect computer files whose contents are broken across several locations on the hard disk, and move the fragments to one location to increase efficiency.
 - **Disk checkers** can scan the contents of a hard disk to find files or areas that are corrupted in some way, or were not correctly saved, and eliminate them for a more efficiently operating hard drive.
 - **Disk cleaners** can find files that are unnecessary to computer operation, or take up considerable amounts of space. Disk cleaner helps the user to decide what to delete when their hard disk is full.
 - **Disk space analyzers** for the visualization of disk space usage by getting the size for each folder (including subfolders) & files in folder or drive. showing the distribution of the used space.
 - **Disk partitioners** can divide an individual drive into multiple logical drives, each with its own filesystem which can be mounted by the operating system and treated as an individual drive.
 - **Backup** utilities can make a copy of all information stored on a disk, and restore either the entire disk (e.g. in an event of disk failure) or selected files (e.g. in an event of accidental deletion).
 - **Disk compression** utilities can transparently compress/uncompress the contents of a disk, increasing the capacity of the disk.
 - **File managers** provide a convenient method of performing routine data management tasks, such as deleting, renaming, cataloging, uncataloging, moving, copying, merging, generating and modifying data sets.
 - **Archive** utilities output a stream or a single file when provided with a directory or a set of files. Archive utilities, unlike archive suites, usually do not include compression or encryption capabilities. Some archive utilities may even have a separate un-archive utility for the reverse operation.

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- **System profilers** provide detailed information about the software installed and hardware attached to the computer.
- **Anti-virus** utilities scan for computer viruses.
- **Text and Hex Editors** directly modify the text or data of a file. These files could be data or an actual program.
- **Data compression** utilities output a shorter stream or a smaller file when provided with a stream or file.
- **Cryptographic** utilities encrypt and decrypt streams and files.
- **Launcher applications** provide a convenient access point for application software.
- **Registry cleaners** clean and optimize the Windows registry by removing old registry keys that are no longer in use.
- **Network managers** check the computer's network, log events and check data transfer.
- **Command line interface (CLI)** and **Graphical user interface (GUI)** Allows the user to contact and make changes to the operating system.

Integrated software

- **Integrated software** is software for personal computers that combines the most commonly used functions of many productivity software programs into one application.
- The integrated software genre has been largely overshadowed by fully functional office suites, most notably Microsoft Office, but at one time was considered the "killer application" type responsible for the rise and dominance of the IBM PC in the desktop business computing world.^[1]
- In the early days of the PC before GUIs became common, user interfaces were text-only and were operated mostly by function key and modifier key sequences. Every program used a different set of keystrokes, making it difficult for a user to master more than one or two programs. Programs were loaded from floppy disk, making it very slow and inconvenient to switch between programs and difficult or impossible to exchange data between them (to transfer the results from a spreadsheet to a word processor document for example). In response to these limitations, vendors created multifunction "integrated" packages, eliminating the need to switch between programs and presenting the user with a more consistent interface.
- The potential for greater ease-of-use made integrated software attractive to home markets as well as business, and packages such as the original AppleWorks for the Apple II and Jane for the Commodore 128 were developed in the 1980s to run on most popular home computers of the day.
- Context MBA was an early example of the genre, and featured spreadsheet, database, chart-making, word processing and terminal emulation functions. However, because it was written in Pascal for portability, it ran slowly on the relatively underpowered systems of the day. Lotus 1-2-3, which followed it, had fewer functions but was written in assembler, providing it with a speed advantage that allowed it to become the predominant business application for personal computers.^[2]
- The integrated software market of today is exemplified by entry-level programs such as Microsoft Works which are often bundled with personal computers as "starter" productivity suites.

DEFINITION 2

- **SOFTWARE that contains several applications rolled into one. Integrated software usually includes word processing, spreadsheet, database, graphics and communication capabilities. Two popular Integrated software packages are Microsoft Works and ClarisWorks:**
- In **integrated software**, you will be able to work on a spreadsheet to perform calculations, store addresses and phone numbers in a database and be able to link information from the spreadsheet & the database when you work with the word processor in the integrated package.

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- Object Linking Embedding [OLE] is a standard that, Allows applications to be linked together.
- OLE-DB allows applications to import virtually any form of data into the database.

Definitions of **integrated software** on the Web:

- A group of applications designed to work together and share data easily.
- Software that combines the features of several different applications in a single program (e.g. word processing, database, spreadsheet, graphics, and communications)..

Advantages of integrated applications

- Easy to use
- Generally costs much less
- Require or needs less RAM
- Ability to pass data from one module to another.
- Takes up much less hard disk space
- Better organization of information. Data is stored in the database, calculations in an electronic spreadsheet & images in a graphical application package

Disadvantages

- Limited portability. The same applications must exist on every computer that you wish to open the document.
- Offer fewer features and less versatility than stand alone versions or software.
- Functional limitation. Its rigid you cannot change it to suit your specific needs.

EXAMPLES OF INTEGRATED SOFTWARE

- Microsoft Office 97
- Microsoft Works
- Claris Works
- Lotus SmartSuite 97

SOFTWARE PACKAGE

- A computer program supplied by computer manufacturers, independent software companies, or other computer users.
- Also known as *canned programs*, *proprietary software* or *packaged programs*.
- A software package is a special method for the distribution and installation of software on computer systems.

Proprietary software/Applications

- Is software offered for sale or license where the vendor controls the source code?
- Much of the software users are accustomed to is known as proprietary software. With proprietary software users are not allowed to see the source code. Nor are they able to modify the code for their own use or to distribute to others. In some cases certain customers are allowed to view the source code - sometimes for an additional fee - but even then they are not able to alter and re-distribute the software. **Examples of proprietary software include Microsoft's Office suite.**
- Proprietary software is a term used to describe software in which the user does not control what it does or cannot study or edit the code, in contrast to free software.

Non-proprietary software

- Non-proprietary software is, in its widest possible sense, software that has no proprietary restrictions attached to it, particularly the restriction about the access to the source code. Non-proprietary software is the opposite of proprietary software, which is software where the source code has not been made available, or it is protected by restrictive licensing clauses.

SOFTWARE PIRACY

- Unauthorized copying of software.
- unauthorized use or reproduction of copyrighted patented material.
- unauthorized copying of software
- The production of illegal copies of software.
- By far, the biggest legal problem affecting the computer industry today is software piracy, which is the illegal copying or use of programs. Piracy is a huge problem mainly

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because it is so easy to do. In most cases, it is no more difficult to steal a program than it is to tape a music CD that you have borrowed from a friend. Software pirates give up the right to receive upgrades and technical support, but they gain the use of the program without paying for it.

Implications of Software piracy

- Loss of business to software companies.
- Lack of technical support, no connection with supplier upgrades.
- Software incompatibility, may require a key, which you might not have, as a result you can't install it.
- Virus Spread
- Lawsuits for copyright, which may be very costly in terms of reputation and money.
- **Piracy harms all software publishers**, Regardless of their size. Software publishers spend years developing software for the public to use. A portion of every dollar spent in purchasing original software is channeled back into research and development so that better, more advanced software products can be produced. When you purchase pirated software, your money goes directly into the pockets of software pirates instead.
- **Software piracy also harms the local and national economies.** Fewer legitimate software sales result in lost tax revenue and decreased employment. Software piracy greatly hinders the development of local software communities. If software publishers cannot sell their products in the legitimate market, they have no incentive to continue developing programs. Many software publishers simply won't enter markets where the piracy rates are too high, because they will not be able to recover their development costs.
- **Software piracy harms everyone in the software community including you**, the end user. How?
 - 1) Piracy results in higher prices for duly licensed users,
 - 2) Piracy reduces levels of support, and
 - 3) Delays in the funding and development of new products, causing the overall selection and quality of software to suffer.

THE DIFFERENT TYPES OF SOFTWARE PIRACY

There are five basic forms of software piracy, and all are damaging both to the software publisher and to you, the end user. The five basic types of piracy are:

- **Soft lifting.** This form of piracy occurs when extra copies of a program are made within an organization for employees to use. It also includes "Disk swapping" among friends and associates.
- **Hard-Disk Loading.** Some computer dealers load unauthorized copies of software onto the hard disks of the computers they offer for sale, as an incentive for an end user to purchase a computer from that particular dealer.
- **Counterfeiting.** This is the illegal duplication and sale of copyrighted software, often in a form designed to make the product appear legitimate. Software counterfeiting can be very sophisticated, including significant efforts to replicate packaging, logos, and anti-counterfeiting techniques such as holograms. It can also be unsophisticated, consisting of inferior or hand-written labels, with disks folded into a plastic bag and sold on street corners. A recent trend in counterfeiting is the emergence of compilation CD-ROMs, where numerous unauthorized software publishers' programs appear on one CD-ROM. In any form, software counterfeiting is very damaging to both the software developer and legitimate end users.
- **Online.** This form of piracy occurs when copyrighted software is downloaded to users connected through a modem to an electronic bulletin board or the Internet without the express permission of the copyright owner. This should not be confused with sharing public domain software or providing "shareware." Shareware is software that may or may not be copyrighted but is generally offered for little or no charge by the author for nearly unrestricted use, including copying or sharing with others. Microsoft distributes promotional products, free software, updates, or enhancements over bulletin boards or online services,

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which may or may not be licensed for use solely with licensed Microsoft products. You should check the EULA accompanying the product to determine how it can be used.

- **License Misuse.** This form of piracy occurs when copyrighted software is distributed outside the restricted legitimate channels it was designed for or the product is used in ways not allowed in the license agreement. Examples of license misuse include:
 - (1) **Academic product misuse**
This form of license misuse occurs when a product that is manufactured, licensed, and specifically marked for distribution to educational institutions and students at reduced prices is diverted into normal commercial channels of distribution. Typically this product will contain a sticker indicating that it is academic product and for use only by educational institutions.
 - (2) **Not for Resale (NFR) product misuse.**
Again, this form of license misuse occurs when a product that has been clearly marked "Not for Resale," and is typically distributed as a promotional or sample product and is not licensed for normal commercial distribution and use, is diverted into normal commercial channels of distribution.
 - (3) **Original equipment manufacturer (OEM) stand-alone product.** This form of license misuse occurs when OEM version software has been unbundled from its designated computer system and distributed as a separate, "stand-alone" product. Microsoft's agreement with computer manufacturers prohibits them from distributing Microsoft products in this fashion, i.e. without accompanying computer hardware. Microsoft products on the retail shelf should **never** include a line on the front cover of the User's Guide that states, "For Distribution Only With New Computer Hardware."
 - (3) **"Microsoft Easy Fulfillment" (MEF) product misuse.** This form of license misuse occurs when a "Microsoft Easy Fulfillment" (MEF) product is diverted into normal commercial channels of distribution. The MEF product may be distributed only to end users who have a Select or Open (MOLP) license and who order the product in accordance with their license terms. The MEF product has a sticker on the jewel case identifying the product as "Microsoft Easy Fulfillment."
 - (4) **Volume licensing misuse.** Volume Licensing enables organizations to acquire the rights to copy and use specific Microsoft software products with agreements tailored to the number of products needed at the particular organization. These volume programs offer a broad range of products and licensing options and reduce administrative overhead and software management costs.

Microsoft offers two types of volume license:

- a. **Select** provides savings for medium and large organizations with significant volume requirements through forecasting software needs over a two-year period.
- b. **Open** provides savings for small and medium organizations for licenses of as few as 20 units of a particular software product, with a simple ordering process.

Misuse under volume licensing occurs when organization copies software purchased through the volume program on more computers than specified in their license agreement. Other violations occur when an organization allows concurrent use of software products (which currently is offered only as an option under the Select and Open programs) without the purchase of the option that allows for this concurrent use.

User licenses show users rights to use pieces of software.

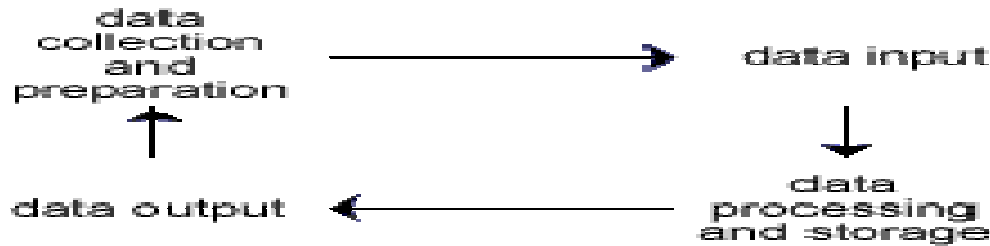
Data processing

- ✓ Refers to a class of programs that organize and manipulate data, usually large amounts of numeric data. Accounting programs are the prototypical examples of data processing applications. In contrast, word processors, which manipulate text rather than numbers, are not usually referred to as data processing applications.
- ✓ Taking the raw data gathered by interviewers, finding and correcting any errors, tabulating it and creating spreadsheets similar types of documents so that response patterns can be read and interpreted.

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The data processing cycle

The data processing cycle is the order in which data is processed. There are four stages:-



1. Data collection and preparation

- Data must be **collected** before it can be input. It can be collected in different ways.
- Often a **source document**, such as an order form, is used to simplify data collection. The data has to be entered into the computer for processing.

2. Data input

The collection of raw data from the outside world so it can be put into an information system. Putting the acquired data into the information system.

Examples:

1. The payroll clerk collects workers' timecards so she knows how many hours each person worked that week and types the hours from the timecards into a spreadsheet
2. Conducting a survey of customer's opinions and then scanning the survey cards with a card reader.
3. Collecting jokes for a joke book and typing jokes into a word processor.
4. Using a form on a website to collect visitors' opinions

Typical input devices:

Keyboards, mice, flatbed scanners, bar code readers, joysticks, digital data tablets (for graphic drawing), electronic cash registers

3. Data processing

- ✓ In this step the input data are changed, and usually combined with other information, to produce data in more useful form. Thus, paychecks may be calculated from the time cards, or a summary of sales for the month maybe calculated from the sales orders. The processing step usually involves a sequence of certain basic processing operations.
- ✓ Most companies keep a master file of customer data which is sorted into customer number order. The company makes up a daily or weekly transaction file on disk, which is sorted into the same order as the master file. The details of each transaction are added to the master file in turn to make a new master file.
- **Interactive processing** When transactions are processed as they are received we say the processing is **interactive**. Often **interactive** processing uses **disks** and has random or direct access. This comes up often and it's important to understand the relationships.

4. Data Output

In information processing, output is the process of transmitting the processing information. Essentially, output is the presentation of any data exiting a computer system. This could be in the form of printed paper, audio, video. In the medical industry this might include CT scans or x-rays.

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Typically in computing, data is entered through various forms (input) into a computer, the data is often manipulated, and then information is presented to a human (output).

Typical output devices:

Monitors

- Pixels: Images are created with dots of lights called picture elements, or pixels. More pixels = higher resolution. Resolution is the maximum number of pixels the monitor can display
- Basic VGA is a pixel grid 640 X 480
- Super VGA is 1,024 X 768
- A video card, (also referred to as a graphics accelerator card, display adapter, graphics card, and numerous other terms), is an item of personal computer hardware whose function is to generate and output images to a display.

Printers

- Laser - laser beams bond toner to paper
- Ink jet - fine stream of ink
- Dot Matrix - pins & ribbon

Attributes of good information

Quality Required	Commentary
Relevant	The information obtained and used should be needed for decision-making. it doesn't matter how interesting it is. Businesses are often criticised for producing too much information simply because their information systems can "do it". A good way of ensuring relevance is to closely define the objectives of any information reports. Another way to improve relevance is to produce information that focuses on "exceptions" - e.g. problems, high or low values, where limits have been exceeded.
Up-to-date	Information needs to be timely if it is to be actioned. For example, the manager of a large retail business needs daily information on how stores are performing, which products are selling well (or not) so that immediate action can be taken. To improve the speed with which information is produced, businesses usually need to look at upgrading or replacing their information systems.
Accurate	As far as possible, information should be free from errors (e.g. the figures add up; data is allocated to the correct categories). The users of information should be informed whenever assumptions or estimates have been used. Accurate information is usually a function of accurate data collection. If information needs to be extremely accurate, then more time needs to be allocated for it to be checked. However, businesses need to guard against trying to produce "perfect" information - it is often more important for the information to be up-to-date than perfect.
Meet the needs of the User	Users of information have different needs. The managing director doesn't have time to trawl through thick printouts of each week's production or sales listings - he or she wants a summary of the key facts. The quality control supervisor will want detailed information about quality testing results rather than a brief one-line summary of how things are going. It is a good idea to encourage users to help develop the style and format of information reporting that they require.
Easy to use and understand	Information should be clearly presented (e.g. use summaries, charts) and not too long. It also needs to be communicated using an appropriate medium (e.g. email, printed report, presentation. Businesses should also consider developing "templates" which are used consistently throughout the organisation - so that users get used to seeing information in a similar style.

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Worth the cost	Often forgotten. Information costs money. Data is costly to collect, analyse and report. Information takes time to read and assimilate. All users should question whether the information they receive/have requested is worthwhile
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Reliable	Information should come from authoritative sources. It is good practice to quote the source used - whether it be internal or external sources. If estimates or assumptions have been applied, these should be clearly stated and explained.
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- 1) Valid
- 2) Fit for use
- 3) Accessible
- 4) Cost effective
- 5) Understandable
- 6) Time

DATA

❖ Relationship between Data and Information

- Data is raw information
- a collection of facts from which conclusions may be drawn; "statistical data"

INFORMATION

- Data that has been processed to add or create meaning and hopefully knowledge for the person who receives it. Information is the output of information systems.
- Organized data which is understood to have significance and meaning

❖ Methods of data collection {see Systems Analysis & Design}

- Questionnaires
- Observation
- Sampling
- Interviews
- Written documents & research

MAKE NOTES ON:

❖ Verification

- The act of reviewing, inspecting, testing, checking, auditing, or otherwise establishing and documenting whether items, processes, services, or documents conform to specified requirements.
- An effort to test the accuracy of the questionnaire response data. The concern is uniquely with data accuracy and deals with neither the reliability nor the validity of measures.

❖ Validation

- The process of evaluating software at the end of the software development process to ensure compliance with software requirements. The techniques for validation are testing, inspection and reviewing.
- Determination of the correctness of the products of software development with respect to the user needs and requirements.

DATABASE

- Is a collection of files?
- An organized collection of interrelated files.
- Is a collection of related files that provides a way of finding information quickly and easily from a chosen reference point?
- Is an integrated collection of logically related records or files?
- A database consolidates records previously stored in separate files into a common pool of data records that provides data for many applications.
- **Examples** of databases are MS Access, MS Works Database, and DBaseIV.
 - **File**-Is a collection of related records.
 - **Record**-Is a collection of fields. It is a collection of fields arranged in a predefined order.

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- **Field**-Is a collection of single items. It is an implementation of the data attribute. It is the smallest unit of data to be stored in a database.
- **Primary key**-Is a field whose value identifies one & only one record in a file.
- **Secondary key**-An alternative index for identifying an entity. Its value can identify single entity occurrences of all entity occurrences. A subset of all entity occurrences.
- **Foreign key**-Are pointers or links to occurrences of a different file. A foreign key in one file must be a primary key in another file.
- **Descriptors**-Are any other fields that describe business entities.

Database Administrator [DBA] tasks

ROLES OF Database Administrator

- The design of the database
- After the initial design, the DBA must monitor the performance of the database, and if problems surface (such as a particular report taking an unacceptably long time to produce), appropriate changes must be made to the database structure.
- Keeping users informed of changes in the database structure that will affect them; for example, if the size or format of a particular field is altered or additional fields added
- Maintenance of the data dictionary for the database, and responsibility for establishing conventions for naming tables, columns, indexes & so on.
- Implementing access privileges for all users of the database; that is, specifying which items can be accessed and / or changed by each user.
- Allocating passwords to each user.
- Providing training to users in how to access and use the database.
- Manage the organizations
 - -data resources
 - -database plans
 - -design
 - -operations
 - -training
 - -user support
 - -security & Maintenance
- Maintain data consistency and security
- Approve access to data stored
- Approve access procedures
- ABILITY to delete, add, modify –existing data must be tightly controlled.

Database Management System (DBMS)

The DBMS is an application program that provides an interface between the operating system and the user in order to make access to the data as simple as possible. It has several other **functions** as well, and these are described below.

1. **Data storage, retrieval and update.** The DBMS must allow users to store retrieve and update information as easily as possible, without having to be aware of the internal structure of the database.
2. **Creation and maintenance of the data dictionary**
3. **Managing the facilities for sharing the database.** The DBMS has to ensure that problems do not arise when two people simultaneously access a record and try to update it
4. **Back up and recovery.** The DBMS must provide the ability to recover the database in the event of system failure.
5. **Security.** The DBMS must handle password allocation and checking, and the 'view' of the database that a given user is allowed.

The data dictionary

The data dictionary is a 'database about the database'. A **data dictionary**, as defined in the *IBM Dictionary of Computing*, is a "centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format."^[1] The term may have one of several closely related meanings pertaining to databases and database management systems (DBMS):

- a document describing a database or collection of databases

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- an integral component of a DBMS that is required to determine its structure
- a piece of middleware that extends or supplants the native data dictionary of a DBMS

It will contain information such as:

1. What tables and columns are included in the present structure?
2. The names of the current tables and columns
3. The characteristics of each item of data, such as its length and data type;
4. Any restrictions on the value of certain columns
5. The meaning of any data fields that are not self-evident; for example, a field such as 'course type';
6. The relationships between items of data
7. Which programs access which items of data, and whether they merely read the data or change it?

Types of databases/Database structures

❑ Hierarchical database

- A logical data structure in which the relationships between records form a hierarchy or tree structure.
- The relationship among records is one to many, since each data element is related only to one element above it.

❑ Network database

- A logical data structure that allows many to many relationships among data records.
- It allows entry into a database at multiple points, because any data element or record can be related to many other data elements.

❑ Relational database

- A logical data structure in which all data elements within the database are viewed as being stored in the form of tables.
- DBMS packages based on the relational model can link data elements from various tables as long as the tables share common data elements.

Advantages and any disadvantages of databases

#	ADVANTAGES	#	DISADVANTAGES
1	Supports data sharing	1	Limitations of databases arise from increased technological complexity
2	Ensures increased data security	2	Developing a large database and installing a DBMS can be difficult and expensive.
3	Ensures data independence	3	More hardware capability is required, since storage requirements for the organization of data, overhead control data, and the DBMS is greater
4	Data integrity is guaranteed. The functions in the DBMS can be used to enforce the integrity rules with minimum programming in the application programs.	4	Problems of data inconsistency can arise if a distributed database approach is used.
5	Reduces or minimizes data redundancy	5	Longer processing times may result from high volume transaction processing applications since extra layer software [the DBMS] exists between application programs and the Operating system.
6	Consistence of data is ensured	6	Security and integrity of data are major concerns.
7	Support data integration. Since related data is stored in one single database, enforcing data integrity is much easier		Centralized databases are vulnerable to errors, fraud and failures.
8	Related data can be shared across programs since the data is stored in a centralized manner		
9	Enforcing of standards in the organization and structure of data files is required and also easy in a Database System, since it is one single set of programs, which is always interacting with data files		
10	The application programmer need not build the functions for handling issues like		

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	concurrent access, security, data integrity, etc. The programmer only needs to implement the application business rules. This brings in application development ease.		
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Q* Limitations of file Processing Systems that the DBM system address

- i. *Data duplication* - Same kind of information is stored in several files
- ii. *Lack of Data integration* - Independent files make it difficult to provide end users with info for ad-hoc requests that require accessing data stored in different files
- iii. *Data dependence* - In a file processing system, the organization of files, their physical location on storage hardware and application software used to access those files depend on each other. Changes in format and structure of data and records in file require changes to all the programs

Q* what is a DBMS (Data base Management System)

Is a collection of software programs that:

- i. Stores data, in a uniform and consistent way
 - ii. Organizes the data, into records in a uniform and consistent way
 - iii. Allows access to the data, in a uniform and consistent way
- ⌚ A collection of software programs that stores organizes and allows access to the data in a uniform and consistent way
 - ⌚ Is a collection of programs that enables users to create and maintain a database?
 - ⌚ A general purpose software system that facilitates the process of defining constructing and manipulating databases for various applications.

Database Environment

- i. Database
- ii. Database users
- iii. DBM System
- iv. Data Dictionary
- v. Database Administrator
- vi. User/system interface

Database Users

- i. Casual users
 - ii. Naïve users – parametric users
 - iii. Sophisticated end users
- I. Occasionally use it, and if they do a HLL such as AQL may be used.
 - II. Those who use Standard queries and updates only e.g. bank cashiers, data entry clerks' e.t.c.
 - III. Know the whole range of facilities on a database environment.

Q a. Explain the 3 Major disadvantages of a file system (6)

b. Give and explain any 7 functions of a DBMSystem

FUNCTIONS OF A Database Management SYSTEM

1. Storage, Retrieval & Update
2. Catalogue
3. Logical transactions
4. Concurrency control
5. Recovery
6. Security
7. Communications
8. Integrity
9. Data Independence
10. Utilities

1. Storage, Retrieval & Update

- Allows users with the ability to store, retrieve and update data in the database.

2. Catalogue

- Allows a catalogue in which descriptions of data items are stored and which is accessible to users

3. Logical Transactions

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- Allows a mechanism which will ensure that either all of the updates corresponding to a given transaction are made or not made. .e.g.
ABORT – ROLLBACK or undo all the changes
COMMIT- Transaction gone thru and is already in the database.
- 4. Concurrency Control**
 - Allows a mechanism to ensure that the database is updated correctly when multiple users are updating the database concurrently
LOCKING OR DEADLOCK situations must be avoided
- 5. Recovery**
 - Allows a mechanism for recovering the database in the event that the database is damaged in any way.
 - A before / after image is usually created
- 6. Security**
 - Allows a mechanism to ensure that only authorized users can access the database.
 - Authorizations, encryption features and limited views can be used as security provisions
- 7. Communication**
 - Must be capable of integrating with communications software on the database environment on a multi-user environment.
- 8. Integrity**
 - Allows a mechanism to ensure that both the data in the database and changes to the data follow certain rules.
- 9. Data Independence**
 - (Has facilities to) support the independence of programs from the actual structure of the database.
- 10. Utilities**
 - Provide a set of utility services e.g. optimized

COMPONENTS OF A DBMS

1. Data dictionary/directory
2. Data language
3. Application development tools
4. Security software
5. Report writers
6. Query language
7. Web server software
8. Teleprocessing monitors
9. Archiving, backup and Recovery systems

1. DATA DICTIONARY/ DIRECTORY

- Contains the names and descriptions of every data element in the database.
- Also contains descriptions of how data elements relate to one another.
- Ensures that data is stored in a uniform and consistent manner hence reducing redundancy.

2. DATA LANGUAGES

- Data description language (DDL) describes the characteristics of data elements.
- DDL requires the application program to use standardized commands to retrieve and process data from a database.
- Data manipulation language (DML) consists of commands such as FIND, GET, INSERT

3. APPLICATION DEVELOPMENT TOOLS

- A program designed to help programmers develop application programs that use the database e.g. CASE TOOLS

4. SECURITY SOFTWARE

- Provides a variety of tools to shield database from unauthorized access and from viruses

5. REPORT WRITERS

- Allows programmers, mgrs and others to design out put reports.

6. QUERY LANGUAGE

- A set of commands used primarily for accessing data from a database.
- You can ask ad-hoc questions of the database interactively without the aid of programmers.
- Examples SQL, NLQ, QBE. Structured Query language, Natural language Queries and Query by Example.

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7. WEB SERVER SOFTWARE

- Turns the computer system housing database into a web server and enables users with web connection to access data from wherever they are located.
- Also convert data into a format that is readable by standard browsers.

8. TELEPROCESSING MONITORS

- A software package that manages communication between the database and remote terminals.

9. ARCHIVING, BACK UP AND RECOVERY SYSTEMS

- Provides the database manager with tools to make copies of the database.
- Restart/ Recovery systems are tools used to restart the database and to recover lost data in the event of a failure.

OBJECTIVES AND BENEFITS OF DATABASE APPROACH

1. Data redundancy is reduced –reduces the duplication of data.
2. Data inconsistency avoided.
3. Sharing of data is promoted –since data is stored in centralized.
4. Standards are enforced, thru the data dictionary /catalogue
5. Base of application development /maintenance is available
6. Uniform security/ privacy /integrity controls are applicable to group of users
 7. SECURITY – Passwords, access controls
 - PRIVACY - Authorized users
 - INTEGRITY – Accurate data
8. Integration of data is achieved - since data is stored in one db
9. Data independence is achieved/ observed - it is the immunity of applications to changes in storage structure and access strategy.
10. Data accessibility and responsiveness.
11. Structured Query Language e.g. select name from customer where town = Harare”

RISK TO DATABASE APPROACH

1. New, specialized personal
2. Backup – important DBMS function (Recovery)
3. Shared data – conflict and concurrency control
4. Technological complexity
5. Expensive to develop a large database and installing a DBMS can be difficult.
6. More hardware capability is required – storage requirements overhead control data and the DBMS required more memory
7. Problems of inconsistency can arise if a distributed data approach is used.
8. Longer processing time may result from high volume transaction processing applications.
9. Security and integrity are major concerns.
10. Centralized database are vulnerable to errors fraud and failures.

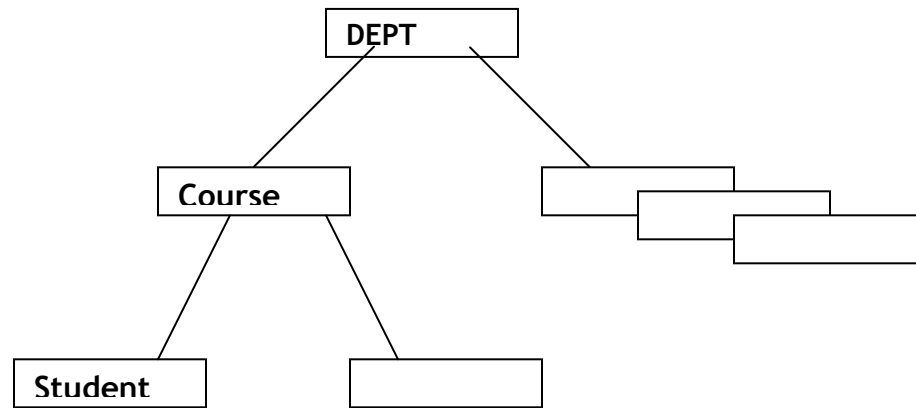
Database Models or types of databases

- i) Hierarchical database model
- ii) Network database model
- iii) Relational database model
- iv) Object-oriented database model

i) Hierarchical database model

- Each record is related to others in a parent-child relationship or tree structure.
- Relationships between among records form a hierarchy or tree structure.
- The relationship among records is one to many since each data element is related only to one element above it.
- Searching a record involves moving progressively downward from a root and along the branches of the tree until the desired record is located.

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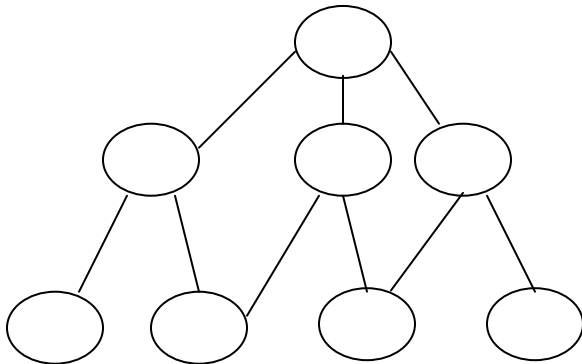


ii) Network database Model

- Relationship among records is many to many
- Allows entry into a database at multiple points because any data element or record can be related to many other data elements.
- Permits a record to be a member of more than one set at a time

Hierarchical and network models are less flexible than other database models because the relationship between records must be determined and implemented before a search can be conducted.

Network database Model



iii) Relational Database model

- All data elements within the database are viewed as being stored in the form of tables
- Database Packages based on this structure link records in different or various tables to provide information to users as long as the share common data elements.
- Data is organized in 2-dimensional tables.

CUSTOMER		
CUSTOMER NO.	CUSTOMER NAME	CUSTOMER ADDRESS
-	-	-
-	-	-

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iv) Object oriented database model

- Each object in an object-oriented database model is bound together with its own data and a set of instructions that describe the behavior and attributes of the objects.
- Objects use messages to interact with one another.
- Every object is described by a set of attributes.
- E.g. the object building in a database or architectural drawing may have the attributes TYPE, SIZE, COLOUR just as in any.
- Every object must have a set of procedures or routines or a set of methods.
- E.G methods for an architectural drawing might include instructions to display, rotate, or explode the drawing on a screen.

FILE PROCESSING

- Is the process of updating the records in a file to accurately reflect changes in the data?
- It can be a manual process where procedures are in place to update forms, which are sent to the respective person responsible for the update.
- It can also be a computerized process where information is first stored in transaction files until the master file is updated.

File processing methods:

1. BATCH PROCESSING is execution of a series of programs ("jobs") on a computer without manual intervention.

- Batch jobs are set up so they can be run to completion without manual intervention, so all input data is preselected through scripts or command-line parameters. This is in contrast to "online" or interactive programs which prompt the user for such input. A program takes a set of data files as input, process the data, and produces a set of output data files. This operating environment is termed as "batch processing" because the input data are collected into batches on files and are processed in batches by the program. **Batch Processing**
 - Transactions are collected and processed in batches [groups].
 - Transaction processing in which transactions are gathered and stored for later execution.
 - E.g. a supermarket chain operates 5 outlets. At the end of the day all information on sales [transaction file] is collected, & the main records [master file] are updated with the sales for that day.
 - Performs a group of tasks all at once, while requiring little [or no] interaction with the user.
 - It is a great time saver because the computer can perform repetitive functions quickly & consistently.

Benefits

Batch processing has these benefits:

- It allows sharing of computer resources among many users and programs,
- It shifts the time of job processing to when the computing resources are less busy,
- It avoids idling the computing resources with minute-by-minute manual intervention and supervision,
- By keeping high overall rate of utilization, it better amortizes the cost of a computer, especially an expensive one.

COMMON BATCH PROCESSING USAGE

Data processing

A typical batch processing procedure is End of day-reporting (EOD), especially on mainframes. Historically systems were designed to have a batch window where online subsystems were turned

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off and system capacity was used to run jobs common to all data (accounts, users or customers) on a system. In a bank, for example, EOD jobs include interest calculation, generation of reports and data sets to other systems, print (statements) and payment processing.

Printing

A popular computerized batch processing procedure is printing. This normally involves the operator selecting the documents they need printed and indicating to the batch printing software when, where they should be output and priority of the print job. Then the job is sent to the print queue from where printing daemon sends them to the printer.

Databases

Batch processing is also used for efficient bulk database updates and automated transaction processing, as contrasted to interactive online transaction processing (OLTP) applications.

Images

Batch processing is often used to perform various operations with digital images. There exist computer programs like Batch Image Processor that let one resize, convert, watermark, or otherwise edit image files.

Converting

Batch processing is also used for converting a number of computer files from one format to another. This is to make files portable and versatile especially for proprietary and legacy files where viewers are not easy to come by.

Job scheduling

UNIX utilizes cron and at facilities to allow for scheduling of complex job scripts. Windows has a job scheduler. Most high-performance computing clusters use batch processing to maximize cluster usage.

2. REAL-TIME PROCESSING.

A **real-time operating system (RTOS)** OR **REAL-TIME PROCESSING** is a multitasking operating system intended for real-time applications. Such applications include embedded systems (programmable thermostats, household appliance controllers), industrial robots, spacecraft, industrial control (see SCADA), and scientific research equipment.

A RTOS facilitates the creation of a real-time system, but does not guarantee the final result will be real-time; this requires correct development of the software. An RTOS does not necessarily have high throughput; rather, an RTOS provides facilities which, if used properly, guarantee deadlines can be met generally or deterministically (known as soft or hard real-time, respectively). An RTOS will typically use specialized scheduling algorithms in order to provide the real-time developer with the tools necessary to produce deterministic behavior in the final system. An RTOS is valued more for how quickly and/or predictably it can respond to a particular event than for the amount of work it can perform over a given period of time. Key factors in an RTOS are therefore a minimal interrupt latency and a minimal thread switching latency.

- An early example of a large-scale **real-time** operating system was Transaction Processing Facility developed by American Airlines and IBM for the Sabre Airline Reservations System.
- **Real-time Processing**
 - The master records are updated as and when the transactions occur.
 - A form of transaction processing in which each transaction is executed as soon as complete data becomes available for the transaction.
 - A supermarket chain would be linked by a network to a main computer.
 - Each time a sale is made the stock information is updated and the sale information is also updated.
 - It is expensive and complicated.
 - Record and file locks are implemented to avoid problems with real-time processing.

3. ONLINE PROCESSING. or **OLP**, refers to a class of systems that facilitate and manage transaction-oriented applications, typically for data entry and retrieval transaction processing. The term is somewhat ambiguous; some understand a "transaction" in the context of computer or

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database transactions, while others (such as the Transaction Processing Performance Council) define it in terms of business or commercial transactions. OLP has also been used to refer to processing in which the system responds immediately to user requests. An automatic teller machine (ATM) for a bank is an example of a commercial transaction processing application.

- The technology is used in a number of industries, including banking, airlines, mailorder, supermarkets, and manufacturing. Applications include electronic banking, order processing, employee time clock systems, e-commerce, and eTrading. The most widely used OLP system is probably IBM's CICS.
- **Online processing**
 - A general term for when one computer is interacting directly and simultaneously with another computer. Many sources of information are available online.
 - Available for immediate use.
 - The master file is updated with the transactions through some form of communication link.
 - All real-time processing involves online processing.
 - Not all online processing needs to be real-time.
 - Require interaction of the user.

Benefits

Online Processing has two key benefits: simplicity and efficiency. Reduced paper trails and the faster, more accurate forecasts for revenues and expenses are both examples of how OLP makes things simpler for businesses.

Disadvantages

As with any information processing system, security and reliability are considerations. Online transaction systems are generally more susceptible to direct attack and abuse than their offline counterparts. When organizations choose to rely on OLP, operations can be severely impacted if the transaction system or database is unavailable due to data corruption, systems failure, or network availability issues. Additionally, like many modern online information technology solutions, some systems require offline maintenance which further affects the cost-benefit analysis.

- **Centralized processing**
 - An information system where all critical data and programs are stored on one main computer, usually a mainframe. Unlike the task-sharing concept of a client/server system, the central computer retains all the brains and brawn.
 - A computing environment in which all processing takes place on a mainframe or central computer.
 - Puts all processing and control authority within one computer to which all other computing devices respond.
- **Distributed Processing**
 - The ability to have several computers working together in a network, where each processor runs different activities for a user, as required.
 - Computers connected together by a communications network. Data processing occurs on each of the computers, unlike a centralized processing system in which terminals are connected to a host computer that performs all of the data processing.
 - Use of computers at various locations, typically interconnected via communication links for the purpose of data access and/or transfer.
 - A computer system in which processing is performed by several separate computers linked by a communications network. The term often refers to any computer system supported by a network, but more properly refers to a system in which each computer is chosen to handle a specific workload, and the network supports the system as a whole.
- **Offline processing**

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- When your computer performs an operation when it is not connected to any other computers, it is working offline.
 - Means that your computer is not currently connected to an Internet server.
 - Not connected (to a network). Not available to the network
 - **Time sharing**
 - A way of sharing out computer facilities between a number of people who want to use the computer at the same time. Each has a separate terminal and gets the impression that they have sole use of the computer with their own 'account'. Similar to NETWORKING.
 - **Transaction processing**
 - A type of computer processing in which the computer responds immediately to User requests, with each User interaction considered to be a transaction. Automatic teller machines for banks are probably the best-known example.
 - In computer science, transaction processing is information processing that is divided into individual, indivisible operations, called transactions. Each transaction must succeed or fail as a complete unit; it cannot remain in an intermediate state.
-  **Master file** is the main information source in any type of organization.
-  **File processing** is the update of the master file [actual records] through manual or computerized methods.

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FILE ORGANIZATION METHODS & ACCESS

- **File Organization**-Is the arrangement of data records on storage media
- It determines the manner in which individual records can be accessed or retrieved

Definitions of **cycle time** on the Web:

- The time usually expressed in seconds for a controller to complete one on/off cycle.

Definitions of **access time** on the Web:

- The average time interval between a storage peripheral (usually a disk drive or semiconductor memory) receiving a request to read or write a certain location and returning the value read or completing the write.
The amount of time it takes a computer to locate an area of memory for data storage or retrieval.
- The time required to retrieve data from a storage device and transmit it to a specified location.

Definitions of **random access** on the Web:

- Storage systems where data may be stored and accessed in any order, independent of the ordinal position of the data when it was originally recorded. This is the opposite of linear (3) access, or linear recording media such as magnetic tape, which necessarily preserves the sequential relation of the data as it is recorded, and depends on this sequential relation for accurate playback. See non-linear recording.
- Able to read any location directly; without having to read sequentially to the location.

Definitions of **sequential access** on the Web:

- Data must be read in the order it appears. This method applies to tape, and to a lesser degree to disks.
Reading data from a file whose records are organized on the basis of their successive physical positions. To reach a specific record, all records previous to that record must be read, in order. Magnetic computer tapes are sequential access storage device.
- Refers to reading or writing data records in sequential order, that is, one record after the other. To read record 10, for example, you would first need to read records 1 through 9. This differs from random access, in which you can read and write records in any order.

TYPES OF QUERIES

- i. DIRECT QUERY
- ii. INVERTED QUERY

i. DIRECT QUERY

- Is one where the user can identify the entity in which he is interested?
- He may want to know some of the particular of attributes of that entity.
- e.g. Where a sales man wants to know the amount and location of a particular product in the organization.
- He is able to identify the product by name or product number and this identification is used as the key for making the query which is also the primary record key.
- The basic purpose of the direct query is to know the attribute values of already identified entities

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ii. **INVERTED ENTITIES**

- is one where the user does not know the identification of entities in which he is interested
- he wants to identify the entities by some criteria or attribute values
- e.g. the purchase department requiring a list of items for which the stock level is below the desired level
- in this case user does not know the identity of items already but want to know it
- only specify a criterion as the basis of the query
- Obviously the primary key cannot be used for retrieval, it is not available
- Some attribute like stock level has to be used for the retrieval of records

MODELS OF FILE ORGANISATION

- i. serial file organization
- ii. sequential file organization
- iii. indexed sequential file organization
- iv. direct file organization
- v. inverted file organization

i. **SERIAL FILE ORGANISATION**

- Is one where records are not stored in any particular order
- They are stored in the order in which they are received
- New records are added at the end of the files
- This method of file organization is used when
 - the usage of the records is not clearly known
 - as a means of recording transaction as they happen
 - as dump files reflecting the physical layout of a file
 - as a prelude to organizing the file in another order i.e. transactions may be punched and stored on a serial file, the file may be subsequently be stored into a desired sequence
 - gives maximum utilization of space
 - no room is left for inserting other records
 - does not cater for direct access to records
 - used on magnetic tapes
 - popular on output files, archive files, security files

i. **SEQUENTIAL FILE ORGANISATION**

- records are arranged in a pre-determined sequence determined by the primary key
- appropriate for files which are store on tape or disk
- nor generally used for online system that demand fast response
- mainly used for batch processing applications
- suitable when there is for little file enquiry
- can be used for variable length records as well as fixed length records
- waste space as some space is left of reserved for other records to be inserted in order to preserve the order or organization
- ✓ **UPDATING**
- ✓ Updating on tape is carried out in batch mode by reading the whole file and rewriting it after necessary modification on to a new file.
- ✓ On disk can be carried out without rewriting the whole file.
- ✓ **RETRIEVAL**

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- ✓ Retrieval of a record from a sequential file on tape media can be made only in the sequential mode by reading all the records in sequence till the desired record is reached.
- ✓ Retrieval of a record from a sequential file on direct access media (DISK) can be made by using sequential reach skip search, binary search.
- ✓ **HIT RATE**
- ✓ In a batch processing run: The proportion of the number of records processed during a run to the total number of records on the file is called a hit rate.
- ✓ The higher the volatility and the lower the frequency of updating,, the higher the hit rate.
- ✓ **PROCESSING OF SEQUENTIAL FILES**
- ✓ Tapes provides for only serial access
- ✓ Disks can be accessed in a serial order or random order.

iii) INDEXED SEQUENTIAL FILES

- Records are stored in sequence but in addition an index is provided to enable individual records to be located directly after reading the index.
- Records are pointed to by indexes.
- Used with disks only not with magnetic tapes.
- Wastes space.
- The index may be ignored and the data is read sequentially
- The file may be accessed sequentially but using the index to skip over unwanted records =selective sequential access
- Index may be read into memory and used to obtain data randomly.
- Records can be accessed sequentially OR randomly.

iv) DIRECT FILE ORGANISATION

- ⌚ Files are stored or haphazardly and an address key is used as reference.
- ⌚ Records are physically located at an address that is calculated from its primary key field.
- ⌚ The calculation of the address from the key field is called hashing.
- ⌚ Records are scattered on the storage disk instead of being arranged next to one another or linked via a pointer field.
- ⌚ Records can rapidly be retrieved and accessed directly
- ⌚ Used on magnetic disks (e.g. floppies, hard disk) and optical disks.
- ⌚ Costly in terms of device used and space.

v) INVERTED FILE ORGANISATION

- ⌚ special case of indexed sequential organization
- ⌚ an index is maintained for every attribute in the record
- ⌚ the main file itself may not be stored if direct queries are not to be handled
- ⌚ the entire file data is, in any case, stored in the indexes
- ⌚ Useful for handling inverted queries.

vi) MULTI-RING FILE ORGANISATION

- ⌚ useful for handling inverted query when sets of records have same attribute value
- ⌚ Each set of records with same attribute values is linked into a ring.
- ⌚ A ring of records is a chain where the last in the chain points to the first.
- ⌚ One of the records is anchored to the index so that it provides an entry point into the ring.
- ⌚ The index of the attribute is read first to retrieve records with a particular attribute value.
- ⌚ All records with that attribute value can be located.

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COMPUTER MEMORY

RAM (Random Access Memory)

Also referred to as Main Memory, Primary Memory or Working Memory is memory that is accessed by the CPU as it processes information. The CPU also places results of processing on the memory chip. The memory has the following characteristics.

- 1) can be read from
- 2) can be changed and can be rewritten
- 3) used for short-term storage of data i.e. it is temporary storage,
- 4) is volatile (data lost when power is turned off).

Types of RAM

- **DRAM (Dynamic RAM)** This is the most common type of computer memory. DRAM needs refreshing and is refreshed hundreds of times each second in order to retain data. During recharging it cannot be accessed by the CPS
- **SRAM (Static RAM)** SRAM is approximately 5 times faster (and twice as expensive, as DRAM). It does not have to be constantly refreshed. Because of its lower cost and smaller size, DRAM is preferred for the main memory, while SRAM is used primarily for cache memory.
- **SDRAM (synchronous dynamic RAM)** is the standard memory offering in the PC industry and is faster than DRAM because it's synchronized to the system clock. . It is designed for mainstream home and business applications, email, and basic audio and video streaming. It provides single-channel memory and is available on our mainstream and value desktop system.
- **DDR SDRAM (double data rate synchronous dynamic RAM)** takes all the features of ordinary SDRAM and increases the frequency bandwidth to improve system performance and speed.
- **RDRAM (Rambus dynamic RAM)** is able to load a new stream of data before the previous stream has completed, resulting in faster access speeds. The RDRAM memory features dual-channel processing which doubles data throughput to further enhance performance. This type of memory is optimized for high-end multimedia tasks involving video and audio. Dual Channel RDRAM is shipped on our performance desktop systems powered by Pentium 4 processors
- **VRAM: Memory** designed for storing the image to be displayed.
- **CACHE RAM** is a small block of high-speed memory located between the Processor and main memory and is used to store frequently requested data and instructions. When the processor requests data, it will check in the cache first

The computer has two types of memory chips within the system unit, known as **RAM** and **ROM**.

ROM (Read Only Memory)

This memory chip has essential programs that start up the computer hardwired on the microchip during the manufacture of the computer. It forms what is known as **FIRMWARE**, being a halfway between hardware and software. The characteristics of this memory are that:-

- 1) can be read from,
- 2) cannot be changed
- 3) is permanent
- 4) non-volatile (data not lost when power is turned off).

❖ ROM

- Stores instructions & data which do not change
- It is written once during manufacture
- It is for holding software such as MS-DOS.Windows 95, 98.XP.

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- It is programmed by the manufacturer and cannot be altered by the user
- It is static & non volatile
- It stores control routines for personal computers

VARIATIONS OF ROM

Allow users to store and in some cases to change data stored in ROM

- **PROM [Programmable ROM]** is not programmed during manufacture but can be programmed by the user. It is programmed according to the needs of the user organization. If you make a mistake the ROM once programmed will become permanent to the effect that any errors will never be corrected thus the user will throw away any chip if errors are made.
- **EPROM [Erasable Programmable ROM]** can be programmed and erased as many times. This time you don't have to worry about errors too much because there is room for correction. The programming is done electronically. Stored data can be lost by exposing it to Ultra Violet light over a period of minutes.
- 1) **EEPROM [Electrically Erasable Programmable ROM]** can be read, erased, rewritten, without removing from the computer. The erasing & rewriting process is very slow compared with reading, which limits their use. These can be erased & programmed with electrical pulses. Can be erased in a few Milliseconds

MEMORY CAPACITY

Everything entering into the computer is converted into digital (number) form, and is stored in that form. When one needs to view the information, it is converted into human sensible text, graphics, audio or video by the operating system. A binary digit is known as a **bit** being either a '0'(zero) or a '1'(one). It is the smallest item that can be stored in a computer's memory.

Memory capacity is measured in **bytes**, a **byte** being a set of eight(8) bits and the byte often represents a single character. Measurement units used for computer memory are:-

- | | |
|------------------------|------------------------------------|
| ○ BIT | 1 binary digit (Either a 0 or a 1) |
| ○ BYTE | Eight bits. |
| ○ NIBBLE | Half byte/4 bits |
| ○ KILOBYTE (KB) | 1024 bytes |
| ○ MEGABYTE (MB) | Million bytes |
| ○ GIGABYTE (GB) | Billion bytes |
| ○ TERABYTE | Trillion bytes |

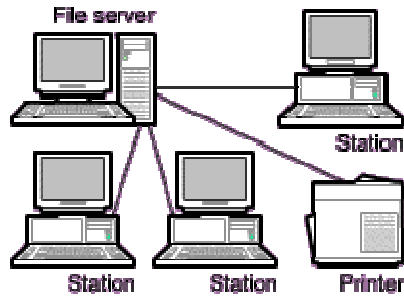
NETWORKING

LANs and WANs

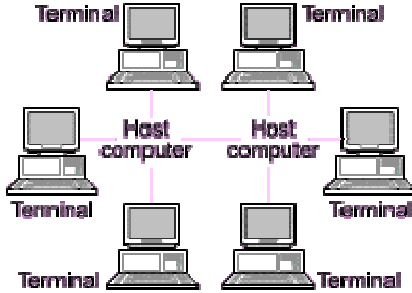
There are two types of network: **LAN** and **WAN**.

A **LAN** is a **Local Area Network** covering a small area such as one building e.g. in a school or a college.

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A WAN is a **Wide Area Network** covering a large geographical area.



- The **Internet** is a WAN.
- A network of bank cash dispensers is a WAN.
- A school network is a LAN.
- LANs are often connected to WANs, for example a school network could be connected to the Internet.

Telephone lines are often used to connect WANs. LANs use dedicated cables or wireless technology.

LAN

- **Local Area Network.**
- Are privately owned network within a single building or even on a single floor.
- LANs are widely used to connect personal computers and workstations in company offices and factories to share **resources e.g.** printers and exchange information.
- Common LAN topologies are Bus, Ring and Star.
- Resources are shared e.g. as above
- Software is shared
- Enhances communication between workstations
- File sharing

Definitions of **Local Area Network** on the Web:

- A computer network limited to the immediate area, usually the same building or floor of a building.
- (1) A computer network located on a user's premises within a communication across the LAN boundary may be subject to some form of regulation. (2) A network in which a set of devices are connected to one another for communication and that can be connected to a larger network.
- Communications network connecting computers by wire, cable, or fiber optics link. Usually serves parts of an organization located close to one another, generally in the same building or within 2 miles of one another. Allows users to share software, hardware and data.

WAN

- **Wide Area Network.**

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- A WAN spans a large geographical area often a country or continent.
- Also called **long-haul network**.
- Transmission lines for a WAN can be leased or can be private lines. I.e. leased telephone lines, fibre optics, coaxial cables, microwave radio links, satellite connections etc.

Definitions of **Wide Area Network** on the Web:

- A wide area network (WAN) is a geographically dispersed telecommunications network. The term distinguishes a broader telecommunication structure from a local area network. A wide area network may be privately owned or rented, but the term usually connotes the inclusion of public (shared user) networks. An intermediate form of network in terms of geography is a metropolitan area network (MAN).

Advantages and disadvantages of networks

ADVANTAGES OF GROUP WORKING & SHARING RESOURCES OVER A NETWORK.

1. It turns isolated computers into integrated systems, providing an environment where resources are shared and capacity problems reduced.
2. Resource sharing is the primary benefit of networking. It allows users on different machines to share modems, printers, tape drives and disk space, for example, users can send network messages requesting to use a central printer, allowing everyone to share that resource.
3. It allows growth. If more computing resources become a need, a new computer can be installed, added to the network, and immediately accessed by other machines and users.
4. High reliability. If applications share data, the data can be replicated across several machines. If one machine goes down, another can take its place and provide the data to the applications.
5. Users realize the benefit of sharing information. Data files can be shared between machines on the network, allowing users to see invoices, results of surveys, company newsletters, and other information.
6. Low cost. Single device shared by several machines reduce the need to buy many peripheral devices. Resource sharing also saves money. An installation can have several low-cost work stations accessing a single file-server. That puts a lot of processing power on the user's desk without the expense of large mainframe systems.
7. Sharing devices such as printers saves money.
8. Site licenses are likely to be cheaper than buying several stand-alone licenses.
9. Files can easily be shared between users.
10. **Network** users can communicate by **email**.
11. Security is good - users cannot see other users' files unlike on stand-alone machines.
12. A **file server** is easy to back up as all the **data** is stored in one place

DISADVANTAGES

1. Security concerns. Some user can gain unauthorized access to private data.
2. Increased administration. There will be need for a systems administrator to tune the network, monitor the network, administer database files, and ensure network integrity (ensure the network runs smoothly)
3. Network failure. As applications increase their use of a network, network failures become catastrophic.
4. Virus attack. A networked system is vulnerable to destructive network messages. For example, a message that sparks damaging activity (like erasing files) may enter the system via the network.
5. Purchasing the network cabling and **file servers** can be expensive.
6. Managing a large **network** is complicated, requires training and a network manager usually needs to be employed.

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7. If the file server breaks down the files on the file server become inaccessible. E-mail might still work if it is on a separate server. The computers can still be used as stand alones.
8. **Viruses** can spread to other computers throughout a computer network.
9. There is a danger of **hacking**, particularly with wide area networks. Security procedures are needed to prevent such abuse.
10. Of all the disadvantages, the greatest drawback is security concerns, therefore, network security is a priority in the development of network applications.

What is the Internet?

- ❖ **The Internet** is the network of networks being a worldwide collection of computers that communicate with one another over cables, satellites and optical fibers. It is literally the whole hardware mass. The Internet is the network used to transport information.
- ❖ When we refer to the Internet we are usually talking about the **World Wide Web (WWW)** which is the most used feature of the Internet.
- ❖ The WWW stores millions of **web pages** on **web servers**. These pages can contain text, pictures, movies, animation and sound. Web pages are written in a language or code called **HTML** (Hypertext Markup Language). A set of web pages is called a **website**.
- ❖ Each web page has its own unique address or **URL**. The URL will have the format "http" and a domain (such as ".co.uk"). What goes in between is arbitrary, but often has the term "www" such as in "**http://www.name.co.uk**", but it doesn't have to (e.g. <http://news.bbc.co.uk>).
- ❖ Most sites have a page that links the user to the other main areas of the site. This is called the **homepage**.



- ❖ Web pages are connected by **hypertext links**. When a link is clicked you will be taken to another page which could be on another server in any part of the world.
- ❖ When you move around web pages you are said to be **surfing** the net. For this you need a **program** to read the pages (called a **browser**), such as Firefox or Internet Explorer.
- ❖ To search for a particular item or topic on the net you use a **search engine**. There are many different kinds of search engine, each using slightly different ways of searching and indexing web content. *Google*, *MSN* and *Alta Vista* are all examples of search engines, while *Yahoo* and *Excite* are web directories (a bit like the Yellow Pages phone book) which have a search function built in.
- ❖ **The World Wide Web** Is software that runs on the internet which includes all documentation that is shared and accessed on the internet.
- ❖ A **Web page** is a document designed to be accessed and read over the WWW. It must have an address in a recognized format.

What are the advantages & disadvantages of the Internet? Discuss

These are far too broad and extending to all realms in life. For instance:-

- 1) Access to a wide range of information from any location worldwide where there is internet access. i.e. airlines, resorts, books, authors.
- 2) Research on any topic i.e. a professor, current affairs.
- 3) Online Shopping i.e. for cars, specific dress.
- 4) Online ordering and carrying out other business transactions e.g.. Ordering a cheque book, advertising a product.
- 5) Online conferencing – communicating across international boundaries.
- 6) Ability to access e-mail from any location worldwide.
- 7) Reading newspapers online.

Advantages of the Internet

The Internet provides opportunities galore, and can be used for a variety of things. Some of the things that you can do via the Internet are:

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- **E-mail:** E-mail is an online correspondence system. With e-mail you can send and receive instant electronic messages, which works like writing letters. Your messages are delivered instantly to people anywhere in the world, unlike traditional mail that takes a lot of time.
- **Access Information:** The Internet is a virtual treasure trove of information. Any kind of information on any topic under the sun is available on the Internet. The 'search engines' on the Internet can help you to find data on any subject that you need.
- **Shopping:** Along with getting information on the Internet, you can also shop online. There are many online stores and sites that can be used to look for products as well as buy them using your credit card. You do not need to leave your house and can do all your shopping from the convenience of your home.
- **Online Chat:** There are many 'chat rooms' on the web that can be accessed to meet new people, make new friends, as well as to stay in touch with old friends.
- **Downloading Software:** This is one of the most happening and fun things to do via the Internet. You can download innumerable, games, music, videos, movies, and a host of other entertainment software from the Internet, most of which are free.

Disadvantages of the Internet

There are certain cons and dangers relating to the use of Internet that can be summarized as:

- **Personal Information:** If you use the Internet, your personal information such as your name, address, etc. can be accessed by other people. If you use a credit card to shop online, then your credit card information can also be 'stolen' which could be akin to giving someone a blank check.
- **Pornography:** This is a very serious issue concerning the Internet, especially when it comes to young children. There are thousands of pornographic sites on the Internet that can be easily found and can be a detriment to letting children use the Internet.
- **Spamming:** This refers to sending unsolicited e-mails in bulk, which serve no purpose and unnecessarily clog up the entire system.
- **There is information overload for the internet.** Sifting through relevant information can be a task in itself.
- If there is tremendous increase in internet use, then **gaining access will be a problem.**

If you come across any illegal activity on the Internet, such as child pornography or even spammers, then you should report these people and their activities so that they can be controlled and other people deterred from carrying them out. Child pornography can be reported to:

- Your Internet service provider
- *Local police station*
- *Cyber Angels (program to report cyber crime)*

Such illegal activities are frustrating for all Internet users, and so instead of just ignoring it, we should make an effort to try and stop these activities so that using the Internet can become that much safer. That said, the advantages of the Internet far outweigh the disadvantages, and millions of people each day benefit from using the Internet for work and for pleasure

The Web uses three standards namely:

- 1 URLs (Uniform Resource Locators) – which are web page addresses
- 2 HTTP (Hypertext Transfer Protocol) – the language used to design web pages
- 3 HTML (Hypertext Markup Language)

To send e-mail access the Internet, both the sender and receiver need:

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WHAT IS NEEDED FOR INTERNET AND E-MAIL SERVICES

- 1) A computer connected to a network,
- 2) The windows operating system installed
- 3) The communication and or browser software such as Ms-Exchange or Ms-Outlook, Internet explorer, Netscape navigator
- 4) A modem
- 5) Telephone line
- 6) Subscription to an ISP (Internet Service Provider).

What do you need to get connected?

To connect to the internet you need:

- 1) A computer
- 2) A telephone line
- 3) A modem - the type of modem you need to use is dependant on the type of connection you have. Some of the choices are:

An analogue **modem** and an ordinary phone line. This type of **modem** links your computer to the phone and converts computer signals to analogue phone line signals - and back again. Typical analogue modem speeds are 56Kbps (**bps** stands for bits per second) which means they can receive about 6000 characters per second. This is the slowest and oldest type of connection and becoming less common with the introduction of broadband.

An **ISDN** line and terminal adaptor. This digital connection is slightly faster than an analogue connection.

An **ADSL** or cable telephone line and broadband modem. *Broadband modems* are much faster than the other two options and their use is increasing.

- 4) An account with an ISP (Internet Service Provider)
- 5) Browser software

ISPs are the companies who provide you with access to the **internet**. Commonly used ISPs include Freeserve, AOL, Virgin, Tesco, BT and many more. Most offer the same basic package of Internet access, **email** addresses, web space for your own pages and local rate call charges.

A **browser** is a **program** that allows you to view the pages on the Web. The most widely used are Internet Explorer and Firefox.

All browsers will have a number of similar features to help you use the Web:

- Forward and back buttons to move between pages
- A history folder which stores details of recently visited web pages
- A stop button if a page is taking too long to load
- Favorites and bookmark options to store often visited pages
- Options to cut, copy, save and print the information viewed

Electronic mail

Electronic mail or **email** is a means of sending messages, text, and computer files between computers via the telephone network. Because the telephone network covers the whole world, email enables you to communicate *world wide*.

Email and web mail

Email

To set up **email** you need:

- A computer
- An internet connection via an analogue modem (ordinary phone line) or terminal adaptor (ISDN) or broadband modem (ADSL and cable)
- An account with an ISP (Internet Service Provider)
- Email software

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Your Internet Service Provider will give you an email account, a password and a mailbox such as **yourname@hostname.co.uk**. You can also set up an email account with a mailbox and passwords with non-ISPs such as Google and Hotmail.

With a dial-up connection, you have to pay the cost of your internet phone calls (local rate), and in most cases a subscription to your provider (though some are free). A broadband connection is 'always on', with only a flat-rate subscription. Very few people pay by the minute nowadays and the majority of people pay a monthly fee for broadband access.

Anti-virus scanning is becoming standard on email accounts and many email providers now offer an anti spam (electronic junk mail) service.

Web mail

Web mail, as its name suggests, is web-based email. To use web mail, you do not need any email software - just a computer connected to the internet via any one of the connection types listed above, and a browser.

Users simply sign up to the web mail service of a web portal such as MSN or Yahoo. They are given a unique user name and a personal mailbox on the portal's email server, and they can then send and receive messages via a special web page.

A basic web mail account is usually free, although this will have a very limited amount of storage. The advantage of web mail is that users can receive and send mail from any computer in the world with internet access. If you have a dial-up connection you can download your emails and then read them offline to avoid staying on-line for long periods.

Some ISPs will enable their regular email customers to access their mailbox via web mail as well as through the email software on their PC.

Features of email

- Automatic reply to messages
- Auto forward and redirection of messages
- Facility to send copies of a message to many people
- Automatic filing and retrieval of messages
- Addresses can be stored in an address book and retrieved instantly
- Notification if message cannot be delivered
- Automatically date and time stamped
- Signatures can be attached
- Files, graphics or sound can be sent as attachments, often in compressed formats
- Web mail and mobile email can be used to receive and send messages while on the move.

Using email

- **Broadband email**
 - Your connection is always on - so simply write and address your message and hit 'send' to send your mail and 'receive' to download any incoming messages from your email provider's server.
- **Web mail**
 - Connect to the internet (if using a dial-up account) and open your browser.
 - navigate to your web mail provider's portal and enter your user-name and password.
 - all incoming mail will now be visible, and you can also compose and send mail, and download attachments to your computer.
 - When you're done, log out and close your connection.
- **Dial-up email**
 - Prepare your message **offline** as typing your message **online** will increase phone charges.
 - Connect to the **internet** and log on to your email account.
 - Send your message and download any incoming mail sitting on your service provider's computer.
 - Log off and close your connection.

Email is evolving...

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- Many mobile phones already allow messages to be sent to the recipient's email inbox while the sender is on the move. The latest generation of mobile phones enables users to send and receive wireless email in exactly the same way as a static computer.
- Email can be sent and received via digital TV, specially adapted phones, public kiosk terminals and the latest generation of games console.
- A spreading network of wireless 'hotspots' in public places allows people to send and receive email via laptop computers.
- A new range of in-car phones will enable motorists to check their email on the road

Benefits and concerns of using email

Email benefits

- Fast delivery of your message
- Available 365 days, 24 hours per day - and, with web mail, wherever you are in the world as long as you have access to the internet.
- Cheap: when using broadband, individual mail transfers are effectively free. When going online from a dial-up account, calls are charged at local rates and (for conventional email) need only last a few seconds.
- Facility to send the same message to more than one person

Email concerns

- It can only be sent to people who themselves have access to the internet.
- **Viruses** are easily spread via email attachments - anti virus measures must be in place to avoid this and are now offered by many e-mail providers.
- **Phishing** - sending an e-mail to a user falsely claiming to be a legitimate company to scam the user into providing information such as personal information and bank account numbers on a bogus website. The details will then be used for identity theft.
- No guarantee the mail will be read until the user logs on and checks their mail.
- Spam! Or Junk mail

NETWORK SYSTEMS

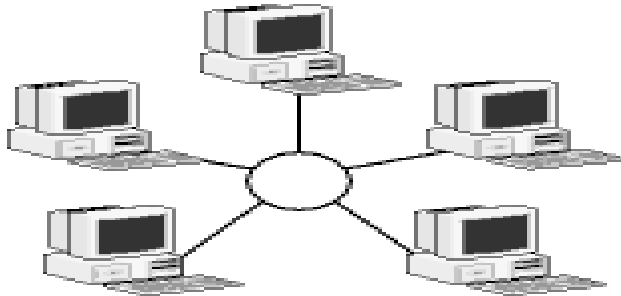
- Is one in which processing is carried out independently in more than one location, but with shared & controlled access to some common facilities, which normally include file storage & information resources?

Definitions of **hub** on the Web:

- ❑ A common connection point for devices in a network. Hubs are commonly used to connect segments of a LAN. A hub contains multiple ports. When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets. A passive hub serves simply as a conduit for the data, enabling it to go from one device (or segment) to another. ...
- ❑ **Node** – *a terminal point in a communications network.*
- ❑ **Workstation** – *A computer terminal or micro- or minicomputer system designed to support the work of one person.*

STAR NETWORK

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- Is a network in which nodes are connected to a central component as its hub?
- The central component can be a switching device like a switch or PABX [*private automatic branch exchange*], a computer [minicomputer to which workstations or PCs are connected] or just a wiring center that is a common termination point for the hub.
- A HUB is a component that serves as a common termination point for multiple nodes and that can **relay** [store & forward] signals along the appropriate paths.

Definitions of **star network** on the Web:

- A network in which all terminals are connected through a single point, such as a star coupler.
- A network topology in the form of a star. At the center of the star is a wiring hub or concentrator, and the nodes or workstations are arranged around the central point representing the points of the star.

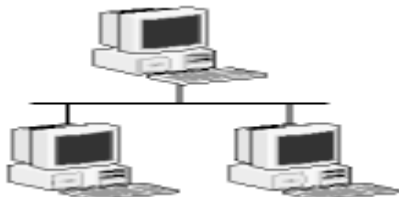
Advantages of star topology

- It is easy to add or remove nodes
- Because each node has its own link to the central node, the star is more reliable than other topologies.
- Full utilization of resources i.e. printer, sharing of processor
- Direct connection of terminals to the main processor, means that if one terminal breaks down it won't affect other terminals.

Disadvantages of star topology

- All traffic between two nodes passes through the central node. If the central component breaks down, the whole network is down.
- The number of ports of the central component limits the number of connected nodes.
- A star topology requires a lot of cabling since all data path connections must be to the central node.
- There is no direct communication between or among workstations
- Sharing of processor delays in the processing of jobs.

BUS/LINEAR NETWORK



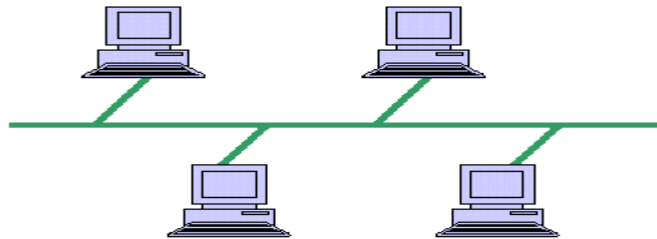
- ✓ Describes a network in which each node is connected to a common line.
- ✓ This is also sometimes called a horizontal, multidrop or multi point network.
- ✓ Though it is preferable to use the last terms for connections with a mainframe/mini computer as in the following diagram.

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- ✓ In a bus messages travel in both directions and do not go through the individual nodes but every node can hear each message as it goes past.
- ✓ When the signal/message reaches an end of the bus, a terminator absorbs it, to keep it from traveling back again along the bus line, to avoid interference with other messages already in the line.
- ✓ The data/signals/messages are transmitted in packets along the bus.
- ✓ Each packet is send with a receiver identification code **that is** an address of destination and all computers on the network are permanently on the alert for any messages coming to them.

Definitions of **bus network** on the Web:

- A single-cable LAN, in which all computers on the network are connected in series to a single cable. On a bus network, each computer hears all of the transmissions going down the line, and selects only those that are addressed to its location. This is the simplest and most common LAN technology.



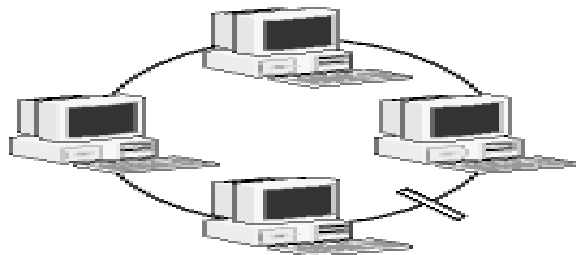
Advantages of Bus Network/topology

- When a node breaks down, the network does not break down.
- A bus uses relatively less cables compared to other topologies
- Direct communication due to direct connection.
- Fast processing since there is no Host computer
- Once a cable has been laid down any new equipment can easily be connected to the network by a simple tapping into the cable.

Disadvantages of Bus Network/topology

- In a bus topology it is not always easy to add a node. [E.g. installation of extra tap in thick Ethernet].
- Diagnosis/trouble shooting can be difficult.
- Expensive to run [several processors] – Detectors.
- Communication between components of the network traveling in opposite direction along the bus can **collide** causing loss of transmitted information.
- Because of this collision a detection system has to be present to control the re-transmission of lost information thereby making it a more expensive option

RING NETWORK



- ✓ A ring network has nodes connected in a circular way.
- ✓ Each node has two neighboring nodes
- ✓ Data flow is **unidirectional**.
- ✓ A message is forwarded in one direction until it reaches the destination with intermediate nodes acting as relay [store & forward] units.
- ✓ The destination node copies the message and passes the message again to the ring.
- ✓ This message then continues to circulate around the ring back to the source.

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Definitions of **ring network** on the Web:

- A network topology in the form of a closed loop or circle. Each node in the network is connected to the next, and messages move in one direction around the system. When a message arrives at a node, the node examines the address information in the message. If the address matches the node's address, the message is accepted; otherwise the node regenerates the signal and places the message back on the network for the next node in the system. ...

Advantages of ring network

- The message return provides the source with a form of **acknowledgement**. The node removes its own message from the ring.
- Little cabling needed compared to Star.
- Each node acts as relay unit.
- Packet collisions do not occur because information is traveling in one direction.
- Relatively cheap to install.
- Efficient and accurate means of communication.
- There is no dependency on the host computer or file server as each node controls transmission to and from itself.

Disadvantages of ring network

- Adding or removing nodes disrupts the network.
- If one of the nodes breaks down, the whole network will be down.
- If the network cable breaks the whole network is affected

HEALTH AND SAFETY

HEALTH PROBLEMS ASSOCIATED WITH COMPUTER USE AND POSSIBLE SOLUTIONS

Are Computers a Health Hazard?

The first thing to know is that the risks are not likely to affect you unless you are a "habitual" computer user. In other words, you're sitting at the computer pretty much all day, every day. Oh yeah... that's all of us! That's not to say that the occasional computer user won't have problems. Everyone's level of sensitivity is unique. The buzz started in the 80's and culminated in the 1992 Health and Safety DSE (Display Screen Equipment) regulations. Then came Carpal-Tunnel Syndrome followed by ergonomics. The hype has subsided, since we all know the computer isn't going to kill us - but we have learned a lot over the past 20 years about potential health risks and more importantly, we've learned ways to avoid being at risk when we're using computers.

Let's look at some of the most common medical problems and what you can do to avoid them.

1. Eye Strain:

- Position your terminal at right angles to the window if possible; avoid facing directly into bright light (coming at you from behind your computer screen).
- Install an anti-glare screen.
- Adjust the brightness controls on the screen until they are comfortable to your eyes.

2. Carpal Tunnel Syndrome:

- Adjust your chair or table height to have your elbow angle at 90-100 degrees.
- Position your keyboard so that you don't have to bend your hands uncomfortably upward to reach the keys; place a raised wrist rest on the table in front of the keyboard if necessary.
- Clench your fists, hold for one second, then stretch your fingers out wide and hold for 5 seconds.
- Organize your workday, if possible, to intersperse other tasks with your computer work so that you're not sitting at the computer for several hours without a break. Variety is key.
- Hold the mouse loosely and click lightly.

3. Neck and Back Strain:

- Check your posture - sit up straight. Thanks Mom.
- The monitor screen surface should be approximately 18-24 inches away from your torso.
- Preferably chairs should be on wheels, have backrest tilt adjustment, and have arms.
- Be sure you have enough desktop space for work papers and other equipment.

4. Conjunctivitis (itchy, bloodshot eyes) and Dermatitis:

- Be sure the screen doesn't flicker or wave - this could indicate that service or adjustment is needed.
- Look away from the screen periodically.

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- Don't forget to blink - your eyes need the moisture.

5. The vision disorder

Due to excessive computer use has been identified as Computer Vision Syndrome. Symptoms are dry eyes, headaches, blurred vision, eyestrain, and shoulder back pain.

- To alleviate the problem it is suggested that computer users take regular breaks, blink their eyes frequently, occasionally close their eyes for a few minutes and every fifteen minutes or so look away from the computer to stare at an object in the distance.

6. For the back pain and other muscular related problems,

It is suggested that people get up every hour, stretch and move around for about five minutes. They should also do an activity which moves each foot and leg.

- A computer user should be seated at least two feet away from the screen with the screen below eye level. Their chair should be comfortable and they should sit up straight in the chair with feet firmly on the floor. They should not cross their legs.

7. Posture-related injuries

Back and neck pain, headaches, and shoulder and arm pain are common computer-related injuries. Such muscle and joint problems can be caused or made worse by poor workstation design, bad posture and sitting for extended periods of time.

Although sitting requires less muscular effort, it still causes fatigue and requires parts of the body to be held steady for long periods of time. This reduces circulation to the muscles, bones, tendons and ligaments and can result in stiffness and pain. If a workstation is not set up properly, these steady positions can put even greater stress on muscles and joints.

Prevention tips – muscle and joint injuries

Suggestions to reduce the risk of muscle and joint problems include:

- Use an adjustable desk designed for use with computers; position the monitor so that it is either at eye level or slightly lower.
- Position your keyboard at a height that allows your elbows to rest comfortably at your side. Forearms should be roughly parallel with the floor and level with your keyboard.
- Adjust your chair so that your feet rest flat on the floor.
- Use a footstool (if your feet do not rest on the floor when the chair is adjusted for good arm position).
- Switch to an ergonomic chair, which helps your spine to naturally hold its curve while sitting.
- Use an ergonomic keyboard to offer your hands and wrists a more natural holding position.
- Take frequent short breaks and go for a walk or perform stretching exercises at your desk. Stand often.

8. Overuse injuries of the upper limbs

Muscles and tendons can become painful with repetitive movements and awkward postures. This is known as 'overuse injury' and these typically occur in the elbow, wrist or hand of computer users. Symptoms of overuse injuries in the upper limbs include pain, swelling, stiffness of the joints, weakness and numbness.

Prevention tips – overuse injuries

Suggestions to reduce the risk of overuse injuries include:

- Keep your mouse at the same height as your correctly positioned keyboard.
- Position the mouse as close as possible to the side of the keyboard.
- Use your whole arm, not just your wrist, when using the mouse.
- Type lightly and gently.
- Mix your tasks to avoid long, uninterrupted stretches of typing.
- Remove the hands from the keyboard when not actively typing, to allow the arms to relax.

9. Eyestrain

Focusing your eyes at the same distance point for extended periods of time causes fatigue. The human eye structurally prefers to look at objects further than six metres away, so any work performed close-up puts extra demands on the eye muscles.

The illuminated computer screen can also contribute to eye fatigue. While there is no evidence that eye fatigue is associated with damage to the eyesight, computer users may experience symptoms such as blurred vision, temporary inability to focus on faraway objects and headaches.

Prevention tips – eyestrain

Suggestions to reduce the risk of eyestrain include:

- Make sure your primary light source (such as a window) is not shining into your face or directly onto the monitor.
- Tilt the monitor slightly to eliminate reflections or glare.
- Make sure your computer screen is not too close to your face.
- Position the screen so that it is either at eye level or slightly lower.
- Reduce the contrast and brightness of your screen by adjusting the controls on the monitor.
- Frequently look away from the screen and focus on faraway objects.

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- Have regular eye examinations to check that blurring, headaches and other associated problems are not caused by any underlying disorders.

Low Back pain

- Usually due to using the wrong type of chair, these have been four legged chairs found in any office for visitors, executive chairs where the back is not separate from the seat and therefore cannot be adjusted to offer any lumbar support.
- If the user has not taken any time to adjust the chair.
- Sitting on the edge of the chair and leaning forward can also cause **problems** and this often happens when there are arms on the chair which are not adjustable in height or when the user is short in stature, the arms on the chair will not move low enough to allow the user to get close enough to the desk.
- Sitting with the legs crossed causes uneven weight on each buttock and strain on back muscles.

I have seen a lot of **problems caused by arms on chairs that cannot be raised or lowered and recommend that all chairs are purchased without arms for this reason.**

Poor desk configuration

If the screen is placed off to one side, the keyboard should still be positioned centrally to avoid twisting the torso, often if the screen is off to one side, the user will support one elbow on the desk putting strain in the muscles of the back.

If odd pieces of furniture are used to make up a workstation ensure that they are the same height. **It is best to draw your proposed desk layout before moving furniture.**

Having a set of drawers that prevent sideways movement

The L shaped desk layout allows more space and freedom of movement if writing and keyboard activity is required but when two desks are butted onto each other, one set of drawers can restrict movement. The set of drawers restricting movement can be removed with a screwdriver.

Ideally desks should be purchased without drawers and then modular drawer units can be placed in convenient locations.

Room temperature

If a room is too cold muscles can go into spasm and injury is more likely.

Dehydration

Many office workers drink too much tea and coffee, which has a diuretic action so water is the best choice. Remember that offices with several computers can make the air feel very dry.

Overstretching

Items used most frequently should be positioned within easy reach

If the hard drive/midi tower is housed under the desk, muscles can be pulled when bending over to switch the **computer** on.

Pain in the neck

- Pain in the neck is often caused by too much up and down or side-to-side movement of the neck, a solution is to use a document holder at the same height as the screen or attached to the screen so that it is only necessary to transfer the gaze.
(These can be purchased from Gaynor in the **computer** shop for approx £4.00 and are much cheaper than in the Lyreco catalogue. You will need to complete a purchase order form for this product. **(Approval for the use of purchase order forms has been sought from Mike Montgomery.)**
- Sitting with the telephone receiver in you neck.
- Using bifocals because the neck is bent backwards to focus on the screen. These people should speak to their Optician about using a special pair of glasses for **computer** use or have their bifocal section on the glasses made wider.
- If the screen is too high. The head should be in the neutral position with the gaze at a slightly downward angle when viewing the screen.

Not taking posture breaks

Posture breaks are important to prevent overuse of one group of muscles, break up periods of keying and mouse work regularly. (See information sheet on posture breaks and wrist exercises)

Pressure on the wrists from a sharp edge on a desk

Some desks have a sharp edge, which can cause pressure on the wrists. If **users** are trained typists, they usually do not rest their wrists whilst keying but most self taught keyboard **users** do, this causes pressure and increased risk of injury if the user is also working with their wrists bent. A gel wrist rest can be used to provide a soft cushion for resting the wrists in between periods of keying or mouse activity. Gel mouse mats are also available.

Painful wrists or fingers

- Painful wrists or fingers often occur when the chair is not adjusted to the correct height to allow a perfect L shape between the shoulder, elbow and wrist, if the wrists are bent up or down then the fingers are not horizontal to the keys or the mouse, the clicking finger is not horizontal to the control mechanism and the wrist is left unsupported.
- Striking the keys too hard or excessive mouse use can also cause painful fingers or by not keeping the mouse close enough to the body. Ideally a mouse should be used directly in front of the shoulder and as close to the body as possible.

Dirty mice are not efficient and require more clicking therefore increasing your risk of overuse syndrome.

Research has shown that intensive use of the mouse is more likely to cause symptoms of overuse syndrome than keyboard activity; you should consider keyboard shortcuts (These can be viewed on a separate page.)

Alternative mice

There are various alternative mice that relieve and prevent symptoms of overuse:

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- Optical mice are more efficient because they do not get dirty balls!
- Cordless optical mice allow more freedom
- Tracker balls offer an alternative movement but one could argue that you are just exchanging one repetitive movement for another and transferring the strain from the fingers to the thumb.
- Joystick mice often have thumb controls.
- Beware of mice that are advertised as being ergonomic because they may not be. Some mice totally fill the hand and can be uncomfortable for **users** with small hands.
- Touch pads can be used in either hand and are ideal for people with one-handed symptoms because they can be used in either hand; the same amount of dexterity is not required for operation.
- The whale mouse is ideal for mouse intensive work and can open out to fit any hand.

There are numerous options on the market but it is better to try before you buy.

You may want to contact other members of staff on the intranet at UWB to see if someone has a different type of mouse for you to try or you can contact the OHSU to see if there is a mouse for you to borrow for a short period so that you can evaluate it's effects on your symptoms

Pain in the buttocks

Pain in the buttocks radiating down the thigh can be caused by a chair seat that is too long and causing pressure behind the knees or from inadequate lumbar support.

Pain in the shoulders

- Pain in the shoulders can be caused by having arms on the chair that cannot be adjusted this causes the user to sit with the arms sticking out.
- Sitting at the desk with one elbow supported on the desk and the other held lower or twisting the torso to look at the monitor will also cause pain.

Pain in the knees

- Pain in the knees can be caused by the fact that when the chair has been adjusted to the correct height to allow the fingers to be horizontal to the keys, the feet are left unsupported, the solution is to purchase a footrest the small step type called (twin plus in the Lyreco catalogue are best because they do not take up much space under the desk, they can be adjusted according to the height of the person using them.)
If transcription is carried out it is better to have a large surface area on the footrest and the ability to keep this level to support the footswitch.
- Pain in the knees can also be caused by a seat that is too long resulting in pressure behind the knees.

Eye strain

- Eyestrain can be caused by glare from the monitor or from overhead lights. Ideally the desk should be placed between lights and not directly under them. If this is not possible you may be able to switch the overhead light off and use an angle poise lamp.
- **Glare from windows without blinds or with incorrectly fitted blinds.**
- Where there are casement window at the top and blinds cannot be used there is often glare from the sun on the screen. Some vertical blinds are not efficient and additional blackout roller blinds would improve efficiency and keep the room cool. **If blinds cannot be fitted it may be possible to fit a tinted transparent film to the window.**

Other solutions:

- See the Optician about middle distance correction
- Ask about a slight tinted coating to your glasses
- Contact IT if there is any flicker on your screen
- If your monitor is old and small (14inch CGA, VGA) it will have poor resolution and should be replaced with a 17inch SVGA or XGA.
- Clean your screen
- Adjust your contrast or brightness
- Make sure that you are not too far away from the screen (correct viewing distance is 18-20 inches)
- Remove the antiglare device. (These can cause **problems** focusing because you have to look thought it to see your screen.)
- Introduce regular task rotation to break up periods of keying.

Choosing the right chair

Tudor Jones in the Estates department will order chairs for you and will be able to advise on models that are appropriate for **computer** use, by using this service you will also benefit from considerable cost savings.

If you use a **computer** you should order a chair that has a height adjustable back and preferably without arms as these often cause poor posture. If you really cannot live without arms on your chair they should be totally height adjustable (They do cost extra).

When you receive your new chair you should read the booklet that is usually attached so that you understand how all the mechanisms work. There is usually a handle or knob to raise and lower the height of the seat and another to enable a rocking action.

It is important to use the rocking action, as this will reduce static loading.

There will be a mechanism for raising and lowering the height of the backrest to ensure adequate lumbar support. (Sometimes you need to release a tension knob at the back of the chair to allow for movement of the backrest.

WORD PROCESSING TERMINOLOGY

Computer Operations & Packages [C.O.P.]

Most computers today come with a **word processing** program which allows the user to write text documents like letters, business correspondence, and even books. The first step in learning how to use a **word** processor is to understand the **terminology** that is unique to **word processing** programs. Here are a few basic terms that will help.

Typeface

A typeface defines the shape of the characters. Some common typefaces are Times New Roman, Arial, and Courier. If you look closely, you will see that the letters in each of these are shaped differently, some narrower, some rounder.

Font

A font is collection of letters and numbers in a particular typeface. The font includes the typeface plus other qualities, such as size, pitch, and spacing. For instance, within the Times Roman family, there are many fonts to choose from, including different sizes, italic, and bold. If you are working in a **word processing** program, the font might be Times New Roman, 10 point, bold. Changing the font can completely change the look and feel of a Web page or document.

Format

In a **word processing** program, you can format any text that you create. Formatting involves specifying the font, alignment, margins, and other properties. The format is the layout of a document. The format determines how the document will appear on the screen and how it will look when printed.

Alignment

The alignment refers to the way lines of text are arranged relative to the edges of a block of text. For example, the text can be flush with the left side of the document (left aligned), flush with the right side of the document (right aligned), or centered horizontally on the document (centered). There is also an alignment option called justified. In text that is justified both the right and left sides of the text will appear to line up evenly.

Tab

The tab function in a **word** processor works much like the tab function on a typewriter. Tab stops can be set in the ruler at the top of the page. When the tab key is pressed, the cursor will move to the next tab location. This allows the user to indent paragraphs and lists.

Margins

Margins are the blank space to the left and right sides and at the top and bottom of a page. The size of the margins can be increased or decreased. In Microsoft **Word**, this is done by choosing File-Page Setup and changing the numbers on the Margin tab.

Header/Footer

The header is one or more lines of text that appear at the top of every page of a document. The header often includes information like names and titles. The footer consists of one or more lines of text that appear at the bottom of every page of a document. The footer often includes the page number.

Line Spacing

Line Spacing refers to the number of blank spaces between lines of type. The most common line spacings are single-spaced and double-spaced, but the line spacing in a **word** processor can be set to almost any amount of line spacing.

Word Wrap

Word wrap is a **word processing** feature that permits information to be keyed on successive lines without having to tap the return/enter key at the key of each line.

Table

A collection of data arranged in rows and columns. Tables are commonly used in **word processing** programs. They are also used in spreadsheets and database programs.

Spell/Grammar Check

Word processing programs contain a spell check program which can verify the correct spelling of words in a document. In some programs you will click on a menu choice or an icon to start the spell checking. Some programs will automatically check your spelling as you go, inserting a squiggly red line under words that may be misspelled. Grammar checking works in much the same way with the program inserting a squiggly green line under words that may be grammatically incorrect. Remember the spell check and grammar checking are not foolproof. A **word** may pass both checks and still be incorrect.

Computer Operations & Packages [C.O.P.]

Indent

An indent is one or more spaces that are used to move the text to the right of the left margin.

Ruler

In a **word** processor, you can use the ruler to set the indent, margin and tab markers and this can allow complex formatting options. Never use the space bar and return key to align text.

Template

Templates establish the initial document setting and formats. A **word processing** program like Microsoft **Word** uses the "normal" template as the basis for all documents. A user can modify the "normal" document, and/or may create other templates to use.

Bullets & Numbering

A **word processing** program can automatically add bullets and/or numbers to the text.

Print Preview

This is a very useful feature. Click on the menu choice or icon for a Print Preview and you will be shown an image of exactly what the printed output will look like.

Spelling checker

- A companion application that you use with a word-processing application to check for misspelled words.
- A word processing function which searches for and corrects misspellings by comparing a document's words with those in a built-in spelling dictionary.
- spell-checker: an electronic dictionary in a word processor that can be used to catch misspelled words
- In computing terms, a spell checker or spelling checker is a design feature or a software program designed to verify the spelling of words in a document, helping a user to ensure correct spelling. A spell checker may be implemented as a stand-alone application capable of operating on an block of text; however, spelling checkers are more often implemented as a feature of a larger document-related application, such as a word processor or an email client.

Macros

- In the context of programming by demonstration, this usually refers to any procedure created by recording the user's actions as they are carried out. Unlike the conventional computer-science definition, the use here does not necessarily have any implications about how the procedure is executed (whether it is expanded in-line using textual substitution or called as a function).
- A macro in computer science is an abstraction, whereby a certain textual pattern is replaced according to a defined set of rules. The interpreter or compiler automatically replaces the pattern when it is encountered. In compiled languages, macro-expansion always happens at compile-time. The tool which performs the expansion is sometimes called a macro-expander. ...

Header

Text that appears at the top of every page of a document when it is printed.

Footer

- Located at the bottom of the page, it usually contains information about the page author or the sponsor. You may also find copyright information and the date the page was last updated.
- Text that appears at the bottom of every page in a document.

Mail merge

- The process by which information (e.g. names and addresses) from a database or other list and inserted into a standard document or letter to produce multiple personalized copies.
- A mass-mail facility that takes names, addresses, and (sometimes) pertinent facts about recipients and merges the information into a form letter or another such basic document.
- The transfer of data from one computer file to address and/or personalize mailing items in another.

JUSTIFICATION

HARD RETURN

END OF PART 2

SPREADSHEET TERMINOLOGY

Microsoft Excel is a spreadsheet program written and distributed by Microsoft for computers using the Microsoft Windows operating system and for Apple Macintosh computers. It features an intuitive interface and capable calculation and graphing tools which, along with aggressive marketing, have made Excel one of the most popular microcomputer applications to date. ...

Workbook

- A spreadsheet document that contains one or more worksheets, a Microsoft Excel document.
- A collection of related worksheets contained within a single file. (in spreadsheet applications).
- The workbook is the main document used in software programs like Excel. A workbook consists of individual worksheets, like an accountant's workbook, each of which can contain data. Basically, a workbook is a very sophisticated ledger.

Worksheet

- A single spreadsheet that contains rows and columns of data. In Excel, a worksheet will have a little white tab at the bottom of the screen. A worksheet is treated like a database table.
- An electronic spreadsheet containing 256 columns by 16,384 rows. (in spreadsheet applications).
- A work area comprised of rows and columns, also known as a spreadsheet.

x-axis label

- A label describing the x-axis of a chart. (in spreadsheet applications)

Cell An intersection of a row and column, the smallest element in which you can enter data.

Column A vertical selection of cells identified by one or more letters, starting with A at the left side of the **spreadsheet**.

Row A horizontal selection of cells identified by a number, starting with 1 at the top of the **spreadsheet**.

Formula An equation that performs an operation on cell data and displays the result. For example, to add the contents of cells A1 and A2, and display the results in A3, cell A3 would contain =A1+A2, a simple formula.

Function A pre-defined formula that allows you to quickly perform a more difficult operation on cell data, such as finding the average of a range of numbers. Functions always take *arguments*, or additional information that enables the computation to occur. For example, the argument to the AVG, or average, function is the range of cells to be averaged: =AVG(A1..C10) averages all cells between A1 (in the upper left) and C10 (in the lower right).

Cell reference

- The column number and the row letter of a cell.

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- The address or name of a specific cell. Cell references may be used in formulas and are relative or absolute. (Spreadsheet application).
- A cell name used in a formula.

Cell pointer

- A highlighted rectangle around a cell that indicates the active cell. (Spreadsheet application)

Relative cell reference

- Used to indicate a relative position in a worksheet. This allows you to copy and move formulas from one area to another of the same dimensions. Excel, for example, automatically changes the column and row numbers to reflect the new position. (in spreadsheet applications).
- Cell references in formulas that change when Excel copies them to another location.

Absolute cell reference on the Web:

- A cell reference in which a dollar sign (\$) precedes both the column and row portions of the cell reference.

Mixed cell reference

- Cell reference in which either the column or the row is never adjusted if the formula containing it is copied to another location.

Cell address

- Usually the intersection of a cell's row and column.
- Unique location identified by intersecting column and row coordinates. (Spreadsheet application)

What if analysis

- The process of playing with numbers to see how they interact in a spreadsheet.

Freeze panes

- Freezing panes prevents the data in the panes from scrolling as you move to different areas in the worksheet.

Goal seek on the Web:

- Goal Seek is a tool that is used to find the value needed in one cell to attain a result you want in another cell.

Formula

- A set of instructions that you enter in a cell to perform numeric calculations (adding, multiplying, averaging, etc.); for example, +A1+B1. (in spreadsheet applications)
- Instructs the spreadsheet to perform a calculation using the contents of a specific cell or range of cells

Filter

- A program that processes individual, personal preferences in an attempt to deliver relevant information to a particular end user.

Solver

- Solver is a tool used to perform what-if analyses to determine the effect of changing values in two or more cells on another cell.

Recalculation

- The act of calculating again (usually to eliminate errors or to include additional data); "recalculation yielded a much larger value"

Excel Terms

Absolute cell reference:

A cell address in a formula that does not change when copied to another cell. An absolute reference has the form \$A\$1.

Active cell:

The selected cell in a worksheet. The active cell is surrounded by a heavy border and is identified by the cell address.

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<u>Argument:</u>	The information that a function uses to produce a new value or perform an action. For example, a new value is displayed when the SUM function adds the argument (A6:A12). An argument consists of numbers, references, text, operators, or error values.
<u>Ascending:</u>	A method of ordering a group of items from lowest to highest, such as from A to Z.
<u>AutoFilter:</u>	Displays all records that meet the criteria as a subset of the database.
<u>AutoFormat:</u>	A feature used to format a range of cells with a predefined set of attributes.
<u>AVERAGE function:</u>	Sums the numbers in the specified range and then divides the sum by the number of non-zero cells in the range.
<u>Cell reference:</u>	An unique address given to a cell; the coordinates of the intersection of a column and a row.
<u>Cell:</u>	Basic unit of a worksheet into which you enter data.
<u>Chart:</u>	A graphic representation of worksheet data. Values from worksheet cells are displayed as bars, lines, or other shapes. Common chart types are pie, bar, line, and area.
<u>Column heading:</u>	The lettered gray area at the top of each column that identifies the letter of the column, such as column B.
<u>Comment:</u>	A note that explains, identifies, or comments on the information in a specific cell or range of cells.
<u>Condition:</u>	Made up of two values and a relational operator, is true or false for each cell in the range.
<u>Conditional formatting:</u>	Allows you to apply formatting that appears only when the value in a cell meets conditions that you specify.
<u>Criteria:</u>	The conditions that control which records to display in a query; the words or values used to determine the data that appears in a data list.
<u>Currency style format:</u>	Displays a dollar sign to the left of the number, inserts a comma every three positions to the left of the decimal point, and displays numbers to the nearest cent (hundredths place).
<u>Data table:</u>	Range of cells that shows the answers generated by formulas in which different values have been substituted.
<u>Data validation:</u>	Restricts the values that may be entered into a cell by the worksheet user.
<u>Debugging:</u>	Process of finding and correcting errors in the worksheet.
<u>Descending:</u>	A method of ordering a group of items from highest to lowest, such as from Z to A.

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<u>Dummy data:</u>	Used in place of actual data to verify formulas in the template.
<u>Embedded chart:</u>	A chart that exists on a worksheet instead of on a separate chart sheet.
<u>Exploded Pie chart:</u>	A Pie chart with one or more slices offset.
<u>Filter:</u>	A set of criteria you can apply to records to show specific tasks, records, or resources. The tasks, records, or resources that match your criteria are listed or highlighted so that you can focus on just the information you want.
<u>Formula:</u>	A sequence of values, cell references, names, functions, or operators that produces a new value from existing values. A formula always begins with an equal sign (=).
<u>Function:</u>	A built-in formula; a named and stored procedure that performs a specific operation and returns a value.
<u>Goal seeking:</u>	Used if you know the result you want a formula to produce by determining the value of a cell on which the formula depends.
<u>Gridlines:</u>	The horizontal and vertical lines on the worksheet.
<u>IF function:</u>	A function that tests the content of the cell, performs a calculation, and displays a value or text based on whether the test is true or false.
<u>Label:</u>	Row title or column title.
<u>Legend:</u>	A box containing the name of each data series in a chart.
<u>MAX function:</u>	Displays the highest value in a range.
<u>MIN function:</u>	Displays the lowest value in a range.
<u>NOW function:</u>	Used to enter the system date in a cell in the worksheet.
<u>PivotChart:</u>	An interactive chart that provides the user with ways to graphically analyze data by varying the fields and categories to present different views.
<u>PivotTable:</u>	An interactive worksheet table that summarizes data using a selected format and calculations. It is called a pivot table because you can rearrange the table structure around the data.
<u>Protected cells:</u>	Cells that you cannot change.
<u>Range:</u>	Series of two or more adjacent cells in a column or row or a rectangular group of cells.

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<u>Reference:</u>	A cell address used in a formula.
<u>Relative cell reference:</u>	In formulas, a reference to the address of another cell in relation to a cell that contains a formula.
<u>Template:</u>	Special workbook or worksheet you can create and then use as a pattern to create new, similar workbooks or worksheets.
<u>Unprotected cells:</u>	Cells whose values you can change at any time.
<u>Workbook:</u>	A Microsoft Excel document in which you can store other documents. A workbook can include multiple worksheets and chart sheets.
<u>Worksheet:</u>	A set of rows, columns, and cells in which you store and manipulate data. Several worksheets can appear in one workbook, and you can switch among them easily by clicking their tabs with the mouse.
<u>X-axis:</u>	A line, usually horizontal, along the bottom of a chart. The x-axis shows the labels for the data series. Also called the horizontal axis.
<u>Y-axis:</u>	A line, usually vertical, along the left side of a chart. The y-axis shows the values for the data series.

Part A

- i. **Worksheet** an electronic spreadsheet containing 256 columns by 16 384 rows. A single spreadsheet that contains rows and columns of data . A set of rows, columns and cells in which you store and manipulate data
- ii. **Cell address** usually the intersection of a row and a column. Is the location of a cell on a worksheet and is defined by the column letter and the row number. E.g. cell A1 is where column A and Row 1 intersect.
- iii. **Formula** a set of instruction that you enter in a cell to perform numeric calculations = A1+B1
- iv. **Cell Pointer** (cursor) A highlighted rectangle around a cell that indicates the active cell.
- v. **Cell** an intersection of a row and column, the smallest element in which you can enter data.

Part B

- i. **Footer** text that appears at the bottom of every page in a document
- ii. **Text** that appears at the top of every page of a document when it is printed.
- iii. **Spell Checker** F1 word processing function which searches for and correct misspellings by comparing a document's words with those in built-in spelling dictionary.
- iv. **Font** is a style and size of type e.g Times New Roman, 12 point is a set of all the characters available in one typeface and size, including uppercase and lowercase letters, punctuation, and numerals.
- v. **Alignment** the position of lines in a paragraph in relation to the document's left and right margins i.e. whether they are left-aligned, centered, right-aligned or justified.

**a) State the functions of the Control Unit & Arithmetic Logic Unit.
[5].**

Control unit functions	A.L.U functions
------------------------	------------------------

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Manages input-output to & from the main memory	Performs arithmetic operations
Interprets instructions in sequence	Performs logical operations
Issue commands to all elements of the computer	Performs computations on data.
Directs sequence of operations	Is where binary data is acted upon?
Directs traffic in the computer [make decisions]	Stores intermediate and final results of arithmetic operations
Interprets coded instructions	
Initiates proper commands to other parts of the computer.	
Performs the functions of fetch, decode, execute, and store.	

Three types of Data Transmission:

1. SIMPLEX

Data in a Simplex transmission is always one way. Simplex transmission are not often used because it is not possible to send back error or control signals to the transmit end.

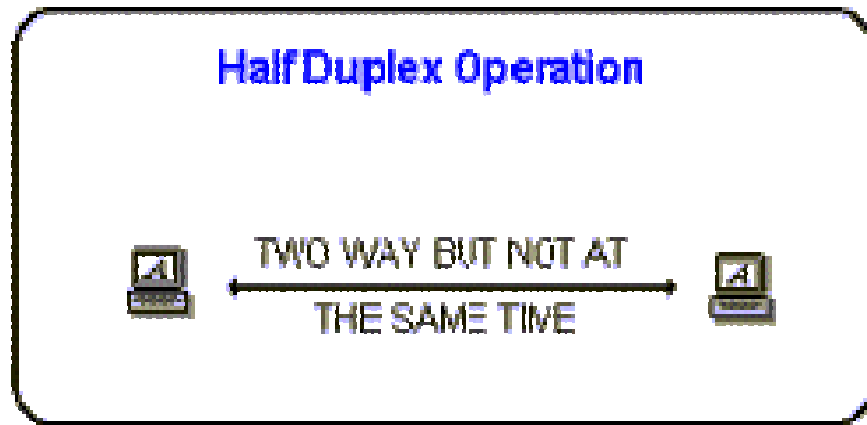
It's like a one-way street. An example of simplex is television, or Radio.



2. HALF DUPLEX

A half-duplex transmission can send and receive in one direction, but not at the same time. It's like a one-lane bridge where two-way traffic must give way in order to cross. Only one end transmits at a time, the other end receives. In addition, it is possible to perform error detection and request the sender to retransmit information that arrived corrupted. In some aspects, you can think of Internet surfing as being half-duplex, as a user issues a request for a web document, then that document is downloaded and displayed before the user issues another request.

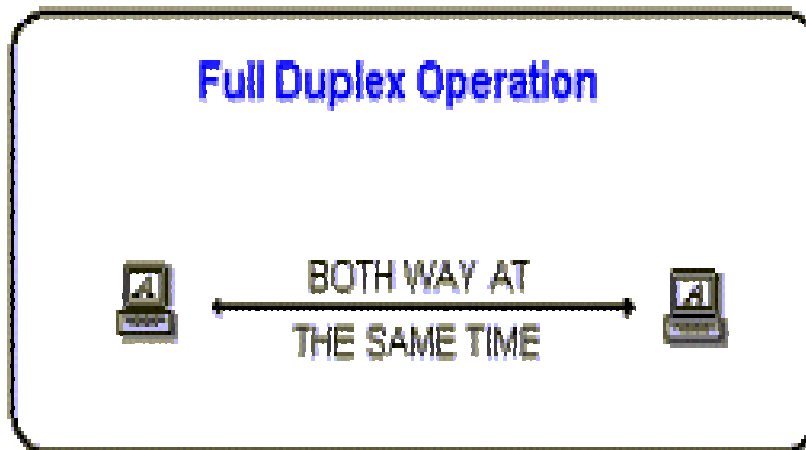
Another example of half-duplex is talkback radio, and CB Radio (Citizens Band). You might have seen movies where a truck driver (drivers of very big trucks) communicates to each other, and when they want the other person to speak they say "over". This is because only one person can talk at a time



3. FULL DUPLEX

Data can travel in both directions simultaneously. There is no need to switch from transmit to receive mode like in half duplex. It like a two lane bridge on a two-lane highway. Have you ever watched these television talk shows where the host has a number of people on the show, and they all try to talk at once. Well, that's full duplex!

Of course, in the world of data communications, full duplex allows both way communications simultaneously. An example can be a consumer, which uses a cable connection not only, receives TV channels, but also the same cable to support their phone and Internet surfing. All these activities can occur simultaneously.



Practical QUESTION

- Name any 2 word processing packages.
- What is page break in word processing and why is it used/
- How do you save a file on a floppy disk using the word processor you have studied?
- How do you underline a block of previously typed text using the word processor you have studied?

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- e. Explain the difference between
- i. **SAVE & SAVE AS**
 - ii. **COPY & CUT**
 - iii. **DELETE & BACK SPACE**
 - iv. **PORTRAIT & LANDSCAPE**
 - v. **TOGGLE CASE & UPPERCASE**
- f. **Briefly outline the procedures for performing the following functions in a windows environment.**
- | | |
|---|-------------------------------|
| i. Creating a folder | xi. Click |
| ii. Opening a folder | xii. Right click |
| iii. Searching folders or files | xiii. Double click |
| iv. Renaming a file or folder | xiv. Drag |
| v. Creating a shortcut for a file or folder | xv. Loading windows |
| vi. Copy file from one folder to another | xvi. Shutting down a computer |
| vii. Move a file from one folder to another | xvii. Emptying recycle bin |
| viii. Deleting a file or folder | xviii. |
| ix. Arrange icons on the desktop | |
| x. Change screen saver | |
- g. **Briefly outline the procedures for performing the following functions using a spreadsheet of your choice**
- i. **Expanding a cell**
 - ii. **Adding a list of figures in the range A10 to D10**
 - iii. **Insert a full border around a document**
 - iv. **Adding and Renaming Worksheets**
 - v. **Resizing Rows and Columns**

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h. Briefly outline the procedures for performing the following functions using a word processor of your choice

- | | | |
|--|--|---|
| 1. Opening a Microsoft word program for typing | 13. Exit MS Word | 28. Adding columns to a table |
| 2. Saving a document | 14. Save a document | 29. Deleting rows of a table |
| 3. Retrieving a document | 15. Copy a document or a word | 30. Deleting columns of a table |
| 4. Setting landscape or portrait orientation | 16. Double line spacing | 31. Shading a table |
| 5. Print preview | 17. Insert a header | 32. Merging cells of a table |
| 6. Formatting text into italics | 18. Insert page number | 33. Splitting cells |
| 7. Setting left and right margins | 19. Save a document onto a floppy disk | 34. Changing text direction |
| 8. Inserting a header or a footer | 20. Mail merge | 35. Inserting drop caps |
| 9. Open a document which has been saved before | 21. JUSTIFYING TEXT | 36. Having 2 columns on a word document |
| 10. Insert a picture | 22. Inserting bullets | 37. Bordering and shading a paragraph |
| 11. Insert a table | 23. Indenting text | 38. Change case of a word |
| 12. Printing a word document | 24. Moving a block of text | 39. Locking a word file with a password |
| | 25. Copying a block of text | |
| | 26. Creating a table | |
| | 27. Adding rows to a table | |

Question 26

- (a) Discuss any 4 possible uses of spreadsheets in a business organization. [8 marks]
- (b) Explain the following terms as used in spreadsheets
- Cell
 - Range. [2 marks each]
- (c) Explain the following terms as used in a word processing
- Soft return
 - Hard return
 - Thesaurus
 - Widow
 - Orphan. [2 marks each]

Question 24

- (a) Define an electronic spreadsheet. [3 marks]
- (b) List 5 facilities provided by a spreadsheet. [5 marks]
- (c) What is word processing? [2 marks]
- (d) Explain the following word processing terms
- Justification
 - Hard return
 - Soft return
 - Header
 - Orientation. [2 marks each]

Question 15

Briefly explain the following terms as used in a spreadsheet

- | | |
|----------------------|-----------------------------|
| (a) Active cell | (l) Relative cell reference |
| (b) Cell | (m) Cell pointer |
| (c) Legend | (n) Absolute cell reference |
| (d) What if analysis | (o) Mixed cell reference |
| (e) Label | (p) Work book |
| (f) FUNCTION | (q) Goal seek |
| (g) GRAPH | (r) Range |
| (h) Formula | (s) Protected cells |
| (i) Cell address | (t) Freeze panes |
| (j) Worksheet | (u) Embedded chart |
| (k) Cell reference | (v) Row & column |
- [1 mark each]

Question 16

Briefly explain the following terms as used in a Word processing

- | | |
|-------------------|--------------------------|
| (a) Orphan | (k) Word wrap |
| (b) Header | (l) Bold |
| (c) Footer | (m) Borders |
| (d) TEMPLATE | (n) Bullets |
| (e) Spellchecker | (o) Hard return |
| (f) FONT | (p) STATUS BAR |
| (g) TAB STOPS | (q) Indentation |
| (h) Mail merge | (r) Soft return |
| (i) Font type | (s) Widow/Orphan Control |
| (j) Justification | (t) Widow |

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- (u) Thesaurus
- (v) Format bar

- (w) Standard bar
- (x) Center alignment

[1 mark each]

Question 6

- (a) Define a word processor & give an example [2 marks]
- (b) List down at least 5 facilities provided by a word processor [5 marks]
- (c) Give a suitable term/ phrase that best describes the following statements in word processing.
 - i) The automatic movement of typed text to the new line, without pressing the enter key.
 - ii) Different "handwritings" in a computer.
 - iii) Line spacing. [1 mark each]
- (d) Define a spreadsheet and give an example [2 marks]
- (e) Give the difference between a spreadsheet & a worksheet [2 marks]
- (f) Give a suitable word/ phrase that best describes the following spreadsheet terms
 - I. Cell [2 marks]
 - II. Hard copy [2 marks]
 - III. Creating a permanent copy of a worksheet on a computer. [2 marks]

QUESTION 31

- (a) Define the term Operating System. [3 marks]
- (b) State any 4 functions of an operating system. [4 marks]
- (c) State any 3 examples of an Operating system (exclude the Windows family). [3 marks]
- (d) Define the following Windows Operating Systems based terms:
 - I. Desktop
 - II. Icon
 - III. Taskbar
 - IV. Window
 - V. Cascade window [2marks each]

Question 6

Expand the following abbreviations.

- | | | | |
|------------|-----------|------------|----------------------|
| a) GIGO | o) DBMS | cc) SRAM | qq) ALU |
| b) WYSIWYG | p) POP 3 | dd) EFT | rr) DOS |
| c) LED | q) HTML | ee) WAN | ss) FORTRAN |
| d) LCD | r) URL | ff) PC | tt) COBOL |
| e) MS DOS | s) TCP-IP | gg) MIS | uu) ISP |
| f) OCR | t) CAD | hh) EBCDIC | vv) CPU |
| g) MICR | u) RAM | ii) MHz | ww) WWW |
| h) CD ROM | v) ROM | jj) OMR | xx) DVD |
| i) ASCII | w) DSS | kk) SDLC | yy) VOIP |
| j) COM | x) LAN | ll) HTTP | zz) ISDN |
| k) WIMP | y) 4GL | mm) CPU | aaa) URL |
| l) GUI | z) UPS | nn) DBA | bbb) PROM [20 marks] |
| m) WIMP | aa) PABX | oo) EPROM | |
| n) CRT | bb) DRAM | pp) VDU | |

Question 7

Explain the following terms in detail, giving examples where applicable.

- a) Computer peripherals
- b) Program
- c) BLUETOOTH
- d) Computer
- e) Hardware
- f) Software
- g) Operating system
- h) Application software
- i) Utility software
- j) Machine language.
- k) Random access memory. [20 marks]

Question 8

Explain the following terms as used in word processing or spreadsheet

- a) Menu
- b) Footer
- c) Soft return
- d) Edit
- e) Cell

- f) Label
- g) Chart
- h) Mail merge
- i) Print preview
- j) Work sheet. [2 marks each]

Questions

- (a) Discuss any 4 possible uses of spreadsheets in a business organization. [8 marks]
- (b) Explain the following terms as used in spreadsheets

I. Cell

- The intersection of a row and a column on a spreadsheet in which data can be entered.

II. Range.

- Series of two or more adjacent cells in a column or row or a rectangular group of cells.

III. [2 marks each]

- (c) Explain the following terms as used in a word processing

VI. Soft return

- A soft return is entered automatically by a word processor's word wrap feature when the text reaches the end of the line. It is contrasted with a hard return which is used primarily to begin a new paragraph. A little known feature of Microsoft Word and OpenOffice. ...

VII. Hard return

- Made by pressing the return or enter key on the keyboard. A hard return, or carriage return, creates a line break and identifies the end of a paragraph.

VIII. Thesaurus

- Provides a listing of words and their synonym/related words.
- Guide to use of terms, showing relationships between them, for the purpose of providing standardized, controlled vocabulary for information storage and retrieval

IX. Widow

- In typesetting, a widow occurs when the majority of a paragraph appears on one page with the last line of that paragraph flowing to the top of the following page. ...
- The last line of a paragraph at the top of a page of text.

X. Orphan.

- In typesetting, an orphan is the first line of a paragraph appearing on its own at the bottom of a page with the remaining portion of the paragraph appearing on the following page. The first line of the paragraph was "left behind" by the remaining portion of text. ...
- Line of type on its own at the bottom of a page.

[2 marks each]

Question 24

- (a) Define an electronic spreadsheet. [3 marks]
- (b) List 5 facilities provided by a spreadsheet. [5 marks]
- (c) What is word processing? [2 marks]
- (d) Explain the following word processing terms

VI. Justification

- a. In typesetting, justification (can also be referred to as 'full justification') is the typographic alignment setting of text or images within a column or "measure" to align along both the left and right margin. Text set this way is said to be "justified".
- b. The alignment of text in a paragraph so that the margins are all straight on the right side, or the left side, or both. E.g. this text is left justified.

VII. Hard return

VIII. Soft return

IX. Header

- Text that appears at the top of every page of a document when it is printed.

X. Orientation.

- Whether page is vertically or horizontally aligned. For example we have portrait and landscape orientation.
- **Portrait orientation.** The mode in which content is viewed where the width is shorter than the height. Portrait orientation mimics the way most reading material is printed, eg, letters, newspapers, books. ...
- **Landscape orientation.** The mode in which content is viewed where the width is longer than the height. Landscape is the traditional orientation for computer viewing and is appropriate when viewing spreadsheets, video games, and movies.

[2 marks each]

Question 15

Briefly explain the following terms as used in a spreadsheet

- | | |
|----------------------|-----------------------------|
| (a) Active cell | (l) Relative cell reference |
| (b) Cell | (m) Cell pointer |
| (c) Legend | (n) Absolute cell reference |
| (d) What if analysis | (o) Mixed cell reference |
| (e) Label | (p) Work book |
| (f) FUNCTION | (q) Goal seek |
| (g) GRAPH | (r) Range |
| (h) Formula | (s) Protected cells |
| (i) Cell address | (t) Freeze panes |
| (j) Worksheet | (u) Embedded chart |
| (k) Cell reference | (v) Row & column |

[1 mark each]

Question 16

Briefly explain the following terms as used in a Word processing

- | | |
|-------------------|---|
| (a) Orphan | (p) STATUS BAR |
| (b) Header | (q) Indentation |
| (c) Footer | (r) Soft return |
| (d) TEMPLATE | (s) Widow/Orphan Control |
| (e) Spellchecker | (t) Widow |
| (f) FONT | (u) Thesaurus |
| (g) TAB STOPS | (v) Format bar |
| (h) Mail merge | (w) Standard bar |
| (i) Font type | (x) Center alignment |
| (j) Justification | (y) Hyphenation – is used to eliminate gaps in justified text or to maintain even line lengths in narrow columns. You can hyphenate manually or automatically. |
| (k) Word wrap | |
| (l) Bold | |
| (m) Borders | |
| (n) Bullets | |
| (o) Hard return | |

[1 mark each]

Question 6

- (g) Define a word processor & give an example [2 marks]
- (h) List down at least 5 facilities provided by a word processor [5 marks]
- (i) Give a suitable term/ phrase that best describes the following statements in word processing.
 - i) The automatic movement of typed text to the new line, without pressing the enter key.
 - ii) Different “handwritings” in a computer.
 - iii) Line spacing. [1 mark each]
- (j) Define a spreadsheet and give an example [2 marks]
- (k) Give the difference between a spreadsheet & a worksheet [2 marks]
- (l) Give a suitable word/ phrase that best describes the following spreadsheet terms
 - IV. Cell [2 marks]
 - V. Hard copy [2 marks]
 - VI. Creating a permanent copy of a worksheet on a computer. [2 marks]

QUESTION 31

- (e) Define the term Operating System. [3 marks]
- (f) State any 4 functions of an operating system. [4 marks]
- (g) State any 3 examples of an Operating system (exclude the Windows family).
[3 marks]
- (h) Define the following Windows Operating Systems based terms:

I. Desktop

- The main workspace in a graphical user interface such as Windows or Macintosh Systems. Users open and work with files and programs on the desktop, and can store files and shortcuts there as well. The user can also customize the look of the desktop with images or wallpaper and custom icons.
- Is the background screen on the computer? It consists of pictures, called icons that show cabinets, files, folders, and various types of documents.

II. Icon

- A picture or graphical representation of an object on a display screen to which a user can point to with a device, such as a mouse, to select a particular operation or perform a certain action.

III. Taskbar

- The bar usually found at the bottom of a Windows screen. The taskbar shows which software programs are open. It also contains the Start button.
- A windows 95/98 screen elements, displayed on the desktops which includes the start button and lists the programs currently running on the computer.

IV. FOLDER

- Folder, a virtual container within a digital file system, in which groups of files and other folders can be kept and organized.

V. WIMP

- **WIMP** (Windows Interface Manipulation Program, also refers to Windows, Icons, Menus and Pointing device - the prehistorical GUI of the 1970's) is a program for creating full-screen visual animations synchronized with sound in real time. As the only visual input WIMP utilizes graphical user interface (GUI) of Windows operation system. Whatever applications windows, icons, images, texts, etc. you have open on your desktop - they become an inspiration source for WIMP and you. Animations are generated by simple 2- and 3-D effects and filters and their superimpositions. As such WIMP can be used as a VJ tool, a screensaver, a cool grafix generator or as a piece of conceptual art.

VI. RECYCLE BIN

- In the Microsoft Windows operating systems, the **Recycle Bin** is a holding area for files and folders that are held before final deletion from a storage device.

VII. GUI

- A **graphical user interface (GUI)** is a type of user interface item that allows people to interact with programs in more ways than typing such as computers; hand-held devices such as MP3 Players, Portable Media Players or Gaming devices; household appliances and office equipment with images rather than text commands. A *GUI* offers graphical icons, and visual indicators, as opposed to text-based interfaces, typed command labels or text navigation to fully represent the information and actions available to a user. The actions are usually performed through direct manipulation of the graphical elements.^[1]

VIII. Window

- In computing, a window is a visual area, usually rectangular in shape, containing some kind of user interface, displaying the output of and allowing input for one of a number of simultaneously running computer processes. ...

IX. Cascade window

- An arrangement of Windows so each window is neatly stacked with only the title bar of each window is showing.
- Arrange (open windows) on a computer desktop so that they overlap each other, with the title bars visible

[2marks each]

Question 7

Explain the following terms in detail, giving examples where applicable.

- a) Computer peripherals
- b) Program
- c) **BLUETOOTH**
 - a. Bluetooth® is the codename for a technology specification for low-cost, short-range radio links between mobile PCs, mobile phones and other portable devices, and connectivity to the Internet. ...
 - b. Bluetooth is the name given to a radio technology making transmission of signals over short distances between telephones, computers and other devices, like household appliances, without the use of wires.
- d) Computer
- e) Hardware
- f) Software
- g) Operating system
- h) Application software
- i) Utility software
- j) Machine language.
- k) Random access memory. [20 marks]

Question 8

Explain the following terms as used in word processing or spreadsheet

k) Menu

- a. A list of options, each of which performs a desired action such as choosing a command or applying a particular format to a part of a document. Menus are commonly used in graphical interfaces.

- l) Footer
- m) Soft return

n) Edit

- o To review a piece of writing, marking and correcting grammatical, spelling, and factual errors. The editing process also often includes shortening or lengthening articles to fit available space, writing headlines and subheads.

- o) Cell
- p) Label
- q) Chart
- r) Mail merge
- s) Print preview
- t) Work sheet. [2 marks each]

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ABBREVIATIONS

- 1) **ALU** Arithmetic Logic Unit
- 2) **ARC net** Attached Resource Computer Network
- 3) **ASCII** American Standard Code for Information Interchange
- 4) **B2B** Business-To-Business
- 5) **BCD** Binary Coded Decimal
- 6) **BIOS** Basic Input/Output System
- 7) **BIS** Business Information System
- 8) **BIT** Binary Digit
- 9) **BLOG** Web Log
- 10) **BSYNC** Binary Synchronous Communications (protocol)
- 11) **C** C Programming Language
- 12) **CAD** Computer Aided Design
- 13) **CAL** Computer Aided Learning
- 14) **CASE** Computer Aided Software Engineering
- 15) **CDMA** Code Division Multiple Access
- 16) **CD-R** Compact Disk - Recordable
- 17) **CD-ROM** Compact Disk - Read Only Memory
- 18) **CD-RW** Compact Disk - Rewritable
- 19) **COBOL** Common Business-oriented Language (See HLL)
- 20) **CODEC** Coder/Decoder + Compression/Decompression
- 21) **COE** Common Operating Environment
- 22) **CPU** Central Processing Unit
- 23) **CRC** Cyclic Redundancy Check
- 24) **CRT** Cathode Ray Tube
- 25) **CSMA/CA** Carrier Sense Multiple Access/with Collision Avoidance
- 26) **CSMA/CD** Carrier Sense Multiple Access/with Collision Detection
- 27) **CSMS** Customer Support Management System
- 28) **DSS** Decision Support System
- 29) **DTE** Data Terminal Equipment + Dumb Terminal Emulator
- 30) **DVD** Digital Video Disk
- 31) **DVD-A** Digital Video Disk-Audio
- 32) **DVD-AR** Digital Video Disk-Audio Recording
- 33) **DVD-R** Digital Video Disk-Recordable
- 34) **DVD-RAM** Digital Versatile Disc-RAM
- 35) **DVD-ROM** Digital Versatile Disc-ROM
- 36) **DVD-SR** Digital Video Disk-Stream Recording
- 37) **DVD-VR** Digital Video Disk-Video Recording
- 38) **DVR** Digital Video/Voice Recorder
- 39) **EBCDIC** Extended Binary Coded Decimal Interchange Code [IBM]
- 40) **EEPROM** Electrically Erasable Programmable Read-Only Memory
- 41) **EFTS** Electronic Funds Transfer System
- 42) **E-MAIL** Electronic Mail
- 43) **EPROM** Electrically Programmable Read Only Memory +
i. Erasable Programmable Read Only Memory
- 44) **FTP** File Transfer Protocol [Internet]
- 45) **FDDI** Fiber Digital Device Interface +
i. Fiber Distributed Data Interface
- 46) **FDISK** Fixed Disk
- 47) **FDM** Frequency Division Multiplexing
- 48) **FDMA** Frequency Division Multiple Access
- 49) **FDX** Full Duplex
- 50) **FEC** Forward Error Correction
- 51) **FM** Frequency Modulation
- 52) **FORTRAN** Formula Translator (Programming Language)(See HLL)
- 53) **FTP** File Transfer Protocol
- 54) **GHZ** Gigahertz
- 55) **GIGO** Garbage In, Garbage Out
- 56) **GIS** Geographic Information System
- 57) **GSM** Global System for Mobile-Communications (network)
- 58) **GUI** Graphical User Interface
- 59) **HDD** Hard Disk Drive
- 60) **HDLC** High-Level Data Link Control
- 61) **HLL** HIGH LEVEL LANGUAGE
- 62) **HDTV** High Definition Television
- 63) **HTML** HyperText Markup Language
- 64) **HTTP** HyperText Transfer Protocol
- 65) **Hz** Hertz
- 66) **IP** Internet Protocol

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- 67) **IQL** Interactive Query Language
- 68) **ISDN** Integrated Services Digital Network
- 69) **IT** Information Technology
- 70) **LAN** Local Area Network
- 71) **LCD** Liquid Crystal Display
- 72) **LED** Light Emitting Diode
- 73) **LPT** Line Printer Terminal
- 74) **LPT1** First Parallel Printer Port
- 75) **LPT2** Second Parallel Printer Port
- 76) **LPT3** Third Parallel Printer Port
- 77) **MBps** Megabytes Per Second
- 78) **Mbps** Megabits Per Second
- 79) **MBR** Master Boot Record
- 80) **MHz** Megahertz
- 81) **MICR** Magnetic Ink Character Recognition
- 82) **MIDI** Musical Instrument Digital Interface
- 83) **MIS** Management Information System
- 84) **MODEM** Modulator/Demodulator
- 85) **MP3** MPEG Audio Layer 3
- 86) **MSAV** Microsoft Anti Virus [Microsoft]
- 87) **MS-DOS** Microsoft - Disk Operating System [Microsoft]
- 88) **PDF** Package Definition File + Portable Document Format +
 - i. Portable Document File + Processor Defined Function +
 - ii. Program Development Facility
- 89) **.PDF** Printer Description (file name extension)[Borland, Lotus]
- 90) **P2P** Peer To Peer + People To People
- 91) **USB** Universal Serial Bus
- 92) **WYSBYGI** What You See Before You Get It
- 93) **WYSIWYG** What You See Is What You Get

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Basic Computer Terminology

Computer has turned from being just a technological gadget to personal PC to a super technological personal gadget. There is nothing in the world that does not depend on computers. You need to understand the basic computer terms and meanings in order to stay up-to-date with the world. If you do not know the basic computer terms and definition, you look like a lost soul in the most happening world of technology. You need to book a movie ticket, so you need Internet access. You have to submit a presentation for your company's project, you need a computer to bail you out. Its out of date to use a video cassette for your recordings. Cassettes are like an ancient relic, its CD's and DVDs that are in vogue today. And to top it all, USB flash drives have created a niche in the minds of tech-savvy people. So before you float your mind in the floating point numbers or scratch your head when learning the queries in databases, let us begin with some simple basic computer terms for beginners.

Basic Computer Terms and Definitions

Computer

A computer is a computing device that is similar to an automated abacus that can execute a program, that is, a sequence of automatic instructions.

Program

A program is a is an executable software that runs on a computer. It contains compiled codes that run directly from the computer's operating system.

Operating System

The operating system (OS) is the software that communicates with the computer hardware on the primary level. No software can be run on the computer without the operating system. It serves as a user interface and helps allocating memory, processes tasks, accesses disks and peripheral.

Command

A command is a series of step by step instructions that helps the computer perform a specific action.

CPU

The central processing unit or CPU, also known as the microprocessor or processor is the brain of the computer. It helps execute instructions in a software program and helps retrieve instructions form the computer's memory, comprehending and executing instructions and helping direct the input and output of the computer.

Desktop

The computer monitor or the screen of the computer against which, the operating system and access application programs, files and documents are displayed is called the desktop.

File

The file is a unit for information storage that includes word-processor, spreadsheet, pictures, music, etc. Each file is differentiated with the help of a unique file name.

Folder

The folder is a unique system that helps in organizing the files, topics, programs and projects on the computer. The similar applications in a folder will be grouped together when an application program is loaded. It is similar to organizing files on a shelf in your cabinet.

Hard Drive

The hard drive is the region where the information is stored and helps the computer to

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permanently retain and store the data.

Hardware

The computer system that comprises of a computer, that is, monitor, keyboard, mouse, printer, computer unit, scanner, etc. is known as the hardware. When the CPU is opened, the components that make a CPU are also the hardware. You can read and learn more on computer hardware basic information.

Hyperlink

An hyperlink is a program that helps one move from one web page to another. The text is underlined and the cursor that moves over this text document, takes the shape of a hand. This means there is a hyperlink added to the text and you can click on it to move over to another page or document.

Internet

The Internet is a world wide network that helps connect millions of users around the world to share and exchange data, information, opinions, etc. The Internet is not similar to World Wide Web (WWW). The WWW is a service that is provided on the Internet for the users.

Monitor

The computer monitor is defined as the computer screen or display unit. The monitor helps in displaying the user interface and programs. It is a way for the user to interact with the computer, using the keyboard and mouse.

Mouse

Do not be baffled, a computer mouse is a hand held device that helps the user move across the flat surface and help control the pointer on the screen. It is a very accessible device, as it helps one move faster across the screen and perform many tasks quickly.

Keyboard

The computer keyboard is the peripheral unit of the computer that helps in typing in the words and numbers along with the symbols and thus communicate with computer. In other words, it is the way, how you can feed in information that needs to be processed into your computer. You can read more on computer keyboard shortcuts.

RAM

Random Access Memory or RAM is the temporary storage space in the computer that helps place information so that it can execute the program and instructions given. Once a program is closed, the data is removed from RAM. The amount of RAM helps in determining the number of programs that can be run.

Browser

The web browser is an application that helps render the source code of World Wide Web or web pages that can be used by the user. The information is rendered in understandable form for the user by the browser. The Microsoft Internet Explorer is an example of the browser.

Peripherals

The peripherals are the input devices that are connected to the machines. The keyboard, mouse and printer are computer peripherals. Some peripherals are important for the working of the machine and some are added components, without which one can work on the computer.

Networks

The network is a physical or logical construction that connects different computers together and helps them communicate. The computer networking is carried out through cabling or through wireless networking. The best example is the Internet that helps sites to connect on different computers.

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Software

The instructions that are provided to the computer with step-by-step actions that help executing a specific task is a software. A computer cannot function without a software.

These were a few basic computer terms for kids that can be helpful for a beginner to understand the computer parts list. Computers have become a part and parcel of our daily lives, without which we cannot function. Man has become so dependent on the computer that a minor computer glitch can send life into a tizzy. One needs to learn using a computer, as much as he needs to learn how to read and write. Today, people breath software and eat hardware (hypothetically speaking). You do not want to be left out in this rat race of technically sound world. So begin by learning and teaching your kids these basic computer terms for beginners.

QUIZ

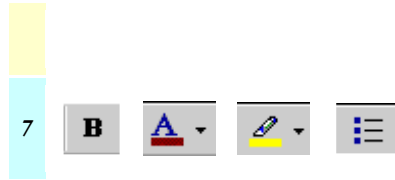
	A	B	C	D	
1					You have completed the document and want to make certain that you have not made spelling or grammar errors. Which button do you select?
2					Text is selected and you want to remove the selection from the active document and place it on the clipboard. Which button do you select?
3					You have made a change to the active document and want to make certain that those changes are saved. Which button do you select?
4					A phrase will appear several times in the document. To save time you have selected the phrase and want to copy it. Which button do you select?
5					The document had been checked for spelling errors, it has been saved, and now you are ready to print. Which button do you select?
6					A document which has been saved into a folder on your computer is to be opened so changes can be made. Which button do you select?
7					The document had been checked for spelling and has been saved. Before sending it to the printer you want to see how it will look. Which button do you select?

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	A	B	C	D	
1					Instead of striking the underline key fifteen times, you plan to draw a single line by clicking and dragging. Which button do you select?
2					You have drawn a line, and now want to change the color of the selected line. Which button do you select?
3					The last line you drew had an arrowhead at the wrong end. You want to change the arrowhead to the other end of the line. Which button do you select?
4					You created an object in a document and you want to tilt it at an angle. Which button do you select?
5					You have drawn a rectangle around a section of a flyer and want to change it to a dashed line. Which button do you select?
6					The heading of a flyer is just plain text and you would rather create special effects with text. Which button do you select?
7					You are going to draw a perfect circle on your document by holding down the shift key, clicking and dragging. Which button do you select?

	A	B	C	D	
1					You wish to call attention to a statement in your document by making the text bold. Which button do you select?
2					A flyer is to be prepared inviting parents to a performance. You want the headline to be centered on the page. Which button do you select?
3					You want to make sure that your instructions are clear and plan to place them in a bulleted list. Which button do you select?
4					To call attention to a block of text, you want place a block of color behind the text to highlight it. Which button do you select?
5					Another way to call attention to a block of text is to place a border around it. Which button do you select?
6					A quoted block of text has been indented. You wish to continue the document but you need to undo the

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indent.

Which button do you select?

You have made a birthday certificate and you want the student's name to be displayed in color.

Which button do you select?